

Sustainability and Business Practices
An Introduction to Project Management (MGMT-3105)

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Welcome

This book is the lecture-note companion for **MGMT-3105 – Sustainability and Business Practices**, a second-year course that introduces the philosophy of *sustainability* and the discipline of *project management* to students in electromechanical and mechanical engineering technology programs.

You are training for careers in the **automotive, food-processing, and defense** industries. In every one of those sectors you will, sooner than you expect, be handed a slice of a project: a line retrofit, a new product introduction, a validation campaign, a plant move. This course is about the structure that keeps such work on scope, on schedule, and on budget — and about doing it in a way that does not quietly borrow from the future to pay for the present.

Why a course that pairs sustainability with project management?

At first glance they look like two subjects stapled together. They are not. Every project consumes resources — materials, energy, money, human effort — and every project leaves something behind: a product, a process, a waste stream, a set of habits in the organization. **Project management is the set of tools that decides how those resources are spent, and sustainability is the lens that asks whether that spend was worth it beyond the next quarter.** A project manager who can build a work breakdown structure but cannot reason about lifecycle impact is only doing half the job.

What Is Project Management? (Association for Project Management): <https://www.youtube.com/watch?v=Jk-JwtScIlw>

What Is Sustainable Development? (FN-sambandet): <https://www.youtube.com/watch?v=7V8oFI4GYMY>

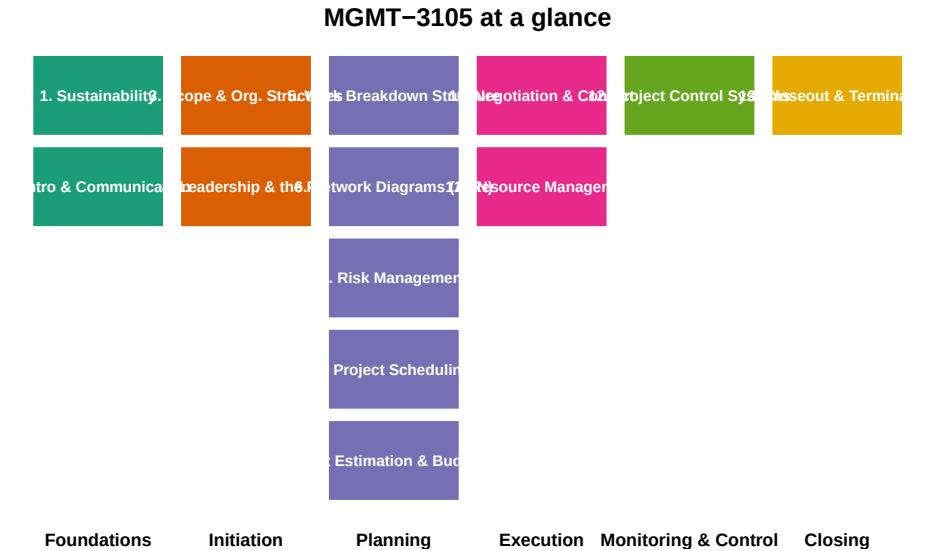


Figure 1: How the thirteen chapters of this book track the MGMT-3105 lecture sequence, grouped by project phase.

Course map

Course learning outcomes

On successful completion of MGMT-3105 you will be able to reliably:

1. Communicate the philosophy of project selection, including the Net Present Value method.
2. Communicate the theory of budgeting and scheduling.
3. Connect project-management concepts to real-world examples.
4. Identify the consequences to society of poor project planning.
5. Communicate the theory of progress management.
6. Discuss the problems of managing people and levelling allocated resources.
7. Communicate the organizational implications of closing a project.
8. Communicate negotiation theory.
9. Communicate effectively in a global environment.
10. Identify the use of sustainability in designs and in project management.

The running project

Throughout the book we return to one worked scenario, taken from the MGMT-3105 term project: your company designs and manages the launch of a **1/18-scale radio-controlled vehicle replica** for the toy market, then wins a contract to manufacture 100 000+ units for the following holiday season. You

choose the vehicle, the body and frame materials (steel, aluminium, or carbon fibre), and one of two Canadian motor/control-system suppliers. Every chapter uses a piece of that project — its scope, its WBS, its schedule, its budget, its risks, its closeout — so the tools connect to a single concrete thing rather than a dozen disconnected examples.

How the book is organized

Each chapter opens with measurable **learning objectives** and a real-world **hook**, develops the theory with plain-language explanations and LaTeX equations, and illustrates it with R-generated figures. Worked examples are shown step by step; wherever an example is a self-contained calculation, it is offered as a **live, editable R cell** you can run and modify in your browser. Collapsible **Practice** and **Discussion** blocks let you test yourself before revealing the answer, and each chapter closes with **curated YouTube videos** for a second explanation in a different voice.

Assessment (for the live course)

Table 1: MGMT-3105 grade breakdown

Component	Timing	Weight
Midterm test	Week 7	25 %
Final test	Week 14	25 %
Weekly quizzes	Weekly	12 %
Weekly case studies	Weekly	12 %
Assignment Phase A	Week 7	10 %
Assignment Phase B	Week 11	10 %
Assignment Phase C	Week 14	6 %

Quizzes and case studies require attendance. Tests are cumulative, closed-book with a handwritten aid sheet, and kit calculators are permitted. To pass, you need an overall grade of 50 % **and** completion of the mandatory work.

A note on academic integrity

Write your reports and answers in your own words. Whenever you take an idea or a passage from another author — including a classmate, a previous year's work, or an AI tool — acknowledge it with quotation marks where appropriate and a proper citation. If you are unsure how to credit shared work, ask **before** you submit. By submitting, you put your name to the integrity of the work.

Chapter 1

Sustainability

1.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Define** sustainability and sustainable design, and explain the three pillars (social, economic, ecological) and the “triple bottom line.”
2. **Identify** the four points of the sustainability platform — recycling, bio-based/biodegradable polymers, material reduction, and design for recyclability — and give an engineering example of each.
3. **Interpret** the seven plastic recycling codes and distinguish post-consumer from post-industrial recycle.
4. **Analyze** the trade-off between recyclability and biodegradability, using the corn-based PLA container case as the worked example.
5. **Calculate** the material, energy, and carbon consequences of a design decision (material substitution, wall-thickness reduction) for a production run.
6. **Evaluate** a product on whole-life cost, and explain why “cheaper to buy” and “cheaper to own” often disagree.
7. **Apply** a sustainability checklist to a project: supplier selection, contract clauses, impact assessment, circular-economy principles, and sustainability KPIs.

Before we begin: think of the last thing you threw away today

Where did its materials come from? How long did you use it before it became waste? Where is it now — landfill, recycling stream, incinerator? Could a different design decision, made years ago by an engineer you will never meet, have changed that answer? Keep the object in mind; we will come back to it.

1.2 Introduction: every gram, multiplied by a hundred thousand

Here is the situation that runs through this whole book. Your company designs and manages the launch of a **1/18-scale radio-controlled vehicle replica** for the toy market. If the prototype is accepted, you win a contract to build **100 000+ units** for the next holiday season. Early on, you must choose the body material — **steel, aluminium, or carbon fibre** — and the frame material — aluminium or carbon fibre.

That single choice is not just a strength-and-weight problem. Multiply it by 100 000 units and it becomes a decision about:

- how many tonnes of raw material leave the ground,
- how many megajoules of energy are spent turning that material into parts,
- how much scrap and packaging waste the line produces every shift,
- whether the finished toy can be recycled when a child outgrows it in three years,
- and whether your customer — a large retailer with its own public sustainability targets — will even put it on the shelf.

This is why **sustainability** belongs at the front of a project-management course, not tacked on at the end. The tools you meet in later chapters — scope, WBS, schedule, budget, risk — all decide how resources get spent. Sustainability is the question of whether that spend was worth it once you look past the launch date.

i Note

Industry snapshot. General Motors' CAMI Assembly plant in Ingersoll, Ontario — an hour from where this course is taught — runs **landfill-free**: every waste stream from daily operations is reused, recycled, or converted to energy. That is not a marketing slogan bolted onto an existing plant; it is a set of thousands of small design and process decisions of exactly the kind this chapter is about.

The Story of Stuff (Annie Leonard) - the materials economy in 20 minutes: <https://www.youtube.com/watch?v=9GorqroigqM>

1.3 What sustainability means

The most quoted definition comes from the 1987 Brundtland report and is used almost word-for-word by the U.S. Environmental Protection Agency:

Sustainability is “meeting the needs of the present without compromising the ability of future generations to meet their own needs.”

The idea is older than the word. Thomas Jefferson expressed the same principle in financial terms: **do not leave a debt for future generations to pay**. Swap “debt” for “depleted resources, degraded ecosystems, and accumulated waste” and you have the modern version.

Sustainable design applies that principle to the things engineers actually produce. Wikipedia’s working definition is useful: *the philosophy of designing physical objects, the built environment, and services to comply with the principles of social, economic, and ecological sustainability*. The intent is to *minimize negative environmental impact through skilful, sensitive design* while holding a **dynamic balance between economy and society**.

1.3.1 The three pillars and the triple bottom line

Sustainability is usually drawn as three overlapping domains. A decision is sustainable only where all three overlap: it has to be **socially** acceptable, **economically** viable, and **ecologically** sound at the same time.

```
circles <- data.frame(
  x0 = c(-0.55, 0.55, 0),
  y0 = c(0.35, 0.35, -0.6),
  r = rep(1.05, 3),
  pillar = c("Social\n(bearable)", "Economic\n(viable)", "Ecological\n(sound)"),
  fill = c("#2E86C1", "#F39C12", "#27AE60")
)

ggplot(circles) +
  ggforce::geom_circle(aes(x0 = x0, y0 = y0, r = r, fill = fill),
    alpha = 0.35, linewidth = 0.8, color = "grey30") +
  scale_fill_identity() +
  annotate("text", x = -0.95, y = 0.9, label = "SOCIAL",
    fontface = "bold", size = 4.5 * diagram_text_scale, color = "#1B4F72") +
  annotate("text", x = 0.95, y = 0.9, label = "ECONOMIC",
    fontface = "bold", size = 4.5 * diagram_text_scale, color = "#9C640C") +
  annotate("text", x = 0, y = -1.55, label = "ECOLOGICAL",
    fontface = "bold", size = 4.5 * diagram_text_scale, color = "#186A3B") +
  annotate("text", x = 0, y = 0.55, label = "bearable",
    size = 3.2 * diagram_text_scale, fontface = "italic") +
  annotate("text", x = -0.7, y = -0.35, label = "equitable",
    size = 3.2 * diagram_text_scale, fontface = "italic") +
  annotate("text", x = 0.7, y = -0.35, label = "viable",
    size = 3.2 * diagram_text_scale, fontface = "italic") +
  annotate("text", x = 0, y = -0.1, label = "SUSTAINABLE",
    fontface = "bold", size = 3.6 * diagram_text_scale, color = "grey15") +
  coord_equal() +
  labs(title = "Sustainability lives in the overlap") +
  theme_void() +
```

```
theme(
  legend.position = "none",
  plot.title = element_text(size = 14 * diagram_text_scale, face = "bold", hjust = 0
)
```

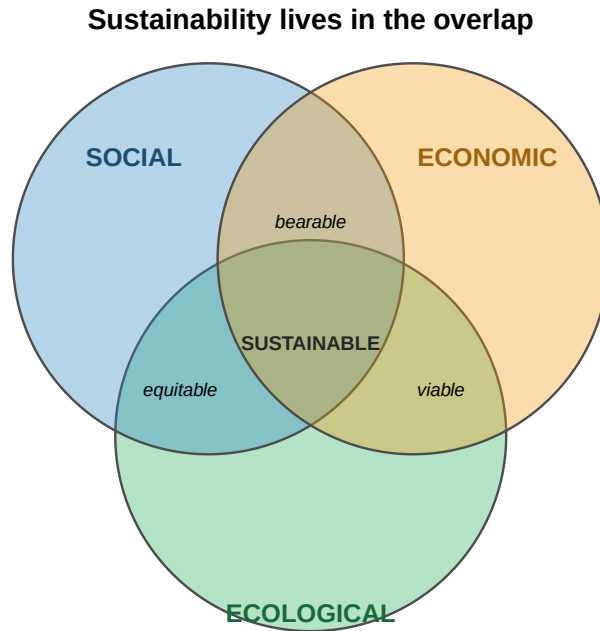


Figure 1.1: The three pillars of sustainability. Only decisions in the central overlap are genuinely sustainable; the pairwise overlaps (bearable, equitable, viable) each miss one dimension.

Discussion: which pillar does each of these decisions serve — and which does it ignore?

- A supplier offers motors 20 % cheaper, made in a plant with a poor safety record. (*economic yes; social no*)
- You switch the body to recycled aluminium, adding \$0.40 per unit. (*ecological yes; economic maybe*)
- You keep an inefficient old assembly line running because retraining staff on a new one is disruptive. (*social short-term yes; ecological and economic long-term no*)

The point of the diagram is that a “win” on one pillar is not a win. You have to check all three.

1.4 The four points of the sustainability platform

For a product engineer, sustainability comes down to four levers:

1. **Recycling** — can the material be recovered and used again?
2. **Bio-based or biodegradable polymers** — can the material come from a renewable source, or return safely to the environment?
3. **Material reduction** — can the product do its job with less material in the first place?
4. **Designing for recyclability** — is the product built so that recovery is actually practical, not just theoretically possible?

The rest of the chapter takes these one at a time, then widens out to what sustainability means when you are managing a whole project rather than designing one part.

1.4.1 Sustainable design principles

The design literature expands the four levers into a set of working principles:

Principle	What it means in practice
Low-impact materials	Non-toxic, sustainably produced or recycled materials that need less energy to process.
Energy efficiency	Choose manufacturing processes — and design products — that require less energy over their life.
Emotionally durable design	Build things people want to keep. A product kept for ten years instead of three cuts consumption and waste by more than any recycling scheme.
Design for reuse and recycling	Design for a commercial “afterlife” — but without compromising performance or service life to get there. It is a balancing act.

Notice the tension already built in: “make it last” and “make it recyclable” pull in opposite directions, and “use less material” can fight “make it durable.” Good sustainable design is the art of resolving those tensions, not pretending they do not exist.

1.5 Recycling codes

Most rigid plastics carry one of seven **resin identification codes** — the number inside the chasing-arrows triangle. The code tells a sorting facility what polymer it is looking at. The three most commonly recycled are **PET (#1)**, **HDPE (#2)**, and **PP (#5)**.

```
recycling_codes <- data.frame(
  Code = 1:7,
  Abbreviation = c("PETE / PET", "HDPE", "PVC / V", "LDPE", "PP", "PS", "OTHER"),
  Polymer = c(
    "Polyethylene terephthalate", "High-density polyethylene",
    "Polyvinyl chloride", "Low-density polyethylene",
    "Polypropylene", "Polystyrene",
    "Acrylic, polycarbonate, PLA, nylon, multi-layer"
  ),
  `Typical products` = c(
    "Drink bottles, food jars",
    "Milk jugs, detergent bottles, piping",
    "Pipe, window frames, cable insulation",
    "Films, bags, squeeze bottles",
    "Bumpers, battery cases, living hinges, tubs",
    "Foam packaging, disposable cups, CD cases",
    "Eyewear, headlamp lenses, RC-car bodies, PLA parts"
  ),
  `Recycled easily?` = c("Yes", "Yes", "Rarely", "Sometimes", "Yes", "Rarely", "Rarely"),
  check.names = FALSE
)

knitr::kable(recycling_codes, align = c("c", "l", "l", "l", "c"),
  caption = "The seven resin identification codes.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) |>
  kableExtra::column_spec(1, bold = TRUE) |>
  kableExtra::row_spec(c(1, 2, 5), background = "#EAF6EC")
```

Table 1.1: The seven resin identification codes.

Code	Abbreviation	Polymer	Typical products
1	PETE / PET	Polyethylene terephthalate	Drink bottles, food jars
2	HDPE	High-density polyethylene	Milk jugs, detergent bottles
3	PVC / V	Polyvinyl chloride	Pipe, window frames, cable
4	LDPE	Low-density polyethylene	Films, bags, squeeze bottles
5	PP	Polypropylene	Bumpers, battery cases, liv

6	PS	Polystyrene	Foam packaging, disposable cups, CD c
7	OTHER	Acrylic, polycarbonate, PLA, nylon, multi-layer	Eyewear, headlamp lenses, RC-car bodi

Two grades of recyclate matter to a manufacturer:

- **Post-industrial** (a.k.a. regrind): clean scrap from your own or a supplier's process — runners, sprues, start-up shots, off-spec parts. Easy to reuse because you know exactly what it is.
- **Post-consumer**: material that has been through a consumer's hands. It must be collected, sorted, washed, ground, and often re-pelletized, and it can carry metal, labels, food residue, and other polymers. Much harder to work with, but it is the stream that actually diverts waste from landfill.

Think about it: your RC-car body is moulded from a glass-filled ABS/PC blend. Which recycling code does it carry, and what does that tell you?

It carries **#7 (OTHER)** — any multi-polymer blend or filled material that is not one of codes 1–6 lands here. In practice, #7 is rarely recycled through curbside programs: the blend cannot be melted back into a single clean polymer. If recyclability matters for this product, that is a signal to reconsider the material *before* tooling is cut, not after.

1.6 Recyclable vs. biodegradable: a real tension

Years ago the plastics industry pursued two waste strategies at once — **recycling** and **biodegradability** — and they turned out to be partly opposed. If you make a part from recyclate and want it to last (think plastic lumber, or a pallet), the last thing you want is biodegradable material mixed in, causing the part to fail early. Contaminate a recycling stream with enough compostable polymer and the whole batch can become unusable.

The industry chose recycling. But with fossil-fuel prices and supply security back in the spotlight, **biopolymers** have returned. Some of the earliest plastics were bio-based (cellulosics, natural rubber); many new ones are built from **corn or potato starch**. **Polylactic acid (PLA)** is made from corn starch and behaves a lot like PET, but it is harder to process and degrades faster at elevated temperature.

And “biodegradable” has an asterisk. Degradation is done by **microbes**, which need **oxygen and moisture**. A modern landfill is compacted, dry, and low-oxygen — almost nothing degrades well there. You can dig up decades-old newspaper and still read it.

! Important

Case study — EcoPlast Packaging (MGMT-3105, Week 1). EcoPlast launched corn-based PLA food containers. Then: customers found them brittle in the cold and warped in heat; recyclers reported PLA contaminating conventional-plastic batches; consumers wanted eco-friendly *and* the same durability as regular plastic; and environmental groups argued the containers mostly ended up in landfill, where they did not meaningfully biodegrade — calling it “green marketing.”

Discussion (Week 1 case): was it unreasonably risky for EcoPlast to trade durability for sustainability, and what would you change?

Model answer. A product that fails in real-world use creates *more* waste, erodes consumer trust, and undermines the point of sustainability — so yes, the trade-off as executed was unreasonably risky. Consumers share *some* responsibility for unrealistic expectations, but the company had a duty to manage those expectations and to test properly before launch. Rushing to market compromised quality.

What to change: (1) real R&D and performance testing across the actual temperature and handling range before launch; (2) an accurate **caution** label (“not suitable for hot liquids”) — a danger label would be excessive; (3) consumer education on correct disposal (industrial compost, not curbside or landfill); (4) stop marketing the product as fully “sustainable” until there is evidence of an environmental benefit in real disposal conditions — otherwise the claim is exposed to a misleading-marketing challenge.

The Life Cycle of a T-Shirt (TED-Ed) - why ‘where does it go?’ is the hard question: https://www.youtube.com/watch?v=BiSYoeqb_VY

1.7 Material reduction

Reducing the material that goes into a product — at the **design stage**, by thinning walls, removing bosses, right-sizing ribs, or substituting a lighter material — pays off several times over:

- the product **costs less** (less material bought per unit),
- there is **less to recycle or dispose of** at end of life,
- **more material is left** for other uses,
- and **less energy** is spent producing *and transporting* every unit.

The catch: you cannot thin your way past the product’s functional requirements. Material reduction is enabled by better material properties, better processes, and better analysis — not by wishful thinking.

1.7.1 Worked Example 1: thinning the RC-car body

Your baseline RC-car body is injection-moulded ABS with a mass of **48 g**. Simulation says you can drop the nominal wall from 2.5 mm to 2.1 mm and still pass the drop test, cutting body mass to **41 g**. The contract run is **120 000 units**. ABS costs **\$3.10/kg**, its production embodied energy is about **90 MJ/kg**, and its cradle-to-gate carbon intensity is about **3.4 kg CO₂e/kg**.

```
mass_before_g <- 48
mass_after_g  <- 41
units         <- 120000
abs_cost_per_kg <- 3.10
abs_energy_MJ_kg <- 90
abs_co2e_kg_kg <- 3.4

saved_kg <- (mass_before_g - mass_after_g) / 1000 * units

cost_saved <- saved_kg * abs_cost_per_kg
energy_saved <- saved_kg * abs_energy_MJ_kg
co2_saved <- saved_kg * abs_co2e_kg_kg

cat("Mass removed per unit :", mass_before_g - mass_after_g, "g\n")
cat("Material saved over the run :", format(round(saved_kg), big.mark = " "), "kg\n")
cat("Material cost saved :", sprintf("$%s", format(round(cost_saved), big.mark = " ")), "\n")
cat("Production energy saved :", format(round(energy_saved), big.mark = " "), "MJ",
    sprintf("(~%s kWh)", format(round(energy_saved / 3.6), big.mark = " ")), "\n")
cat("Cradle-to-gate CO2e avoided :", format(round(co2_saved), big.mark = " "), "kg",
    sprintf("(~%.1f t)", co2_saved / 1000), "\n")
```

```
Mass removed per unit : 7 g
Material saved over the run : 840 kg
Material cost saved : $2 604
Production energy saved : 75 600 MJ (~21 000 kWh)
Cradle-to-gate CO2e avoided : 2 856 kg (~2.9 t)
```

A 7 g decision, taken once on a screen, moves almost a tonne of material and several tonnes of CO₂e once it is multiplied by the production run — before counting the lighter shipping weight.

1.7.2 Try it yourself: the material-choice decision

Now compare the three candidate body materials head to head. The live cell below holds rough figures for **steel**, **aluminium**, and **carbon-fibre composite** bodies of the mass each material would need for equal stiffness. Change the numbers, add a recycled-content option, or change the run size and see what happens to cost, energy, and carbon.

```

#| label: material-choice-compare-01
# Rough per-unit body mass for equal stiffness, and cradle-to-gate factors.
material <- c("Steel", "Aluminium", "Carbon-fibre")
mass_g    <- c(140,      70,      38)      # grams per body
cost_kg    <- c(1.20,    3.00,    28.00)    # $ per kg
energy_kg  <- c(25,     155,     300)      # MJ per kg (primary)
co2_kg     <- c(2.0,    11.0,    24.0)     # kg CO2e per kg

units <- 120000

kg_total    <- mass_g / 1000 * units
cost_total  <- kg_total * cost_kg
energy_total <- kg_total * energy_kg
co2_total   <- kg_total * co2_kg

result <- data.frame(
  Material      = material,
  Body_mass_g   = mass_g,
  Material_t    = round(kg_total / 1000, 1),
  Cost_usd      = round(cost_total),
  Energy_MWh    = round(energy_total / 3600, 1),
  CO2e_t        = round(co2_total / 1000, 1)
)
print(result, row.names = FALSE)

cat("\nLightest body:", material[which.min(mass_g)], "\n")
cat("Lowest material cost:", material[which.min(cost_total)], "\n")
cat("Lowest cradle-to-gate CO2e:", material[which.min(co2_total)], "\n")

```

What the numbers usually show — and why the “greenest” answer is not obvious

Carbon fibre gives the lightest body, which cuts shipping energy and, in a real vehicle, fuel over the whole service life. But its *production* energy and carbon per kilogram are the highest of the three, and it is the hardest to recycle (code #7, thermoset matrix). Steel is cheap and highly recycled but heavy. Aluminium sits in between and is excellent if you can get **recycled** stock — secondary aluminium uses roughly 5 % of the energy of primary.

For a *toy* that is shipped once and never driven under its own power for thousands of kilometres, the use-phase fuel saving from carbon fibre never arrives — so the decision tilts back toward recycled aluminium or steel. The lesson: **the right material depends on where the impact actually lands in the life cycle.**

1.8 Designing for recyclability

Recyclability is designed in, not hoped for. The checklist a plastics engineer can actually influence:

- Make the assembly from a **single material**, or from **compatible** materials that can be recycled together.
- Make it **easy to disassemble**.
- Prefer **snap-fits and press-fits** over screws and metal fasteners.
- **Reduce the part count**.
- **Avoid labels** (or use same-polymer labels / in-mould labelling).
- Keep **thermosets** out of thermoplastic assemblies — use thermoplastic elastomers instead of rubber seals where you can.

You may not have the final word on a design, but you can always **offer the greener alternative**. And the mindset has shifted: products used to be designed for *end of life* — last for period X, then done. Now good products are designed for **next life**: what can this material become *after* its first use?

Try it: redesign the RC-car battery door for recyclability

Current design: ABS door, two stainless M2 screws, a printed polyester label, a moulded-in TPE gasket (thermoset-like behaviour), a steel spring.

One redesign:

- Replace the two screws with a **living hinge + snap latch** moulded into the door (same ABS) — removes two dissimilar-metal fasteners and a driver operation on the line.
- Replace the adhesive polyester label with **in-mould labelling** in an ABS-compatible film, or mould the text into the part.
- Replace the steel spring with a **moulded ABS cantilever spring** if the cycle life allows.
- Keep the TPE gasket but make it a **removable press-in** part in a recycling-compatible grade, or switch to a moulded-in flexible ABS lip if the seal spec allows.

Result: the door goes from five materials and two fasteners to essentially one polymer, snaps apart by hand, and carries no adhesive.

1.9 The consumer's perspective

Sustainable design has to survive contact with real buyers. Market research is consistent on two points:

1. Environmentally conscious consumers **will pay marginally more** and **accept marginally lower fit and finish** for a product that is genuinely more environmentally friendly.

2. They have **zero tolerance** for a drop in **performance/functionality** or in **service life**. They bought the product to do a job. If a “sustainable” design wears out sooner, it just gets replaced sooner — consuming more resources and defeating the purpose.

```
tol <- data.frame(
  attribute = c("Higher price", "Cosmetic fit & finish",
               "Core performance", "Service life / durability"),
  tolerance = c(0.55, 0.45, 0.03, 0.05)
)
tol$attribute <- factor(tol$attribute, levels = rev(tol$attribute))
tol$kind <- ifelse(tol$tolerance > 0.2, "Some tolerance", "Near-zero tolerance")

ggplot(tol, aes(x = tolerance, y = attribute, fill = kind)) +
  geom_col(width = 0.65) +
  geom_text(aes(label = scales::percent(tolerance, accuracy = 1)),
            hjust = -0.15, size = 3.8 * diagram_text_scale) +
  scale_x_continuous(labels = scales::percent, limits = c(0, 0.7),
                    expand = expansion(mult = c(0, 0.05))) +
  scale_fill_manual(values = c("Some tolerance" = "#27AE60",
                              "Near-zero tolerance" = "#C0392B")) +
  labs(title = "Where consumers will --- and will not --- give ground",
       x = "Approximate share of consumers willing to accept a trade-off",
       y = NULL, fill = NULL) +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )
```

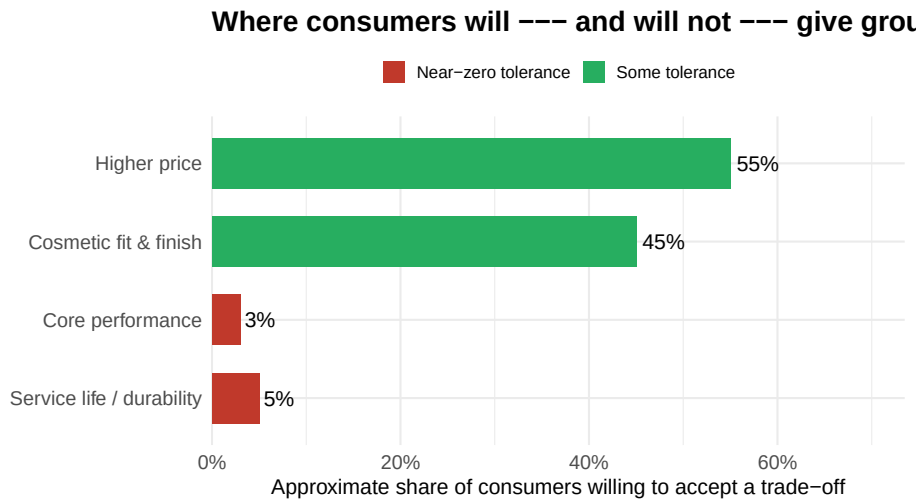


Figure 1.2: Consumer tolerance for trade-offs made in the name of sustainability: some room on price and cosmetics, essentially none on how well the product works or how long it lasts.

1.9.1 Whole-life cost: cheaper to buy vs. cheaper to own

This is where **planned obsolescence** enters. A business may have a commercial reason to limit a product's life — it protects repeat sales. Sustainable design pushes the other way: make it last. The buyer's own wallet is often on the side of durability, even when the sticker price is higher.

The live cell compares a **cheap** version of a product (low price, short life, replaced repeatedly) with a **durable** version (higher price, lasts the whole period). Change the prices, lifetimes, and horizon and watch which one wins on **total cost of ownership** — and on **units sent to landfill**.

```
#| label: whole-life-cost-01
horizon_years <- 10

# Cheap option
cheap_price    <- 18
cheap_life_yrs <- 2

# Durable option
durable_price  <- 42
durable_life_yrs <- 10

cheap_purchases <- ceiling(horizon_years / cheap_life_yrs)
durable_purchases <- ceiling(horizon_years / durable_life_yrs)
```

```

cheap_tco    <- cheap_purchases    * cheap_price
durable_tco  <- durable_purchases * durable_price

cat(sprintf("Over %d years:\n", horizon_years))
cat(sprintf("  Cheap   : %d purchases x $%d = $%d    (%d units to landfill)\n",
            cheap_purchases, cheap_price, cheap_tco, cheap_purchases))
cat(sprintf("  Durable: %d purchase(s) x $%d = $%d    (%d unit to landfill)\n",
            durable_purchases, durable_price, durable_tco, durable_purchases))

better <- if (durable_tco < cheap_tco) "Durable" else "Cheap"
cat(sprintf("\nLower total cost of ownership: %s (saves $%d)\n",
            better, abs(cheap_tco - durable_tco)))

```

Where does the cheap option actually win?

Shorten the horizon (you only need the product for a year), or make the durable version's premium extreme relative to its extra life, and “cheap” wins on cost. But the **waste count** almost always favours durable: five landfilled units versus one. Whole-life thinking means putting *both* numbers — dollars and units — in front of the decision-maker, not just the sticker price.

1.10 Sustainability in project management

Everything above was about designing a *part*. When you are running a *project*, sustainability becomes a set of management actions spread across the whole life cycle. The MGMT-3105 checklist:

Procurement and contracts

- Choose **eco-friendly and ethical suppliers**; audit them for compliance.
- Put **sustainability clauses** in contracts.
- Perform an **Environmental & Social Impact Assessment (ESIA)**.

Design and materials

- Minimize raw-material use; prioritize **recyclable, renewable, and local** resources.
- Design for **energy and water efficiency**.
- Apply **circular-economy** principles: reuse, refurbish, recycle.
- Plan the **product/service lifecycle** — what happens to it after the project ends.

Operations

- Reduce **carbon emissions** from transport and operations.
- Use **digital tools** to cut paper and travel.
- Run proper **waste management**: segregation, recycling, safe disposal.

People and governance

- Safeguard **health, safety, and well-being** of the team and the community.
- Support **social responsibility**: local hiring, fair wages, training.
- Track **sustainability KPIs**: carbon footprint, waste-reduction %, energy saved.
- Monitor suppliers and contractors; **identify and mitigate** environmental and social risks; run **sustainability audits** when needed.

1.10.1 The circular economy

The circular economy is the alternative to the linear **take — make — use — dispose** model. Instead, material is kept in use through loops of increasing size: maintain and reuse first, then refurbish and remanufacture, then recycle, with disposal as the last resort.

```
stages <- data.frame(
  label = c("Design", "Make", "Distribute", "Use", "Maintain / Reuse",
            "Refurbish / Remanufacture", "Recycle"),
  angle = seq(90, 90 - 360 + 360/7, length.out = 7) * pi / 180
)
stages$x <- cos(stages$angle)
stages$y <- sin(stages$angle)

seg <- data.frame(
  x = stages$x,
  y = stages$y,
  xend = stages$x[c(2:7, 1)],
  yend = stages$y[c(2:7, 1)]
)

ggplot() +
  geom_curve(data = seg, aes(x = x, y = y, xend = xend, yend = yend),
            curvature = -0.25, linewidth = 0.9, color = "#27AE60",
            arrow = arrow(length = unit(0.18, "cm"), type = "closed")) +
  geom_point(data = stages, aes(x = x, y = y), size = 20, color = "#EAF6EC") +
  geom_point(data = stages, aes(x = x, y = y), size = 20, shape = 1, color = "#27AE60") +
  geom_text(data = stages, aes(x = x, y = y, label = label),
            size = 3 * diagram_text_scale, fontface = "bold",
            lineheight = 0.9, color = "grey15") +
  annotate("text", x = 0, y = 0, label = "keep materials\nin use",
            size = 3.2 * diagram_text_scale, fontface = "italic", color = "grey40") +
  coord_equal(xlim = c(-1.4, 1.4), ylim = c(-1.4, 1.4)) +
  labs(title = "Design out waste; circulate materials") +
  theme_void() +
  theme(plot.title = element_text(size = 14 * diagram_text_scale,
```

```
face = "bold", hjust = 0.5))
```

Design out waste; circulate materials

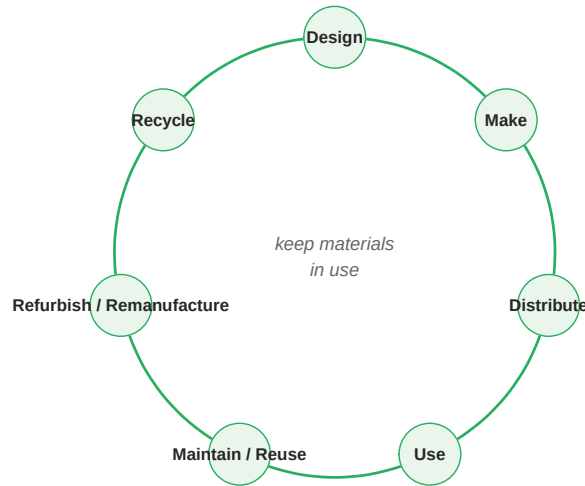


Figure 1.3: The circular economy: keep materials circulating through the tightest loop that still works. Inner loops (maintain, reuse) retain the most value; recycling is the outermost loop before disposal.

Explaining the Circular Economy (Ellen MacArthur Foundation):
<https://www.youtube.com/watch?v=zCRKvDyyHmI>

1.10.2 Try it yourself: a sustainability KPI snapshot

Sustainability only gets managed if it gets measured. The live cell computes a few of the KPIs from the checklist for the RC-vehicle project: carbon footprint per unit, waste-diversion rate, and recycled-content share. Adjust the inputs to your own project assumptions.

```
#| label: sustainability-kpi-01
units      <- 120000

# Carbon
material_co2_t  <- 71      # tonnes CO2e, cradle-to-gate materials
process_co2_t   <- 18      # tonnes CO2e, moulding + assembly energy
```

```

transport_co2_t <- 26      # tonnes CO2e, inbound + outbound freight

# Waste
process_waste_t <- 9.0     # tonnes of scrap + packaging generated
waste_diverted_t <- 8.1    # tonnes reused / recycled / to-energy

# Materials
recycled_input_t <- 15.5   # tonnes of recycled feedstock
total_input_t    <- 62.0   # tonnes of material input

total_co2_t <- material_co2_t + process_co2_t + transport_co2_t

cat(sprintf("Carbon footprint      : %.1f t CO2e total,  %.3f kg CO2e per unit\n",
            total_co2_t, total_co2_t * 1000 / units))
cat(sprintf("Waste diversion rate  : %.1f %% (target: >= 90 %%)\n",
            100 * waste_diverted_t / process_waste_t))
cat(sprintf("Landfilled waste      : %.2f t\n", process_waste_t - waste_diverted_t))
cat(sprintf("Recycled content      : %.1f %% of material input\n",
            100 * recycled_input_t / total_input_t))

status <- if (100 * waste_diverted_t / process_waste_t >= 90) "PASS" else "BELOW TARGET"
cat(sprintf("\nWaste KPI status: %s\n", status))

```

1.11 Key Takeaways

- **Sustainability** = meeting present needs without compromising future generations' ability to meet theirs. **Sustainable design** applies that across social, economic, and ecological pillars at once — a “win” on one pillar is not a win.
- The four levers for a product engineer: **recycle, bio-source/biodegrade, reduce material, design for recyclability**.
- The **seven resin codes** tell a sorter what a plastic is; PET (#1), HDPE (#2), and PP (#5) recycle well, #7 (blends/PLA/filled) rarely does. **Post-consumer** recyclate diverts real waste but is hard to process; **post-industrial** regrind is easy.
- **Recyclable and biodegradable can conflict**, and “biodegradable” does little in a dry, compacted landfill — the EcoPlast case shows what happens when that asterisk is ignored.
- A small **material-reduction** decision, multiplied by the production run, moves tonnes of material, energy, and CO2e. The “greenest” material depends on **where in the life cycle the impact lands** — use phase vs. production vs. end of life.
- Consumers give ground on **price and cosmetics** but not on **performance or durability**; **whole-life cost** usually favours the durable de-

sign on both dollars and landfill count.

- In **project management**, sustainability is a checklist across procurement, design, operations, and governance — backed by **circular-economy** loops and tracked with **KPIs** (carbon per unit, waste diversion, recycled content).

1.12 Practice Problems

Problem 1: Packaging substitution (food industry)

A food-processing client ships 2.4 million meal trays a year. The current tray is 22 g of black CPET (#1, but often rejected by sorters because black plastic is invisible to optical sorting). A proposed natural-coloured PP (#5) tray weighs 19 g, costs \$0.006/g versus \$0.005/g for the CPET, and is accepted by the client's regional recycler.

- What is the annual change in material mass and material cost?
- Give two non-cost arguments for switching, and one argument against.

Solution.

```
trays      <- 2.4e6
mass_cpet_g <- 22
mass_pp_g  <- 19
cost_cpet  <- 0.005
cost_pp    <- 0.006

mass_change_kg <- (mass_pp_g - mass_cpet_g) / 1000 * trays
cost_change    <- (mass_pp_g * cost_pp - mass_cpet_g * cost_cpet) * trays

cat("Annual material mass change :", format(round(mass_change_kg), big.mark = " "), "kg\n",
    "(a reduction)\n")
cat("Annual material cost change : ",
    sprintf("%s", format(round(cost_change), big.mark = " ")),
    " (an increase)\n", sep = "")
```

Annual material mass change : -7 200 kg (a reduction)

Annual material cost change : \$9 600 (an increase)

- About 7 200 kg less plastic per year, but roughly \$3 700/yr more spent on material because PP costs more per gram here and the 3 g saving does not fully offset it.
- For:* the natural-coloured PP is actually recovered by the recycler (black CPET usually is not), and PP trays tolerate reheating better. *Against:* higher material spend, and a tooling change is needed — capital plus a qualification run.

Problem 2: Recycled-aluminium body (automotive/toy)

Switching the RC-car body from primary to **recycled** aluminium keeps the mass at 70 g/unit but changes the cradle-to-gate energy from 155 MJ/kg to 8 MJ/kg and the carbon from 11 kg CO₂e/kg to 0.6 kg CO₂e/kg. Recycled stock costs \$0.25/kg more. Run size 120 000 units.

Estimate the energy saved, the CO₂e avoided, and the extra material cost.

Solution.

```
mass_kg    <- 70 / 1000
units      <- 120000
kg_total   <- mass_kg * units

energy_saved <- kg_total * (155 - 8)
co2_saved    <- kg_total * (11 - 0.6)
extra_cost   <- kg_total * 0.25

cat("Aluminium in the run :", format(round(kg_total), big.mark = " "), "kg\n")
cat("Production energy saved :", format(round(energy_saved), big.mark = " "), "MJ",
    sprintf("(~%s kWh)\n", format(round(energy_saved / 3.6), big.mark = " ")))
cat("CO2e avoided :", format(round(co2_saved), big.mark = " "), "kg",
    sprintf("(~%.1f t)\n", co2_saved / 1000))
cat("Extra material cost :",
    sprintf("$%s\n", format(round(extra_cost), big.mark = " ")))
```

```
Aluminium in the run : 8 400 kg
Production energy saved : 1 234 800 MJ (~343 000 kWh)
CO2e avoided : 87 360 kg (~87.4 t)
Extra material cost : $2 100
```

Roughly 1.2 million MJ (about 340 000 kWh) and 87 t of CO₂e avoided for about \$2 100 of extra material cost — one of the highest-leverage sustainability moves available on this project.

Problem 3: Design-for-recyclability audit (defense)

A defense client requires that field-returned electronic enclosures be **materially recoverable**. You are handed this enclosure: glass-filled nylon (#7) housing, 6 zinc-plated steel screws, a bonded aluminium RFI shield, a polyurethane potting compound over the PCB, and two adhesive asset-tracking labels.

List, in priority order, the four changes that would most improve material recovery, and note the functional risk of each.

Model answer.

1. **Replace the potting compound** with a removable conformal coating or a gasketed enclosure. *Risk*: reduced shock/moisture protection — must re-qualify to the environmental spec.

2. **Eliminate the bonded aluminium shield**; use a conductive coating on the nylon or a snap-in shield of the same-family metal. *Risk*: RFI performance must be re-verified.
3. **Swap the 6 steel screws** for moulded snap features or same-metal fasteners. *Risk*: fewer re-assembly cycles; captive-hardware requirement may forbid it.
4. **Drop adhesive labels** for moulded-in or laser-etched marking. *Risk*: contrast/legibility and traceability sign-off.

The housing polymer itself (#7 glass-filled) is hard to change if the structural spec depends on it — note it as a known limitation rather than a quick fix.

Problem 4 (discussion): planned obsolescence vs. sustainable design

“Sustainable design requires that our products last as long as possible. But business decisions may deliberately shorten product life to protect repeat sales — planned obsolescence. Which do you prefer, and how would you reconcile the two in a company that has to stay profitable?”

There is no single right answer — this is a position you argue. Strong answers usually land on **service and circular business models**: the manufacturer keeps revenue through repair, upgrades, refurbishment, take-back, or product-as-a-service, so “make it last” and “keep earning” stop being opposites. Weak answers just pick a side and ignore the constraint on the other.

1.13 Further Resources

- **U.S. EPA — Learn About Sustainability**: <https://www.epa.gov/sustainability/learn-about-sustainability>
- **Brundtland Report (*Our Common Future*, 1987)**: <https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf>
- **Ellen MacArthur Foundation — Circular Economy Introduction**: <https://www.ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview>

The Story of Stuff: <https://www.youtube.com/watch?v=9GorqroigqM>

Explaining the Circular Economy (Ellen MacArthur Foundation): <https://www.youtube.com/watch?v=zCRKvDyyHmI>

What Is Sustainable Development?: <https://www.youtube.com/watch?v=7V8oFI4GYMY>

Chapter 2

Introduction to Project Management and Communication

2.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Distinguish** a project from a process, and explain why project work needs its own discipline.
2. **Describe** the triple constraint (scope, time, cost) and its relationship to quality and resources.
3. **Identify** the five process phases of a project and what happens in each.
4. **Apply** the project manager's decision hierarchy and name the four pillars of the PMI Code of Ethics.
5. **Build** a project communication plan and choose the right method (formal/informal, written/verbal) for a message.
6. **Calculate** the number of communication channels on a team and explain why they grow faster than headcount.
7. **Apply** active- and effective-listening techniques and recognize the listening filters that distort a message.
8. **Select** the appropriate performance report (status, progress, trend, forecast, variance, earned value) for a stakeholder need.

Before we begin: think of a group assignment that went badly

Was the problem that nobody knew *what* to do, or that people knew but never told each other? Most project failures are communication failures wearing a different coat. Keep that example in mind.

2.2 Introduction: only one project in four

Here is a sobering number that opens this course: by common industry estimates, only about **26 % of projects** are delivered successfully while respecting the triple constraint — on scope, on time, on budget. The other three in four slip, overspend, descope, or fail outright.

That is why project management exists as a discipline. Running the RC-vehicle launch from the last chapter is not the same as running a production line. A line is a **process**: repetitive, permanent, tuned once and then kept stable. A **project** is a one-time push toward a unique goal, under a deadline and a budget, and it needs different tools and a different mindset.

! Important

Case study — Orion Tech (MGMT-3105, Week 2). A software company built a healthcare app. The team was skilled, but: developers were never clear on requirements and reworked constantly; the PM ran everything through email and buried the team; stakeholders got no status reports; a few voices dominated meetings while others went silent; and tone and urgency were misread between the local team and offshore developers. Deadlines slipped, the budget grew, and the client's satisfaction collapsed. Leadership asked whether the PM had followed any communication protocol at all.

What Is Project Management? (Association for Project Management): <https://www.youtube.com/watch?v=Jk-JwtScIlw>

2.3 Project versus process

	Project	Process
Duration	Temporary, one-time	Permanent, ongoing
Work	Unique, complex, connected activities toward one goal	Repetitive, designed to be repeated
Ends when	The goal is met (by a specific time, within budget, to spec)	It doesn't — it is improved and kept running
Example on our project	<i>Designing and launching the RC-vehicle replica</i>	<i>Moulding 100 000 bodies once the line is running</i>

Projects help companies cope with **shortened product life cycles**, **narrow launch windows**, **increasingly complex products**, and **global markets**.

The RC-vehicle brief has all four: a holiday launch window, a technical product, and suppliers on two continents.

2.4 The triple constraint

Every project is boxed in by **scope**, **time**, and **cost**, with **quality** sitting in the middle as the measure of whether stakeholders are satisfied. Resources (people, equipment, facilities) are the fourth practical limit.

```
tri <- data.frame(
  x = c(0, 2, 4, 0),
  y = c(0, 3.4, 0, 0)
)
ggplot(tri, aes(x, y)) +
  geom_polygon(fill = "#5DADE2", alpha = 0.25, color = "#2E86C1", linewidth = 1.1) +
  annotate("text", x = 1, y = 1.9, label = "SCOPE", angle = 60,
    fontface = "bold", size = 4.2 * diagram_text_scale, color = "#1B4F72") +
  annotate("text", x = 3, y = 1.9, label = "TIME", angle = -60,
    fontface = "bold", size = 4.2 * diagram_text_scale, color = "#1B4F72") +
  annotate("text", x = 2, y = -0.28, label = "COST",
    fontface = "bold", size = 4.2 * diagram_text_scale, color = "#1B4F72") +
  annotate("text", x = 2, y = 1.15, label = "QUALITY",
    fontface = "bold", size = 4 * diagram_text_scale, color = "#B03A2E") +
  annotate("text", x = 2, y = 0.72,
    label = "(stakeholder satisfaction)", fontface = "italic",
    size = 3 * diagram_text_scale, color = "grey35") +
  coord_equal(xlim = c(-0.6, 4.6), ylim = c(-0.7, 3.8)) +
  labs(title = "Pick two to fix; the third must flex") +
  theme_void() +
  theme(plot.title = element_text(size = 14 * diagram_text_scale,
    face = "bold", hjust = 0.5))
```

Pick two to fix; the third must flex



Figure 2.1: The triple constraint. Scope, time, and cost form the sides; quality is the area they enclose. Change one side and at least one other must move, or quality suffers.

Think about it: the retailer moves the RC-vehicle launch four weeks earlier. Which constraint gives?

Time just shrank. To hold the launch you must either **cut scope** (fewer body options, defer a feature), **add cost** (overtime, expedited tooling, a second supplier), or **accept lower quality** (less testing) — and that last one is how projects like Orion Tech’s die. The PM’s job is to make that trade explicit and get a decision, not to quietly absorb it.

2.5 The five process phases

```
phase_x <- seq(0, 10, length.out = 200)
effort <- dnorm(phase_x, mean = 6, sd = 2.2)
df <- data.frame(x = phase_x, effort = effort / max(effort))
```

```

ggplot(df, aes(x, effort)) +
  geom_area(fill = "#5DADE2", alpha = 0.4) +
  geom_line(color = "#2E86C1", linewidth = 1.1) +
  annotate("segment", x = c(2, 4, 8, 9.2), xend = c(2, 4, 8, 9.2),
           y = 0, yend = 1.05, linetype = "dotted", color = "grey55") +
  annotate("text", x = c(1, 3, 6, 8.6, 9.7), y = 1.12,
           label = c("Initiating", "Planning", "Executing", "Controlling", "Closing"),
           size = 3.2 * diagram_text_scale, fontface = "bold") +
  coord_cartesian(ylim = c(0, 1.2)) +
  labs(title = "Project management process life cycle",
       x = "Project timeline", y = "Resources and effort") +
  theme_minimal() +
  theme(
    axis.text = element_blank(),
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale)
  )

```

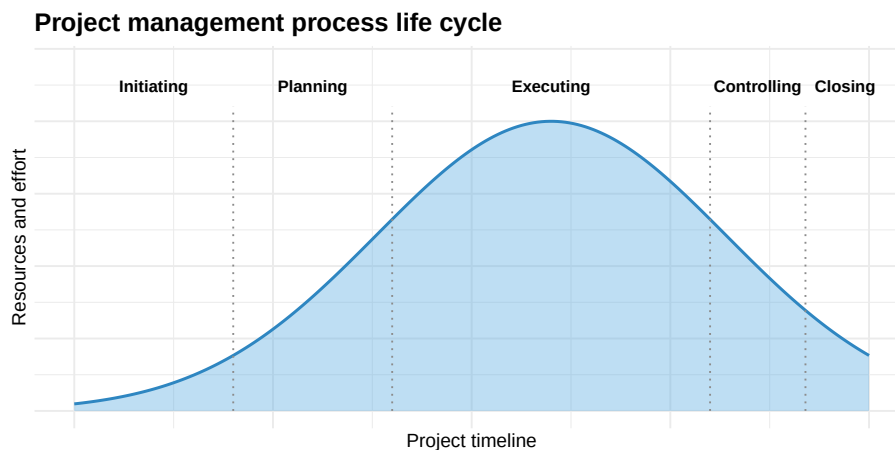


Figure 2.2: Effort and resource use across the project life cycle. Little happens during initiation, effort peaks through execution and control, and it falls away at closing.

Phase	Key work
1. Initiating	Business case, feasibility, project sponsors, project charter + sign-off.

Phase	Key work
2. Planning	Scope statement, WBS, network diagram (AON), budget, risk plan, quality standards, control system, communication protocols. <i>This phase decides success or failure.</i>
3. Executing	Execute the plan, monitor progress, quality assurance. The PM shifts from doing to leading.
4. Controlling	Scope/change/schedule/budget/risk control become daily; overlaps execution.
5. Closing	Final review of budget, schedule, scope; customer sign-off; lessons learned ; release resources; final payments. The hardest phase to finish well.

2.5.1 The project manager's decision hierarchy

When decisions conflict, this is the order of consideration:

1. **Safety** (always first)
2. What is best for **the company I represent**
3. What is best for **the project sponsor**
4. What is best for **the customer**
5. What is best for **the suppliers**

And the **PMI Code of Ethics** rests on four pillars: **responsibility, respect, fairness, and honesty**. You can violate ethics without breaking a law — but as an accredited PM you would swear to uphold the code, and breaking it puts your professional standing at risk.

2.6 Project communications

Communication is not a side task. **About 90 % of a project manager's time is spent communicating.** The three communication processes are **plan, manage, and control**.

2.6.1 The communication plan

A written communication plan is built from the stakeholders' information needs. Typical elements:

- What information is collected, and when

- Who receives it
- How it is gathered and stored
- Limits, if any, on who may tell whom
- Reporting relationships
- Contact information for all stakeholders
- The distribution schedule for each type of communication

Orion Tech fix: what should have been in the plan on day one?

A stakeholder list with each party's information needs; the report types (status, cost, risk, performance) and their frequency; the mix of channels (email for the record, chat for quick updates, dashboards for tracking, video for collaboration); and **feedback loops** so every team member confirms they understood the requirements. None of that existed — so requirements were guessed at and stakeholders were left in the dark.

2.6.2 The communication model

Every communication has three elements: a **sender**, a **message**, and a **receiver**. The sender **encodes** the message based on the receiver's education, experience, language, and culture, picks the right method, and **confirms it was understood**. The receiver **decodes** carefully and checks their understanding. **Feedback** ("Does that make sense?", a nod, a clarifying question) closes the loop.

2.6.3 Communication channels grow faster than the team

Add one person to a team and you do not add one relationship — you add one for every person already there. The number of two-way channels on a team of n people is:

$$\text{channels} = \frac{n(n-1)}{2}$$

```
n <- 2:15
ch <- n * (n - 1) / 2
df <- data.frame(n, ch)

ggplot(df, aes(n, ch)) +
  geom_col(fill = "#5DADE2", width = 0.7) +
  geom_text(aes(label = ch), vjust = -0.4, size = 3.3 * diagram_text_scale) +
  scale_x_continuous(breaks = n) +
  labs(title = "Communication channels = n(n - 1) / 2",
       x = "People on the team (n)", y = "Two-way channels") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
```

```
axis.title = element_text(size = 11 * diagram_text_scale),
axis.text = element_text(size = 10 * diagram_text_scale)
)
```

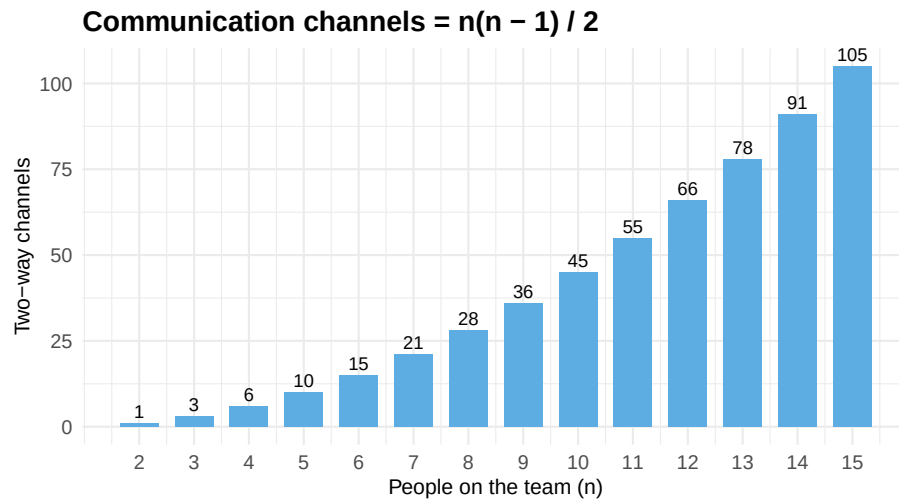


Figure 2.3: Two-way communication channels versus team size. Doubling the team roughly quadruples the channels that have to be kept coherent.

2.6.4 Worked Example 1: channels on the RC-vehicle project

The core project has the PM, 4 team members, an accountant, and a sponsor liaison — **7 people**. The extended stakeholder circle adds 2 supplier contacts, a retailer contact, and a compliance officer — **11 people**.

```
core <- 7
extended <- 11

ch_core <- core * (core - 1) / 2
ch_extended <- extended * (extended - 1) / 2

cat("Core team channels      :", ch_core, "\n")
cat("Extended circle channels:", ch_extended, "\n")
cat("Added by widening the circle from 7 to 11:", ch_extended - ch_core, "\n")
```

```
Core team channels      : 21
Extended circle channels: 55
Added by widening the circle from 7 to 11: 34
```

Four extra people did not add four conversations — they added 34. That is the

2.7. LISTENING: THE MODE WE USE MOST AND UNDERSTAND LEAST³³

argument for a **hub-and-spoke** communication plan (routes defined, the PM as the hub) instead of letting everyone talk to everyone about everything.

2.6.5 Try it yourself: channel growth and the cost of one more stakeholder

```
#| label: channels-calc-02
team_size <- 9

channels <- function(n) n * (n - 1) / 2

for (extra in 0:4) {
  n <- team_size + extra
  cat(sprintf("Team of %2d -> %3d channels", n, channels(n)))
  if (extra > 0) cat(sprintf(" (+%d vs. baseline)", channels(n) - channels(team_size)))
  cat("\n")
}

cat(sprintf("\nAdding the 10th person alone creates %d new channels.\n",
           channels(team_size + 1) - channels(team_size)))
```

2.7 Listening: the mode we use most and understand least

The four modes of communication, by share of a typical day and by how well they are scientifically understood:

Mode	Share	Notes
Listening	~55 %	The most used, the least studied and taught
Speaking	~23 %	
Reading	~13 %	
Writing	~9 %	

And most of the meaning is **nonverbal**: roughly **93 %** of person-to-person communication is carried by body language (~55 %) and tone of voice (~38 %) — and tone's share rises toward 86 % on the phone, where body language is gone.

- **Active listening**: the receiver signals engagement — agreeing, asking for clarification.
- **Effective listening**: watching the speaker for gesture and expression, thinking before responding, asking questions, repeating back, giving feedback.

2.7.1 The listening filters

Every message passes through three filters before it lands:

- **Selective attention** — we choose what to focus on.
- **Selective interpretation** — we impose our own meaning (shaped by gender, generation, geography, culture, education, occupation).
- **Selective retention** — we choose what to keep, and for how long.

There is also a raw arithmetic problem. People **listen** at about 500 words per minute but **speak** at about 150. That leaves ~350 words per minute of spare capacity — room to tune out.

2.7.2 Try it yourself: the tune-out budget

```
#| label: tune-out-02
listen_wpm <- 500
speak_wpm  <- 150
briefing_minutes <- 12

spare_wpm    <- listen_wpm - speak_wpm
spare_words  <- spare_wpm * briefing_minutes
# Time the listener could "leave" if they mentally processed at listening speed:
tune_out_seconds_per_min <- spare_wpm / listen_wpm * 60

cat(sprintf("Spare listening capacity: %d words/min\n", spare_wpm))
cat(sprintf("Over a %d-minute briefing: %s spare words to fill or drift\n",
            briefing_minutes, format(spare_words, big.mark = " ")))
cat(sprintf("Equivalent 'tune-out' time available: %.0f seconds per minute\n",
            tune_out_seconds_per_min))
cat("\nCauses of tuning out: purpose mismatch, under- or over-stimulation,\n")
cat("pace and delivery. The fix: give the listener a reason to stay ('what's in it for
```

Listening de-railers — which one is Orion Tech's problem?

- **Listening without focus** — mind multitasking.
- **Listening without respect (impatience)** — deciding the other person is wrong and interrupting.
- **Listening without respect (ego)** — hearing only from your own perspective, not theirs.
- **Listening with bias (defensive / adverse)** — building your rebuttal while they talk, or listening only for points to attack.

On Orion Tech it was mostly the first and last: dominant voices building responses instead of listening, quieter voices tuned out, and cross-cultural tone read through the listener's own filter rather than the speaker's intent.

2.8 Choosing a communication method

	Written	Verbal
Formal	Complex problems, project plans, charters, long-distance — <i>formal written</i>	Presentations and speeches — <i>formal verbal</i>
Informal	Memos, emails (careful: email is also a contract), texts, notes — <i>informal written</i>	Meetings and conversations — <i>informal verbal</i>

Should a PM try to control **all** communication between team members and outsiders? **Yes** — and can they? **No**. The plan defines the important routes; it cannot police every hallway conversation.

2.9 Performance reporting

Different stakeholder questions need different reports:

Report	Answers
Status	Where does the project stand right now?
Progress	What has been accomplished?
Trend	Is performance improving or deteriorating over time?
Forecast	What will status and performance be in the future?
Variance	How do actual results compare to the plan?
Earned value	Integrating scope, cost, and schedule into one performance picture (see Chapter 12).

Orion Tech’s stakeholders were right to expect regular **status, progress, and forecast** reports — they fund and approve the project, and reports give them transparency, accountability, and early warning.

2.10 Key Takeaways

- A **project** is temporary and unique; a **process** is permanent and repetitive. Only ~26 % of projects hit the full triple constraint — which is why

the discipline exists.

- The **triple constraint** is scope–time–cost, enclosing **quality**. Move one side and another must move, or quality pays.
- Five phases: **initiating, planning, executing, controlling, closing**. Planning decides the outcome; closing is the hardest to finish.
- Decisions are weighed **safety** → **company** → **sponsor** → **customer** → **suppliers**; ethics rest on **responsibility, respect, fairness, honesty**.
- ~90 % of a PM's time is communication. Build a written **communication plan** from stakeholder needs, and close every loop with **feedback**.
- Channels grow as $n(n-1)/2$ — faster than headcount. Plan routes; don't let everyone talk to everyone.
- **Listening** is ~55 % of communication and the least trained; ~93 % of meaning is nonverbal. Watch for the **selective attention/interpretation/retention** filters and the de-railers.
- Match the **method** (formal/informal × written/verbal) to the message, and the **report** (status/progress/trend/forecast/variance/EVM) to the stakeholder's question.

2.11 Practice Problems

Problem 1: Channel count and a reorganization

The RC-vehicle project currently routes all communication through a 6-person core team. A proposal would merge in 3 supplier engineers as full team members.

- How many channels before and after?
- By what percentage do channels increase versus the 50 % increase in headcount?

Solution.

```
before <- 6; after <- 9
ch <- function(n) n * (n - 1) / 2
cat("Before:", ch(before), "channels\n")
cat("After :", ch(after), "channels\n")
cat(sprintf("Headcount up %.0f%%, channels up %.0f%%\n",
            100 * (after / before - 1),
            100 * (ch(after) / ch(before) - 1)))
```

Before: 15 channels

After : 36 channels

Headcount up 50%, channels up 140%

Headcount rises 50 %, channels rise 125 %. Merging them in as full members triples the coordination load of those three people; a defined liaison route would add far less.

Problem 2: Pick the method (automotive, food, defense)

Choose formal/informal and written/verbal for each:

1. Notifying a defense customer that a delivered lot failed an acceptance test.
2. A daily 10-minute stand-up with the assembly team.
3. The project charter for the RC-vehicle launch.
4. Reminding a teammate that a drawing is due Friday.

Answers. 1 — **formal written** (contractual weight, complex, long-distance). 2 — **informal verbal** (meeting). 3 — **formal written** (plan/charter). 4 — **informal written** (a text or short email; keep a light record).

Problem 3 (discussion): rebuild Orion Tech's communication system

If you were the PM, redesign the tools, reporting, and meetings to prevent the failure.

Model answer. Write a **communication plan before launch**. Use a mix: **email** for records, **Teams/Slack** for quick updates, **dashboards** for visual tracking, **video calls** for collaboration (so tone and context survive the offshore gap). Run **regular status meetings with an agenda and action items**, and have offshore developers present their own weekly updates so no group dominates. Send stakeholders a **weekly progress report** (status, cost, risk). Build **feedback loops**: every developer confirms, in their own words, what a requirement means before coding it. Use plain language, avoid idioms, and summarize decisions in writing.

2.12 Further Resources

5 Ways to Listen Better (Julian Treasure, TED): <https://www.youtube.com/watch?v=cSohjIYQI2A>

10 Ways to Have a Better Conversation (Celeste Headlee, TED): <https://www.youtube.com/watch?v=R1vskiVDwl4>

Chapter 3

Project Scope, Charter, and Organizational Structures

3.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Evaluate** a candidate project with payback period and net present value (NPV).
2. **Write** a project charter and explain what authority its sign-off grants.
3. **Identify** the five inputs to a charter and the benefits of using one.
4. **Build** a stakeholder register and place stakeholders on a power-interest grid.
5. **Develop** a scope statement — deliverables, acceptance criteria, exclusions, constraints, assumptions — and distinguish product scope from project scope.
6. **Recognize** scope creep and run a change-control process.
7. **Compare** functional, projectized, and matrix (weak/balanced/strong) structures and the trade-offs each imposes on the project manager.

Before we begin: who told you it was okay to start?

On your last group project, was there a moment where everyone agreed *this is what we are building and this is who signed off*? Or did work just... start? The charter is that moment, written down.

3.2 Introduction: nothing official without a charter

Your company must get **Senior Management to approve the Scope and Charter before work begins** on the RC-vehicle replica. That is not bureaucracy — it is the difference between a project and a hobby. *“There is no official project without an approved Project Charter.”*

Two questions come first: **should we do this project at all** (selection), and **exactly what are we committing to** (scope). Then the organization’s shape — functional, projectized, or matrix — decides how much authority you actually have to deliver it.

i Note

Industry snapshot. In automotive new-product introduction, a project cannot pass its first gate without a signed charter and a business case showing positive NPV. A food company scoping a new line writes a **statement of work** that names the throughput, the allergen controls, and what is explicitly *out*. A defense contractor’s **SOW** is a contract exhibit — every deliverable and exclusion in it is legally enforceable.

What Is Scope in Project Management?: <https://www.youtube.com/watch?v=DZr3ZMWGGwc>

3.3 Project selection: is it worth doing?

3.3.1 Payback period

The simplest screen: how long until the project’s net cash inflows repay the initial outlay?

$$\text{Payback} = \text{years until cumulative cash flow} \geq 0$$

3.3.2 Worked Example 1: payback on the RC-vehicle tooling investment

The project needs **\$450 000** up front (tooling, setup, approvals). Once the 100 000-unit contract runs, it returns an estimated **\$180 000** of net cash per year.

```
initial    <- 450000
annual_cf  <- 180000
horizon    <- 5

cum <- cumsum(rep(annual_cf, horizon)) - initial
```

```

year_paid <- which(cum >= 0)[1]
# Fractional year within the payback year:
prev <- if (year_paid == 1) -initial else cum[year_paid - 1]
frac <- -prev / annual_cf

cat("Cumulative cash flow by year:", paste0("$", format(cum, big.mark = " ")), "\n")
cat(sprintf("Payback period: %.2f years\n", (year_paid - 1) + frac))

```

Cumulative cash flow by year: \$-270 000 \$ -90 000 \$ 90 000 \$ 270 000 \$ 450 000
 Payback period: 2.50 years

Payback is quick to compute but ignores the **time value of money** and anything that happens after the payback point. NPV fixes both.

3.3.3 Net present value

NPV discounts every future cash flow back to today's dollars at the organization's required rate of return r , then subtracts the initial outlay. A positive NPV means the project earns more than that hurdle rate.

$$NPV = -C_0 + \sum_{t=1}^n \frac{CF_t}{(1+r)^t}$$

3.3.4 Try it yourself: NPV of the project

```

#| label: npv-calc-03
initial <- 450000
cash_flows<- c(180000, 180000, 180000, 120000, 90000) # years 1..5
r <- 0.12 # required rate of return

t <- seq_along(cash_flows)
pv <- cash_flows / (1 + r)^t
npv <- sum(pv) - initial

cat("Year  Cash flow  Present value\n")
for (i in t) {
  cat(sprintf(" %d    $%s    $%s\n", i,
              format(cash_flows[i], big.mark = " "),
              format(round(pv[i]), big.mark = " ")))
}
cat(sprintf("\nSum of PVs : $%s\n", format(round(sum(pv)), big.mark = " ")))
cat(sprintf("Less outlay: $%s\n", format(initial, big.mark = " ")))
cat(sprintf("NPV      : $%s  -> %s\n",
              format(round(npv), big.mark = " "),
              ifelse(npv > 0, "ACCEPT (beats the 12% hurdle)", "REJECT")))

```

What happens to NPV if the hurdle rate rises to 18 %?

Raise r in the cell and re-run. Higher discount rates punish cash flows that arrive later, so NPV falls — a project that clears a 12 % hurdle may fail an 18 % one. This is why a company with expensive capital (or a risky project that demands a risk premium) is choosier about which projects it starts. Course Learning Outcome 1 is exactly this: *communicate the philosophy of project selection using the NPV method.*

3.4 The project charter

PMI: developing a document that formally **authorizes** a project and documents the initial requirements that satisfy the stakeholders' needs and expectations. **Approval of the charter formally initiates the project.**

3.4.1 Five inputs to a charter

1. **Project Statement of Work** — the high-level “project brief”: business need and product scope, fewer details than the scope statement.
2. **Business Case** — why the project is being done (the NPV/payback story).
3. **Contract** (if the client is external).
4. **Enterprise Environmental Factors (EEFs)** — conditions you cannot control (market, regulation, weather).
5. **Organizational Process Assets (OPAs)** — the company's plans, processes, policies, and accumulated knowledge.

3.4.2 What a charter contains

Title and description; business case; pre-assigned resources; stakeholders and their known requirements; product deliverables and the end result; measurable objectives tied to strategy; a **summary** milestone schedule; a **summary** budget (gross only — the real budget comes after planning); approval requirements and who signs off; high-level risks; and **signatures**.

3.4.3 Why bother

- Stakeholders are clearly defined
- Communication channels are defined
- Roles and responsibilities are outlined
- Scope is defined — helping prevent **scope creep**
- Better project processes in later phases
- Higher probability of success
- **Buy-in** from the team

The charter is a **living document**: update it as the project moves. Used well, it is “*the single most important tool the project manager uses to manage the expectations of the sponsor and all other stakeholders.*”

3.5 Stakeholders

A **stakeholder** is any individual or group with an active interest in the project who can affect it.

Internal	External
Senior management, accountant, team members, IT department	Client, competitors, suppliers, environmental agencies, government agencies, research labs

Managing them is a loop: **identify** → **gather information** → **determine strengths and weaknesses** → **predict behaviour** → **build an engagement strategy** → **implement**. The point of engagement is to **increase support and reduce resistance** — by listening, informing, and helping stakeholders along.

```
sh <- data.frame(
  who = c("Senior mgmt / sponsor", "Retailer (client)", "Motor supplier",
          "Carbon-fibre supplier", "Compliance officer", "Assembly team",
          "Accountant", "Competitors", "Local community"),
  power = c(0.92, 0.80, 0.55, 0.45, 0.70, 0.40, 0.50, 0.30, 0.20),
  interest = c(0.75, 0.85, 0.60, 0.55, 0.65, 0.90, 0.55, 0.65, 0.35)
)

ggplot(sh, aes(interest, power)) +
  annotate("rect", xmin = 0.5, xmax = 1, ymin = 0.5, ymax = 1, fill = "#27AE60", alpha = 0.10) +
  annotate("rect", xmin = 0, xmax = 0.5, ymin = 0.5, ymax = 1, fill = "#F39C12", alpha = 0.10) +
  annotate("rect", xmin = 0.5, xmax = 1, ymin = 0, ymax = 0.5, fill = "#5DADE2", alpha = 0.10) +
  annotate("rect", xmin = 0, xmax = 0.5, ymin = 0, ymax = 0.5, fill = "grey70", alpha = 0.12) +
  geom_point(color = "#1B4F72", size = 3) +
  geom_text(aes(label = who), size = 2.9 * diagram_text_scale, vjust = -0.8) +
  annotate("text", x = 0.75, y = 0.97, label = "MANAGE CLOSELY",
          fontface = "bold", size = 3 * diagram_text_scale, color = "#186A3B") +
  annotate("text", x = 0.25, y = 0.97, label = "KEEP SATISFIED",
          fontface = "bold", size = 3 * diagram_text_scale, color = "#9C640C") +
  annotate("text", x = 0.75, y = 0.03, label = "KEEP INFORMED",
          fontface = "bold", size = 3 * diagram_text_scale, color = "#1B4F72") +
  annotate("text", x = 0.25, y = 0.03, label = "MONITOR",
          fontface = "bold", size = 3 * diagram_text_scale, color = "grey30") +
  scale_x_continuous(limits = c(0, 1), breaks = c(0.1, 0.9), labels = c("low", "high")) +
```

```

scale_y_continuous(limits = c(0, 1), breaks = c(0.1, 0.9), labels = c("low", "high"))
labs(title = "Stakeholder power vs. interest", x = "Interest", y = "Power") +
theme_minimal() +
theme(
  plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
  axis.title = element_text(size = 11 * diagram_text_scale),
  axis.text = element_text(size = 10 * diagram_text_scale)
)

```

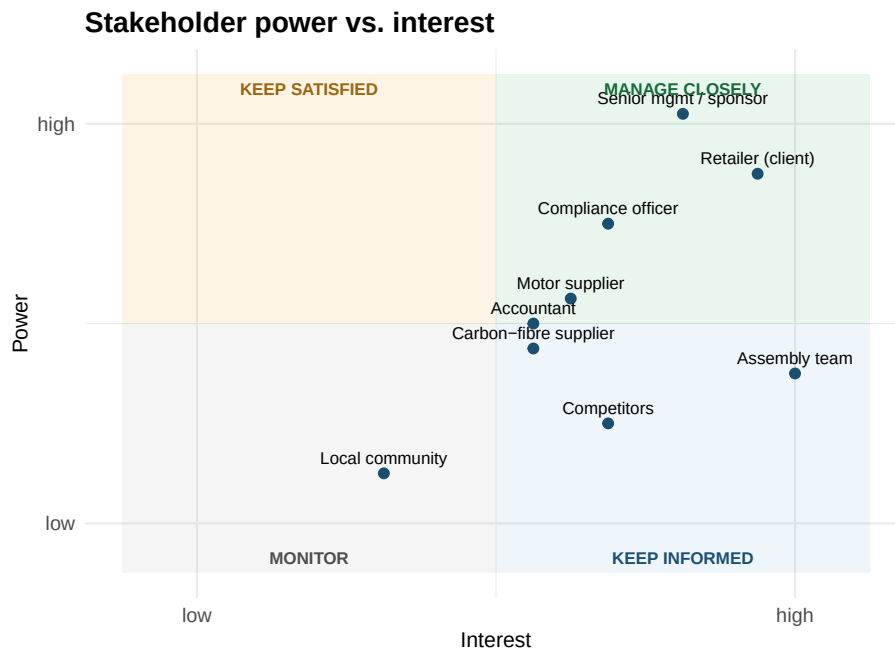


Figure 3.1: Power-interest grid for the RC-vehicle project. Effort goes where power and interest are both high; low-low stakeholders are monitored, not managed.

3.6 The scope statement

Written early in **planning**, the scope statement is more detailed than the charter and should include:

- The **project statement** (what will be done)
- **Acceptance criteria** and the **validation/testing plan**
- **Deliverables**
- **Exclusions** (what is explicitly *not* in scope)
- **Constraints** and **assumptions**

- Project coordination
- **Signed-off** review with the customer and with corporate management

Product scope = the features and functions of the thing you are building. **Project scope** = the work needed to deliver it. A scope statement covers both, plus boundaries and references to specs. *As the project progresses, scope should get clearer and more specific — and changes here should be avoided, though not forbidden.*

3.6.1 The scope management processes

Plan scope → **collect requirements** → **define scope** → **create WBS** → **validate scope (at closing)** → **control scope (throughout)**. The scope management plan is often run as an extension of **Management by Objectives (MBO)**: set unambiguous, realistic objectives; check periodically whether they are being met; take corrective action.

3.7 Scope creep

Scope creep is the uncontrolled growth of requirements over the project's life:

- one deliverable quietly becomes five
- three “essential” features become ten
- mid-project, the customer's needs change and requirements are reopened

Its usual causes are **stakeholders changing requirements** and **internal miscommunication**. It is *not always bad* — customer needs genuinely evolve, and refusing all change can deliver the wrong product. The problem is **uncontrolled change**. A good PM **expects it and plans for it** with a **change-control process**: every change request is logged, its impact on scope, schedule, and cost is assessed, and it is approved or rejected before work changes.

3.7.1 Try it yourself: the cost of an “innocent” change

```
#| label: scope-change-03
# A stakeholder asks to add a second body colourway "while we're at it".
extra_design_hours    <- 60
extra_tooling_cost    <- 28000
designer_rate          <- 46
indirect_rate         <- 0.35
schedule_days_added   <- 9
penalty_per_day_late  <- 1500      # retailer contract clause

direct <- extra_design_hours * designer_rate + extra_tooling_cost
loaded <- direct * (1 + indirect_rate)
sched  <- schedule_days_added * penalty_per_day_late
```

```

cat(sprintf("Direct cost of the change          : $%s\n", format(round(direct), big.mark = ",")))
cat(sprintf("Fully burdened (x%.2f)            : $%s\n", 1 + indirect_rate,
            format(round(loaded), big.mark = " ")))
cat(sprintf("Schedule slip: %d days -> penalty : $%s\n", schedule_days_added,
            format(sched, big.mark = " ")))
cat(sprintf("-----\nTotal impact of one 'small' change: $%s\n",
            format(round(loaded + sched), big.mark = " ")))
cat("\nThrough change control this gets a number and a decision.\n")
cat("Without it, it just shows up as an overrun later.\n")

```

3.8 Organizational culture

Every organization develops its own outlook, policies, habits of thought, and norms — “*as unique as a fingerprint.*” Culture is *the solution to problems that has worked consistently for a group and is therefore taught to new members as the correct way to perceive and think.* It is shaped by **technology, environment, geographic location, reward systems, rules and procedures, key organizational members, and critical incidents.**

Culture affects projects through:

- **Department interaction** — cooperative cultures run projects far better than adversarial ones.
- **Employee commitment to goals** — teams need to understand the *why* to be motivated.
- **Project planning** — some cultures tolerate padding estimates but not being late.
- **Performance evaluation** — a culture that rewards innovation breeds aggressive, high-payoff project management (with occasional big losses).

3.9 Organizational structures

How the lines of authority and communication are arranged decides **when people are available and how you get them.**

```

org <- data.frame(
  type = factor(c("Functional", "Weak matrix", "Balanced matrix",
                  "Strong matrix", "Projectized")),
  levels = c("Functional", "Weak matrix", "Balanced matrix",
             "Strong matrix", "Projectized")),
  authority = c(0.08, 0.28, 0.5, 0.72, 0.95)
)
ggplot(org, aes(type, authority, fill = authority)) +
  geom_col(width = 0.65) +
  geom_text(aes(label = scales::percent(authority, accuracy = 1)),

```

```

      vjust = -0.4, size = 3.6 * diagram_text_scale) +
    scale_fill_gradient(low = "#AED6F1", high = "#1B4F72", guide = "none") +
    scale_y_continuous(labels = scales::percent, limits = c(0, 1.05),
      expand = expansion(mult = c(0, 0.02))) +
    labs(title = "How much authority does the PM actually have?",
      x = NULL, y = "PM authority over resources") +
    theme_minimal() +
    theme(
      plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
      axis.title = element_text(size = 11 * diagram_text_scale),
      axis.text = element_text(size = 10 * diagram_text_scale)
    )

```

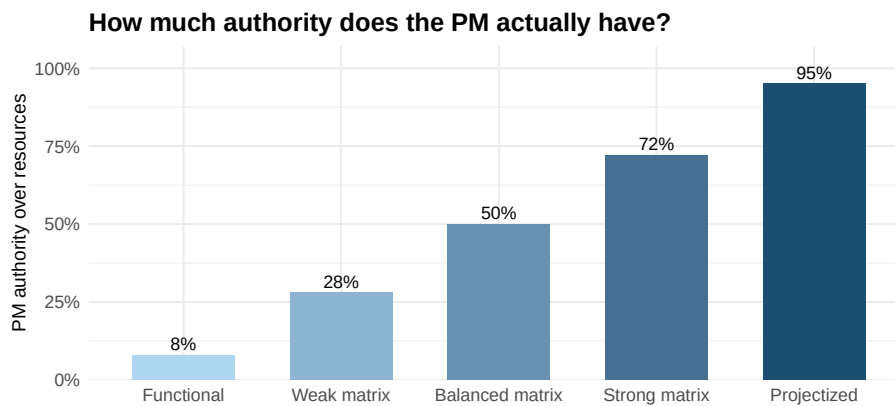


Figure 3.2: Project manager authority across organizational structures. It rises from almost none in a functional organization to near-total in a projectized one, with the matrix band in between.

Structure	Strengths for the PM	Weaknesses for the PM
Functional (grouped by specialty; one clear boss; each department runs its own projects)	No restructuring needed; deep expertise; everyone knows their place	Many layers to reach people; weak customer focus; projects drag because staff have “day jobs” and other priorities

Structure	Strengths for the PM	Weaknesses for the PM
Projectized (the company just does projects; team co-located)	One reporting line; easy communication; fast decisions; strong PMs; rapid market response	Teams are expensive to set up and maintain; loyalty to the project over the organization; hard to keep a shared pool of expertise; team anxiety about “what next” when the project ends
Matrix (blend; classified weak / balanced / strong by PM influence)	Suited to dynamic environments; values both project and functional efficiency; coordinates across units; maximizes scarce shared resources	Two bosses ; heavy time spent negotiating shared resources; frustrating for staff caught between project and functional demands

Which structure is the RC-vehicle project running in?

Almost certainly a **matrix** — the company has an existing manufacturing line and functional departments (design, quality, purchasing), and it borrows people into the project. That means the PM shares the design engineers with their functional manager and must negotiate their time. Expect the “administrative conflict” of Chapter 10: every matrix team member has two bosses.

3.10 Key Takeaways

- **Select** projects on merit: **payback** is quick but ignores the time value of money; **NPV** discounts every future cash flow and is the course’s standard (CLO 1). Positive NPV = beats the hurdle rate.
- The **charter** *formally initiates* the project. Inputs: SOW, business case, contract, EEFs, OPAs. It defines stakeholders, roles, and scope, and earns team **buy-in**. No approved charter, no official project.
- Manage **stakeholders** on a **power–interest grid**: manage the high-power/high-interest closely, monitor the rest.
- The **scope statement** covers **product and project scope**, plus acceptance criteria, exclusions, constraints, and assumptions, and is signed off.
- **Scope creep** is uncontrolled requirement growth. Change is normal; *uncontrolled* change is the enemy — run every request through **change control** for a cost/schedule impact and a decision.
- **Culture** shapes project outcomes through department cooperation, commitment, planning honesty, and what performance evaluation rewards.

- **Structure** sets the PM's real authority: near zero in **functional**, near total in **projectized**, negotiated in the **matrix** (weak/balanced/strong) — where everyone has two bosses.

3.11 Practice Problems

Problem 1: Payback vs. NPV disagree

Project A: \$300 000 outlay, then \$150 000/yr for 3 years, nothing after. Project B: \$300 000 outlay, then \$90 000/yr for 6 years. Hurdle rate 10 %. Which has the shorter payback, and which has the higher NPV?

Solution.

```
npv <- function(c0, cfs, r) -c0 + sum(cfs / (1 + r)^seq_along(cfs))
A <- c(150000, 150000, 150000)
B <- rep(90000, 6)
cat(sprintf("A: payback 2.0 yr, NPV $%s\n", format(round(npv(3e5, A, 0.10)), big.mark = " ")))
cat(sprintf("B: payback ~3.3 yr, NPV $%s\n", format(round(npv(3e5, B, 0.10)), big.mark = " ")))
```

A: payback 2.0 yr, NPV \$73 028

B: payback ~3.3 yr, NPV \$91 973

A pays back faster; B has the higher NPV because its cash flows keep coming for three more years. Payback alone would pick the worse project.

Problem 2: Charter or scope statement?

Which document does each belong in?

1. “Gross budget: approximately \$650 000.” → **Charter** (summary budget).
2. “Body colourways: Racing Red and Gunmetal only; no custom colours.” → **Scope statement** (exclusions).
3. “Sponsor: VP Operations. Signs off at Gate 1.” → **Charter** (approval/signatures).
4. “Drop test: survives 10 drops from 1.5 m onto concrete, no functional damage.” → **Scope statement** (acceptance criteria).

Problem 3 (discussion): the functional manager won't release the good engineer

In a weak matrix, your best-choice design engineer's functional manager keeps assigning them to department work and offers you a junior instead. Using the culture and structure ideas in this chapter, explain why this happens and what you can do.

Discussion points. The functional manager is **evaluated on their department's output**, so they hoard their strongest people — a reward-system effect. In a weak matrix you have little formal authority to override it. Levers: escalate to the sponsor (whose NPV depends on the project), negotiate a part-time or time-boxed assignment, trade a deliverable the functional manager cares about,

or get the resource commitment written into the **charter** so it is a documented agreement rather than a favour.

3.12 Further Resources

What Is Scope Creep? (Explained in 3 Minutes): <https://www.youtube.com/watch?v=30a6zZP5EVI>

Scope Creep in Project Management: Causes, Examples, Prevention: <https://www.youtube.com/watch?v=HHey0EXZR0w>

Chapter 4

Leadership and the Project Manager

4.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Distinguish** leadership from management in a project context.
2. **Explain** the four foundations of a project partnership and Bennis's four competencies of the "new project leader."
3. **Match** a leadership style (command & control, relating, democratic, hands-on, coaching) to a team situation, and state each style's downside.
4. **Score** a leadership self-assessment and interpret the result.
5. **Describe** the stages of team development and what the leader does in each.
6. **Apply** motivation theory — Maslow, Herzberg, and expectancy — to diagnose a performance problem.
7. **Identify** the five components of emotional intelligence and why they matter to a project leader.

Before we begin: recall a team you would follow anywhere

Who led it, and what did they *do* — not what title they held? Most people describe behaviour, not authority. That is the whole chapter in one observation.

4.2 Introduction: the project manager as mini-CEO

Leading the RC-vehicle project means more than tracking a Gantt chart. The brief has your team learning a **new assembly line** with a real learning curve,

working with an **unproven supplier**, and racing a **holiday deadline**. The PM “*encapsulates the role of a mini-CEO*” — expected to manage holistically from start to finish: acquiring resources, motivating and building the team, holding a vision *and* fighting fires, and communicating relentlessly. “*A PM cannot get up in the morning and think ‘I don’t want to communicate today.’*”

i Note

Industry snapshot. On an automotive launch, the line lead who can only give orders stalls the moment a novel problem appears; the one who has built a team that raises issues early hits the job-one date. In food processing, a sanitation crew that understands *why* the changeover procedure matters follows it at 3 a.m. when no one is watching. Leadership is what fills the gap between the procedure and reality.

Why Good Leaders Make You Feel Safe (Simon Sinek, TED): <https://www.youtube.com/watch?v=ImyZMtPVodo>

4.3 Leadership versus management

Leadership is the ability to inspire confidence and support among the people needed to achieve organizational goals. A **manager** holds a title that lets them plan, organize, direct, and control others. Leadership may be part of the manager’s job, but the other roles are more administrative.

Leaders create a sense of **partnership**. Four things build it:

Foundation	What it means
Exchange of purpose	Every member knows the project’s goals and vision.
A right to say “no”	Members can voice professional disagreement. <i>Losing an argument is acceptable; losing the right to disagree is not.</i>
Joint accountability	Every member owns the project’s outcomes — good or bad.
Absolute honesty	All information becomes community information.

```
lm <- data.frame(
  aspect = c("Focus", "Asks", "Relies on", "Time horizon",
             "Handles change by", "Success is"),
  manager = c("Systems & structure", "How & when", "Control", "Short term",
              "Administering", "Doing things right"),
```

```

leader = c("People", "What & why", "Trust", "Longer term",
           "Setting direction", "Doing the right things")
)
lm_long <- tidyr::pivot_longer(lm, c(manager, leader),
                              names_to = "role", values_to = "text")
lm_long$aspect <- factor(lm_long$aspect, levels = rev(lm$aspect))
lm_long$role <- factor(lm_long$role, levels = c("manager", "leader"),
                      labels = c("MANAGER", "LEADER"))

ggplot(lm_long, aes(role, aspect)) +
  geom_tile(aes(fill = role), color = "white", linewidth = 1.5) +
  geom_text(aes(label = text), size = 3.1 * diagram_text_scale, lineheight = 0.9) +
  scale_fill_manual(values = c("MANAGER" = "#AED6F1", "LEADER" = "#A9DFBF"),
                    guide = "none") +
  labs(title = "Two hats the project manager wears", x = NULL, y = NULL) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.text.x = element_text(size = 11 * diagram_text_scale, face = "bold"),
    axis.text.y = element_text(size = 10 * diagram_text_scale)
  )

```

Two hats the project manager wears

Focus	Systems & structure	People
Asks	How & when	What & why
Relies on	Control	Trust
Time horizon	Short term	Longer term
Handles change by	Administering	Setting direction
Success is	Doing things right	Doing the right things
	MANAGER	LEADER

Figure 4.1: Manager and leader emphases. Neither column is ‘bad’ — a project manager needs both, but leans on the right-hand behaviours during execution.

4.3.1 Bennis’s four competencies of the new project leader

1. **The power of appreciation** — recognize and reward others’ talent; your job is not to outshine the team but to let it develop.

2. **Keep reminding people what's important** — amid the noise of problems, keep everyone's eye on the prize.
3. **Generate and sustain trust** — through authenticity, fairness, honesty, and caring. Honesty is the trait people look for most.
4. **The leader and the led are intimate allies** — leadership supports the team's effort; it does not dominate it.

4.4 The five leadership styles

Effective PMs switch styles to fit the person and the situation.

Style	Use it when	The downside if overused
Command & control	You know the best way and the other person doesn't; beginners; a crisis	Team feels devalued; learned helplessness; kills creativity and initiative
Relating	Building rapport, bonding the team, focusing people on common goals	Can't make tough calls or take charge in a crisis; seen as weak
Democratic	<i>Planning</i> — getting buy-in and reality-based plans from those who will execute	<i>Execution</i> often needs autocracy; consensus-building wastes a critical deadline
Hands-on	You must step in and model the way to get a stuck task done	Disempowering; low morale; you become the bottleneck
Coaching	Long-term development; asking questions that lead people to their own solutions	Slow for short-term objectives — but it is the style most managers most need to build

```
sl <- data.frame(
  x = c(0.8, 0.35, 0.65, 0.2),
  y = c(0.8, 0.8, 0.3, 0.3),
  label = c("COACHING\n(high direction,\nhigh support)",
            "SUPPORTING\n(low direction,\nhigh support)",
            "DIRECTING\n(high direction,\nlow support)",
            "DELEGATING\n(low direction,\nlow support)")
)
ggplot(sl, aes(x, y)) +
  geom_point(size = 42, shape = 21, fill = "#D6EAF8", color = "#2E86C1") +
```

```

geom_text(aes(label = label), size = 2.6 * diagram_text_scale, lineheight = 0.9) +
annotate("segment", x = 0, xend = 1, y = 0, yend = 0,
        arrow = arrow(length = unit(0.2, "cm")), color = "grey40") +
annotate("segment", x = 0, xend = 0, y = 0, yend = 1,
        arrow = arrow(length = unit(0.2, "cm")), color = "grey40") +
annotate("text", x = 0.5, y = -0.08, label = "Directive behaviour ->",
        size = 3 * diagram_text_scale) +
annotate("text", x = -0.06, y = 0.5, label = "Supportive behaviour ->",
        angle = 90, size = 3 * diagram_text_scale) +
coord_equal(xlim = c(-0.12, 1.05), ylim = c(-0.12, 1.05)) +
labs(title = "Match the style to team readiness") +
theme_void() +
theme(plot.title = element_text(size = 14 * diagram_text_scale,
                                face = "bold", hjust = 0.5))

```

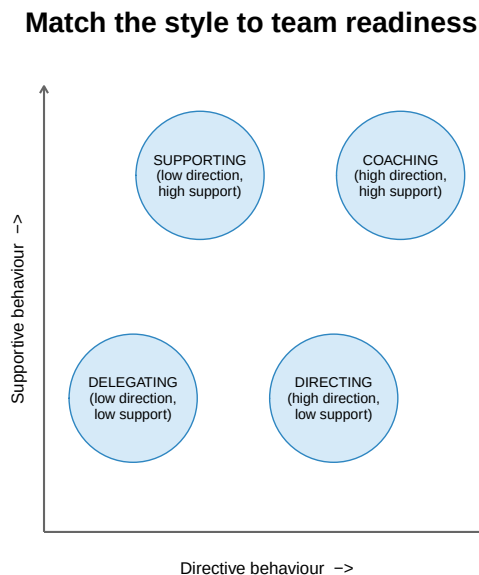


Figure 4.2: Situational leadership: match the style to how ready the team is. New or struggling teams need direction; capable, committed teams need delegation.

4.4.1 Try it yourself: the leadership self-assessment

The course quiz has 18 statements rated 1 (strongly disagree) to 5 (strongly agree). Items 1, 4, 7, 10, 13, 16 sum to a **command & control** score; items 2, 5, 8, 11, 14, 17 to a **democratic** score; items 3, 6, 9, 12, 15, 18 to a **relating** score. Enter your 18 answers and run.

```

#| label: leadership-score-04
# Replace with your own 1-5 answers to items 1..18, in order:
responses <- c(2, 4, 3, 2, 5, 4,      # items 1-6
               2, 4, 4, 2, 4, 4,      # items 7-12
               2, 5, 2, 2, 4, 3)      # items 13-18

cc <- sum(responses[c(1, 4, 7, 10, 13, 16)])
dem <- sum(responses[c(2, 5, 8, 11, 14, 17)])
rel <- sum(responses[c(3, 6, 9, 12, 15, 18)])

scores <- c("Command & Control" = cc, "Democratic" = dem, "Relating" = rel)
for (nm in names(scores)) cat(sprintf("%-18s %2d / 30\n", nm, scores[nm]))
cat(sprintf("\nYour dominant style: %s\n", names(which.max(scores))))
cat("Remember: the best PMs deliberately use all three depending on the situation.\n")

```

4.5 Building the team

Teams develop through predictable stages, and the leader's job changes with each:

```

stage_x <- seq(0, 10, length.out = 200)
perf <- 0.4 + 0.15 * sin(stage_x) +
        pmin(0.5, 0.06 * stage_x) -
        0.22 * exp(-((stage_x - 2.2)^2) / 0.6)
df <- data.frame(x = stage_x, perf = pmin(pmax(perf, 0.1), 1))

ggplot(df, aes(x, perf)) +
  geom_line(color = "#2E86C1", linewidth = 1.2) +
  annotate("text", x = c(1, 2.4, 4.6, 7.2, 9.4), y = 1.03,
         label = c("Forming", "Storming", "Norming", "Performing", "Adjourning"),
         fontface = "bold", size = 3.1 * diagram_text_scale) +
  annotate("segment", x = c(1.7, 3.5, 6, 8.6), xend = c(1.7, 3.5, 6, 8.6),
         y = 0.1, yend = 1, linetype = "dotted", color = "grey60") +
  coord_cartesian(ylim = c(0.1, 1.1)) +
  labs(title = "Tuckman's stages of team development",
       x = "Time together", y = "Team performance") +
  theme_minimal() +
  theme(
    axis.text = element_blank(),
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale)
  )

```

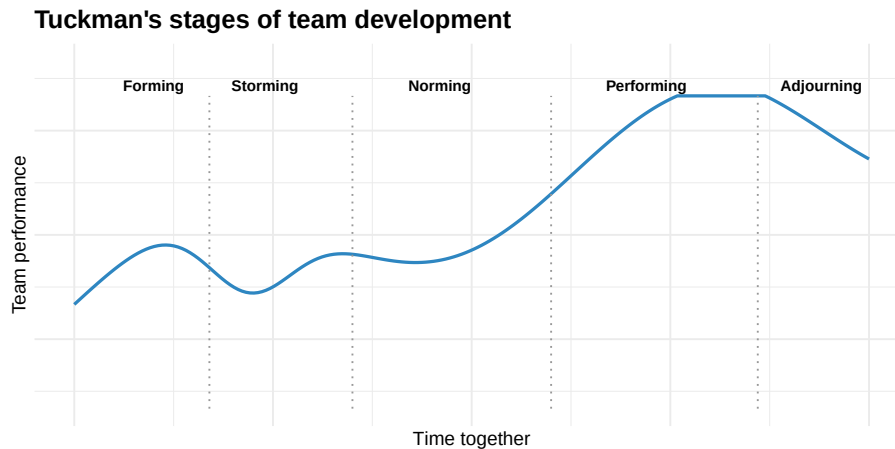


Figure 4.3: Team performance through Tuckman's stages. Performance dips during storming before rising; a leader who expects the dip manages it instead of panicking.

- **Forming** — polite, uncertain. Leader: **direct** — clarify goals, roles, the charter.
- **Storming** — conflict over approach and authority; performance dips. Leader: **coach** — surface conflict, don't suppress it.
- **Norming** — norms settle, trust builds. Leader: **support**.
- **Performing** — the team runs itself. Leader: **delegate**.
- **Adjourning** — the project ends; recognize contributions and manage the transition (see Chapter 13).

4.6 Motivating the team

4.6.1 Worked Example 1: Herzberg — is this a hygiene problem or a motivator problem?

Herzberg splits job factors into **hygiene factors** (pay, conditions, policy, supervision — their *absence* causes dissatisfaction, but their presence does not motivate) and **motivators** (achievement, recognition, the work itself, responsibility, growth — these drive engagement).

```
complaints <- data.frame(
  issue = c("Cramped, hot project room", "No recognition for hitting the drop-test milestone",
            "Confusing expense policy", "Same repetitive CAD cleanup every week",
            "Parking is far from the building"),
  type = c("Hygiene", "Motivator", "Hygiene", "Motivator", "Hygiene")
)
print(complaints, row.names = FALSE)
```

```
fix_first <- complaints$issue[complaints$type == "Hygiene"]
cat("\nFix the hygiene issues to stop dissatisfaction:\n - ",
    paste(fix_first, collapse = "\n - "), "\n", sep = "")
cat("\nThen add motivators (recognition, varied/owned work) to actually raise engagement")
```

	issue	type
	Cramped, hot project room	Hygiene
No recognition for hitting the drop-test milestone	Motivator	
	Confusing expense policy	Hygiene
Same repetitive CAD cleanup every week	Motivator	
	Parking is far from the building	Hygiene

Fix the hygiene issues to stop dissatisfaction:

- Cramped, hot project room
- Confusing expense policy
- Parking is far from the building

Then add motivators (recognition, varied/owned work) to actually raise engagement.

4.6.2 Try it yourself: expectancy theory

Vroom's expectancy model says motivation is the product of three beliefs, each 0–1: **Expectancy** (effort → performance), **Instrumentality** (performance → reward), and **Valence** (how much the person wants the reward).

$$\text{Motivation} = E \times I \times V$$

Because it is a product, **any factor near zero kills motivation**.

```
#| label: expectancy-04
team_member <- data.frame(
  name      = c("Priya", "Sam", "Dev"),
  expectancy = c(0.9, 0.8, 0.4), # "if I try, I can hit the target"
  instrumentality = c(0.8, 0.3, 0.9), # "hitting it gets me the reward"
  valence     = c(0.7, 0.9, 0.8)  # "I want that reward"
)
team_member$motivation <- with(team_member, expectancy * instrumentality * valence)

print(team_member, row.names = FALSE)
weakest <- team_member$name[which.min(team_member$motivation)]
cat(sprintf("\nLowest motivation: %s\n", weakest))
cat("Look at which of E, I, V is dragging the product down -- that is where the leader")
cat("(Sam doesn't believe performance leads to the reward; Dev doesn't believe he can")
```

4.7 Emotional intelligence

Goleman: *the most effective leaders all have a high degree of emotional intelligence* — IQ and technical skill are just the entry ticket. Five components:

- **Self-awareness** — a clear view of your strengths, weaknesses, drives, and ego needs.
- **Self-regulation** — keeping yourself under control; think before you act.
- **Motivation** — driven to your own potential *and* to drawing the best from each team member.
- **Empathy** — weighing team members' feelings when making a decision.
- **Social skill** — managing relationships with purpose.

4.8 Key Takeaways

- **Management** administers systems; **leadership** inspires people. A PM needs both and leans on leadership during execution.
- Partnership rests on **exchange of purpose, the right to say no, joint accountability, and absolute honesty**. Bennis: **appreciate, remind what matters, build trust, be an ally**.
- Switch among the **five styles** — command & control, relating, democratic, hands-on, coaching — by situation. Plan democratically; often execute autocratically. Every style overused has a cost.
- Teams move through **forming → storming → norming → performing → adjourning**; performance dips in storming — expect it and coach through it.
- Diagnose motivation with **Herzberg** (fix hygiene to stop dissatisfaction, add motivators to raise engagement) and **expectancy** ($E \times I \times V$ — any factor near zero kills it).
- **Emotional intelligence** — self-awareness, self-regulation, motivation, empathy, social skill — separates effective project leaders from merely competent ones.

4.9 Practice Problems

Problem 1: Pick the style

Which leadership style fits each moment on the RC-vehicle project?

1. Kicking off planning for the WBS with the whole team. → **Democratic** (buy-in, reality-based plan).
2. A moulding press fails at 2 a.m. and the launch date is at risk. → **Command & control** (crisis; take charge).
3. A capable senior designer is bored and thinking of leaving. → **Coaching** (long-term development; varied, owned work).

4. A brand-new co-op student has never used the company's PLM system.
→ **Command & control** initially, moving to **coaching**.
5. The team has gelled and is hitting milestones on its own. → **Delegating**
/ **supporting** — stay out of the way.

Problem 2: Expectancy diagnosis

A skilled technician has stopped putting in effort on validation testing. In a one-on-one you learn they are certain the extra effort won't change the delivery date because "purchasing will be late anyway." Which expectancy factor is broken, and what do you do?

Answer. Instrumentality — they no longer believe performance leads to the outcome. Restoring it means making the link visible again: show how the test results actually gate the next milestone, remove the purchasing blocker or re-sequence around it, and close the loop when their work does move the date.

Problem 3 (discussion): the hands-on trap

You are the strongest CAD modeller on the team, and every time someone struggles you take the model over and finish it yourself. The work gets done and is high-quality. Why is this a leadership failure, and what should you do instead?

Discussion points. Taking over is the **hands-on** style overused: it is disempowering, tanks morale, stops people learning, and makes you the bottleneck — the project's throughput is now capped by your personal hours. The **coaching** move is to ask questions that lead the person to the fix, sit with them while they do it, and follow up — slower once, faster forever.

4.10 Further Resources

Why Leaders Eat Last (Simon Sinek, 99U): <https://www.youtube.com/watch?v=ReRcHdeUG9Y>

Chapter 5

Work Breakdown Structures (WBS)

5.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Explain** what a WBS is and how it links scope, schedule, and budget.
2. **Decompose** a project into deliverables, work packages, and activities.
3. **Apply** the 100 % rule, the mutually-exclusive rule, and “plan outcomes, not actions.”
4. **Choose** an appropriate level of decomposition using the 8/80 and single-reporting-period rules of thumb.
5. **Number** a WBS and write a WBS dictionary entry for a work package.
6. **Roll up** cost and effort from work packages to the total project.
7. **State** the six requirements of a valid project contract and the purpose of work authorization.

Before we begin: how do you eat an elephant?

One bite at a time. “*Project managers cannot manage projects. Project managers manage pieces of projects.*” The WBS is how you cut the elephant into bites.

5.2 Introduction: from a scope statement to something you can manage

You have a signed scope statement for the RC-vehicle replica. It is still far too big to schedule, budget, or assign. The **work breakdown structure** is the bridge: it takes the total scope and breaks it into **deliverables**, then **work**

packages, then **activities** small enough that one person can own each one and you can put a real number on it.

Definition. A WBS is a **deliverable-oriented** grouping of project components that organizes and defines the **total scope** of the project.

A well-built WBS lets the team **plan every step, collect and organize** the information each task needs, and **assign** manageable chunks to team members. It also enables the **WBS dictionary**, which pins down the requirements for each task.

i Note

Industry snapshot. An automotive program WBS separates *body, powertrain integration, electrical, validation, and launch* — each a deliverable, not an action. A food-plant install WBS breaks out *utilities, process equipment, controls, sanitation qualification, and commissioning*. A defense contract WBS follows a mandated standard (MIL-STD-881) so the customer can track cost against the exact same structure the contractor uses.

Work Breakdown Structure (WBS) Explained: <https://www.youtube.com/watch?v=B8q4uz5XiZc>

5.3 The hierarchy

```
nodes <- data.frame(
  id = c("root", "d1", "d2", "d3", "d4",
        "w11", "w12", "w21", "w22", "w31", "w41", "w42"),
  label = c("1.0 RC-Vehicle Replica Project",
            "1.1 Body & Frame", "1.2 Drive & Controls", "1.3 Assembly & Test", "1.4 La",
            "1.1.1 Outer body\ndesign", "1.1.2 Inner frame\ndesign",
            "1.2.1 Motor/controller\nprocurement", "1.2.2 Wiring\nharness",
            "1.3.1 Prototype\nassembly", "1.4.1 Packaging\ndesign", "1.4.2 Compliance\n"),
  x = c(5, 1.5, 4, 6.5, 9, 0.6, 2.4, 3.3, 4.7, 6.5, 8.2, 9.8),
  y = c(3, 2, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1)
)
edges <- data.frame(
  x = c(5,5,5,5, 1.5,1.5, 4,4, 6.5, 9,9),
  y = c(3,3,3,3, 2,2, 2,2, 2, 2,2),
  xend = c(1.5,4,6.5,9, 0.6,2.4, 3.3,4.7, 6.5, 8.2,9.8),
  yend = c(2,2,2,2, 1,1, 1,1, 1, 1,1)
)
ggplot() +
  geom_segment(data = edges, aes(x, y, xend = xend, yend = yend),
```

```

    color = "grey55", linewidth = 0.6) +
  geom_label(data = nodes, aes(x, y, label = label),
    size = 2.5 * diagram_text_scale, lineheight = 0.9,
    fill = c("#1B4F72", rep("#5DADE2", 4), rep("#D6EAF8", 7)),
    color = c("white", rep("white", 4), rep("black", 7)),
    fontface = "bold", label.size = 0) +
  annotate("text", x = -0.2, y = c(3, 2, 1),
    label = c("Level 1", "Level 2", "Level 3"),
    fontface = "italic", size = 3 * diagram_text_scale, hjust = 1) +
  coord_cartesian(xlim = c(-1.2, 10.6), ylim = c(0.5, 3.5)) +
  labs(title = "WBS: project -> deliverables -> work packages") +
  theme_void() +
  theme(plot.title = element_text(size = 14 * diagram_text_scale,
    face = "bold", hjust = 0.5))

```

Warning: The `label.size` argument of `geom\_label()` is deprecated as of ggplot2 3.5.0.
 i Please use the `linewidth` argument instead.

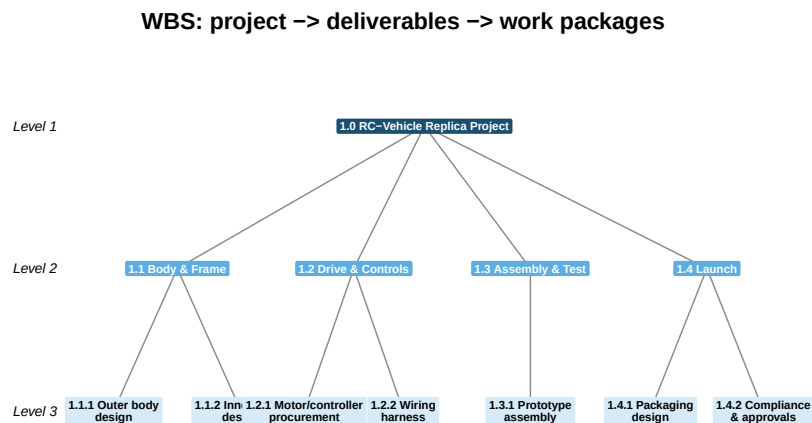


Figure 5.1: A partial WBS for the RC-vehicle project. Level 1 is the project; Level 2 the major deliverables; Level 3 the work packages. Numbering encodes the hierarchy.

- **Level 1** — the project. *What is the project?*
- **Level 2** — the summary level: the project's major deliverables.
- **Level 3** — the start of the **work package** level (more levels if the package is complex).

The same content can be shown as a **tree** (good for communicating structure) or a **tabular / indented list** (good for working). Both carry the numbering.

5.4 Design principles

5.4.1 The 100 % rule

The WBS includes **100 % of the work** in the scope — all deliverables, internal, external, and interim — and **no more**. At every level, the **sum of the children equals the parent**, and nothing outside the scope appears.

5.4.2 Worked Example 1: does this WBS roll up to 100 %?

The “1.1 Body & Frame” deliverable is estimated at **340 hours**. Its four work packages are estimated at 120, 90, 70, and 60 hours.

```
parent_hours    <- 340
children_hours  <- c("1.1.1 Outer body design" = 120,
                    "1.1.2 Inner frame design" = 90,
                    "1.1.3 Material selection" = 70,
                    "1.1.4 Design review & release" = 60)

child_sum <- sum(children_hours)
cat("Children:\n"); print(children_hours)
cat(sprintf("\nSum of children : %d h\n", child_sum))
cat(sprintf("Parent estimate : %d h\n", parent_hours))

if (child_sum == parent_hours) {
  cat("OK - the children account for exactly 100% of the parent.\n")
} else if (child_sum < parent_hours) {
  cat(sprintf("GAP - %d h of the parent's scope is not captured by any work package.\n",
              parent_hours - child_sum))
} else {
  cat(sprintf("OVER - the children exceed the parent by %d h (>100% -- scope crept in\n",
              child_sum - parent_hours))
}
```

Children:

1.1.1 Outer body design	1.1.2 Inner frame design
120	90
1.1.3 Material selection	1.1.4 Design review & release
70	60

Sum of children : 340 h

Parent estimate : 340 h

OK - the children account for exactly 100% of the parent.

Here the children sum to 340 — the deliverable is fully decomposed with nothing missing and nothing extra.

5.4.3 Mutually exclusive

No overlap between WBS elements. Overlap causes duplicated work, confusion over who is responsible, and messy cost accounting. If element names are ambiguous, the **WBS dictionary** disambiguates them.

5.4.4 Plan outcomes, not actions

Define elements as **results** (“Training module completed”), not **actions** (“write script”, “record video”, “upload”). Action lists always capture too many or too few steps — breaking the 100 % rule — and they over-prescribe *how* the work is done, killing the team’s ingenuity.

5.5 Level of detail

A work package at the **activity level** is a task that:

- can be **realistically and confidently estimated**;
- makes no practical sense to break down further;
- produces a **measurable deliverable**; and
- forms a unique package that **could be outsourced or contracted out**.

Rules of thumb for how far to decompose:

- **80-hour rule** — no work package (single deliverable at the lowest level) exceeds ~80 hours of effort.
- **One reporting period** — no package is longer than a single progress-reporting period (if you report monthly, nothing runs longer than a month).
- **8-hour floor** — generally do not define a task below ~8 hours; below that you are micromanaging.
- **“If it makes sense”** — apply common sense.

5.5.1 Try it yourself: decomposition check

```
#| label: wbs-decomp-05
packages <- data.frame(
  id    = c("1.1.1", "1.2.1", "1.3.1", "1.3.4", "1.4.2"),
  name  = c("Outer body design", "Motor/controller procurement",
            "Prototype assembly", "Tighten one fastener",
            "Full regulatory approval campaign"),
  hours = c(64, 40, 72, 1.5, 260)
)

flag <- ifelse(packages$hours > 80, "TOO BIG (> 80 h -- split it)",
              ifelse(packages$hours < 8, "TOO SMALL (< 8 h -- roll it up)",
                    "ok"))
```

```
out <- cbind(packages, verdict = flag)
print(out, row.names = FALSE)

cat(sprintf("\n%d of %d packages need re-sizing.\n",
            sum(flag != "ok"), nrow(packages)))
```

```
hub <- data.frame(x = 0, y = 0)
spokes <- data.frame(
  x = c(0, 1.3, 0, -1.3),
  y = c(1.3, 0, -1.3, 0),
  label = c("Estimating\n(cost & time)", "Network diagram\n(schedule)",
    "Risk\nidentification", "Staffing\n& responsibility")
)
ggplot() +
  geom_segment(data = spokes, aes(x = 0, y = 0, xend = x, yend = y),
    color = "grey55", linewidth = 0.7) +
  geom_point(data = hub, aes(x, y), size = 46, color = "#1B4F72") +
  annotate("text", x = 0, y = 0, label = "WBS", color = "white",
    fontface = "bold", size = 5 * diagram_text_scale) +
  geom_label(data = spokes, aes(x, y, label = label),
    size = 3 * diagram_text_scale, lineheight = 0.9,
    fill = "#D6EAF8", label.size = 0, fontface = "bold") +
  coord_equal(xlim = c(-2.2, 2.2), ylim = c(-2.1, 2.1)) +
  labs(title = "Everything downstream depends on the WBS") +
  theme_void() +
  theme(plot.title = element_text(size = 14 * diagram_text_scale,
    face = "bold", hjust = 0.5))
```

Everything downstream depends on the WBS

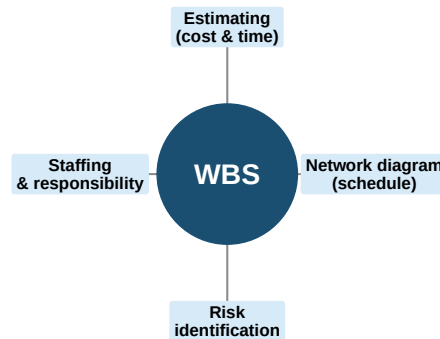


Figure 5.2: The WBS feeds four downstream planning activities. Get it wrong and the estimate, the schedule, the staffing plan, and the risk register are all built on sand.

5.7 The WBS dictionary

The dictionary is used to **control what work is done and when, prevent scope creep, and capture the effort** for each task. Physically, teams post the work packages on a wall in logical work-flow order; **removing them in numerical order generates the dictionary**. Each entry — a **task description** — records:

- Task description (what work is authorized)
- Quantified objectives and deliverables (cross-referenced to the charter)
- Product description; acceptance criteria (how the owner checks their own work)
- Assumptions; resources assigned; duration; cost; due date
- Interdependencies (before this task / after this task)
- Project manager approval and date

5.8 Work authorization and contracts

Once the scope is decomposed into work packages, stakeholders formally agree on project content — the **work authorization**. If the client is external, this includes the **contract sign-off**.

A valid project-management contract requires all six of:

1. The legal **right** to contract
2. Legal **age** (18 in Ontario and Quebec; 19 or 21 elsewhere in Canada)
3. **Sound mind** at signing
4. Valid, legal **consideration** for both parties
5. A **legal scope of work**
6. **Offer and acceptance**

Contracts also commonly contain: excusable delays; allowable costs; **liquidated damages** (non-performance / penalty clauses); inspection criteria; responsibility for correcting defects; and dispute-resolution mechanisms.

5.8.1 Try it yourself: cost roll-up from the WBS

```
#| label: wbs-cost-rollup-05
# Work-package costs (direct), grouped by Level-2 deliverable.
wp <- data.frame(
  deliverable = c(rep("1.1 Body & Frame", 3),
                  rep("1.2 Drive & Controls", 2),
                  rep("1.3 Assembly & Test", 3),
                  rep("1.4 Launch", 2)),
  package = c("Outer body", "Inner frame", "Material selection",
              "Motor/controller", "Wiring harness",
              "Prototype assembly", "Drop test", "EMC test",
              "Packaging design", "Approvals"),
  cost = c(18000, 14000, 6000, 140000, 9000, 21000, 12000, 15000, 8000, 6000)
)

by_deliverable <- tapply(wp$cost, wp$deliverable, sum)
cat("Roll-up to Level 2:\n")
for (d in names(by_deliverable))
  cat(sprintf("  %-22s $%s\n", d, format(by_deliverable[d], big.mark = " ")))

cat(sprintf("\nLevel 1 (total project direct cost): $%s\n",
          format(sum(wp$cost), big.mark = " ")))
cat("\nThis roll-up only works because the WBS obeys the 100% and\n")
cat("mutually-exclusive rules -- every dollar is counted once, and all of them.\n")
```

5.9 Key Takeaways

- A **WBS** is a **deliverable-oriented** decomposition of the **total scope** into deliverables → work packages → activities. PMs manage the pieces.
- **Numbering** encodes the hierarchy: Level 1 project, Level 2 deliverables, Level 3+ work packages.
- Obey the three design rules: **100 % rule** (children sum to the parent, never more), **mutually exclusive** (no overlap), and **plan outcomes**,

not actions.

- Size work packages with the **8/80 rule** and the **one-reporting-period** rule; below ~8 h you are micromanaging, above ~80 h you cannot estimate confidently.
- The WBS feeds **estimating, scheduling, staffing, and risk** — errors propagate to all four.
- The **WBS dictionary** locks down each task’s description, acceptance criteria, resources, and interdependencies, and helps prevent scope creep.
- **Work authorization** formalizes agreement on content; a valid contract needs all **six** legal elements and usually carries **liquidated-damages** and dispute clauses.

5.10 Practice Problems

Problem 1: Fix the 100 % rule violation

The “1.3 Assembly & Test” deliverable is estimated at 300 h. Its packages are listed as: Prototype assembly (110 h), Drop test (60 h), EMC test (70 h), Prototype assembly rework (40 h), Write the test report (30 h), *and* “General project management (50 h).”

What is wrong, and how do you fix it?

Solution. The children sum to 360 h — 60 h **over** 100 % of the parent. Two problems: “Prototype assembly rework” overlaps “Prototype assembly” (not mutually exclusive — fold rework into the assembly package or make it a contingency), and “General project management” is **not part of this deliverable’s scope** — PM effort belongs in its own Level-2 element (e.g. “1.0.1 Project Management”), not smuggled into a technical deliverable.

Problem 2: Outcome or action?

Rewrite these action-style entries as outcome-style WBS elements:

1. “Call three carbon-fibre suppliers and get quotes”
2. “Run the mould 20 times and measure the parts”
3. “Email the drawings to the customer for approval”

Model answers. 1 → “Carbon-fibre supplier selected and quoted.” 2 → “Moulding process capability established.” 3 → “Customer design approval obtained.” Each is a result you can mark done or not done, and it does not dictate the method.

Problem 3 (discussion): how deep should the RC-vehicle WBS go?

Your draft WBS has a work package “Design the outer body” estimated at 180 h and spanning 6 weeks, and another, “Deburr the frame jig”, estimated at 2 h. Using the rules of thumb, what do you do with each?

Discussion points. “Design the outer body” breaks both the 80-hour rule and (if you report every 2 weeks) the single-reporting-period rule — split it into concept, detailed CAD, and design-release packages. “Deburr the frame jig” is below the 8-hour floor — roll it into the jig-fabrication package rather than tracking it as its own line.

5.11 Further Resources

What Is a Work Breakdown Structure? (Online PM Courses): <https://www.youtube.com/watch?v=akO2Lf1fHmM>

Work Breakdown Structure Basics: https://www.youtube.com/watch?v=_7R_jjxVpK4

Chapter 6

Network Diagrams and Activity-on-Node (AON)

6.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Build** an activity-on-node (AON) network from a precedence table.
2. **Apply** the rules for constructing a valid network diagram.
3. **Perform** the forward pass to find early start (ES) and early finish (EF).
4. **Perform** the backward pass to find late start (LS) and late finish (LF).
5. **Calculate** total float and identify the **critical path**.
6. **Compute** a three-point (PERT) activity duration.
7. **Choose** a strategy to shorten the critical path when the project is too long.

Before we begin: which task can slip without moving the deadline?

On any project some tasks have slack and some do not. Guessing wrong — pouring effort into a task with float while a zero-float task quietly slips — is one of the most common ways schedules die. The network diagram tells you which is which.

6.2 Introduction: when can we launch?

The RC-vehicle WBS gives you a list of activities and their durations. It does *not* tell you how long the whole project takes, because the activities are not all independent — you cannot assemble a prototype before the body is designed. A **network diagram** captures those dependencies and answers the question the sponsor actually cares about: **what is the earliest the project can finish**,

and which activities decide that date?

i Note

Industry snapshot. Every automotive program plan is a critical-path network: the “job one” date is the length of the longest dependent chain. A food-plant commissioning schedule uses the same math to decide whether sanitation qualification can overlap controls checkout. A defense contractor reports critical-path status to the customer monthly — slipping a critical task is a contractual event.

Forward and Backward Pass -> Project Duration and Critical Path (Engineer4Free): <https://www.youtube.com/watch?v=4oDLMs11Exs>

6.3 From WBS to precedence table

For each activity you need its **duration** and its **immediate predecessors** — the activities that must finish before it can start. Rules for a valid AON network:

1. Determine precedence ordering **before** drawing.
2. Networks flow **left to right**.
3. An activity cannot begin until **all** connected predecessors are complete.
4. Arrows show **precedence and logical flow** (they may cross, but minimize it).
5. Every activity has a **unique identifier**.
6. **No looping** back to an earlier activity.
7. It is common (not required) to start from a **single start node**...
8. ...and end at a **single end node**.

6.4 AON node anatomy

Each activity is a box with six numbers:

```
ggplot() +
  annotate("rect", xmin = 0, xmax = 3, ymin = 0, ymax = 2, fill = "#D6EAF8",
    color = "#2E86C1", linewidth = 1) +
  annotate("segment", x = 0, xend = 3, y = 1, yend = 1, color = "#2E86C1") +
  annotate("segment", x = 1, xend = 1, y = 0, yend = 2, color = "#2E86C1") +
  annotate("segment", x = 2, xend = 2, y = 0, yend = 2, color = "#2E86C1") +
  annotate("text", x = c(0.5, 1.5, 2.5), y = 1.5,
    label = c("ES", "DUR", "EF"), fontface = "bold",
    size = 4 * diagram_text_scale) +
  annotate("text", x = c(0.5, 1.5, 2.5), y = 0.5,
    label = c("LS", "FLT", "LF"), fontface = "bold",
    size = 4 * diagram_text_scale) +
```

6.5. WORKED EXAMPLE: THE RC-VEHICLE SUB-PROJECT NETWORK73

```

annotate("text", x = 4.6, y = 1.5,
        label = "ES = earliest start\nDUR = duration\nEF = earliest finish",
        hjust = 0, size = 3 * diagram_text_scale, lineheight = 0.95) +
annotate("text", x = 4.6, y = 0.5,
        label = "LS = latest start\nFLT = float (slack)\nLF = latest finish",
        hjust = 0, size = 3 * diagram_text_scale, lineheight = 0.95) +
coord_equal(xlim = c(-0.3, 8.5), ylim = c(-0.4, 2.4)) +
theme_void()

```

ES	DUR	EF
LS	FLT	LF

ES = earliest start
DUR = duration
EF = earliest finish

LS = latest start
FLT = float (slack)
LF = latest finish

Figure 6.1: The six-number activity node. Forward pass fills the top row (ES, DUR, EF); backward pass fills the bottom (LS, FLT, LF).

An activity with 2+ predecessors flowing **in** is a **merge**; one with 2+ successors flowing **out** is a **burst**.

6.5 Worked Example: the RC-vehicle sub-project network

Ten activities (a simplified slice of the project):

```

acts <- data.frame(
  id = LETTERS[1:10],
  dur = c(5, 8, 3, 4, 5, 7, 6, 8, 6, 5),
  pred = c("", "A", "B", "A", "C", "C", "D,E", "F", "G", "H,I"),
  stringsAsFactors = FALSE
)
knitr::kable(acts, col.names = c("Activity", "Duration (days)", "Predecessors"),
  align = c("c", "c", "l")) |>

```

```
kableExtra::kable_styling(bootstrap_options = c("striped", "condensed"),
                           full_width = FALSE)
```

Activity	Duration (days)	Predecessors
A	5	
B	8	A
C	3	B
D	4	A
E	5	C
F	7	C
G	6	D,E
H	8	F
I	6	G
J	5	H,I

6.5.1 Forward pass (ES, EF)

Move left to right. $EF = ES + DUR$. Carry each EF forward as the successor's ES — **except at a merge, where the ES is the largest incoming EF**.

6.5.2 Backward pass (LS, LF)

Move right to left from the project finish. $LS = LF - DUR$. Carry each LS back as the predecessor's LF — **except at a burst, where the LF is the smallest outgoing LS**.

6.5.3 Float and the critical path

$$FLT = LS - ES = LF - EF$$

Activities with **zero float** cannot slip without delaying the whole project — together they form the **critical path**.

```
# A small critical-path solver (base R + data.frame).
cpm <- function(df) {
  df$pred <- strsplit(df$pred, ",")
  n <- nrow(df); id <- df$id
  ES <- EF <- LS <- LF <- setNames(rep(NA_real_, n), id)

  # Forward pass (ids are already in a valid topological order here)
  for (i in seq_len(n)) {
    p <- df$pred[[i]]; p <- p[p != ""]
    ES[i] <- if (length(p) == 0) 0 else max(EF[p])
    EF[i] <- ES[i] + df$dur[i]
```

```

}
proj <- max(EF)

# Backward pass
for (i in rev(seq_len(n))) {
  succ <- id[sapply(df$pred, function(pp) id[i] %in% pp)]
  LF[i] <- if (length(succ) == 0) proj else min(LS[succ])
  LS[i] <- LF[i] - df$dur[i]
}

data.frame(Activity = id, Dur = df$dur,
           ES = ES, EF = EF, LS = LS, LF = LF,
           Float = LS - ES,
           Critical = ifelse(LS - ES == 0, " <- critical", ""))
}

res <- cpm(acts)
print(res, row.names = FALSE)
cat(sprintf("\nProject duration: %d days\n", max(res$EF)))
cat("Critical path:", paste(res$Activity[res$Float == 0], collapse = " - "), "\n")

```

Activity	Dur	ES	EF	LS	LF	Float	Critical
A	5	0	5	0	5	0	<- critical
B	8	5	13	5	13	0	<- critical
C	3	13	16	13	16	0	<- critical
D	4	5	9	17	21	12	
E	5	16	21	16	21	0	<- critical
F	7	16	23	18	25	2	
G	6	21	27	21	27	0	<- critical
H	8	23	31	25	33	2	
I	6	27	33	27	33	0	<- critical
J	5	33	38	33	38	0	<- critical

Project duration: 38 days

Critical path: A - B - C - E - G - I - J

The whole forward/backward/float sequence is one connected calculation — change any duration and every downstream number moves — so work it as a block, not as isolated steps.

6.6 Three-point (PERT) duration estimates

When a duration is uncertain, estimate three values — **optimistic** *o*, **most likely** *m*, **pessimistic** *p* — and combine them:

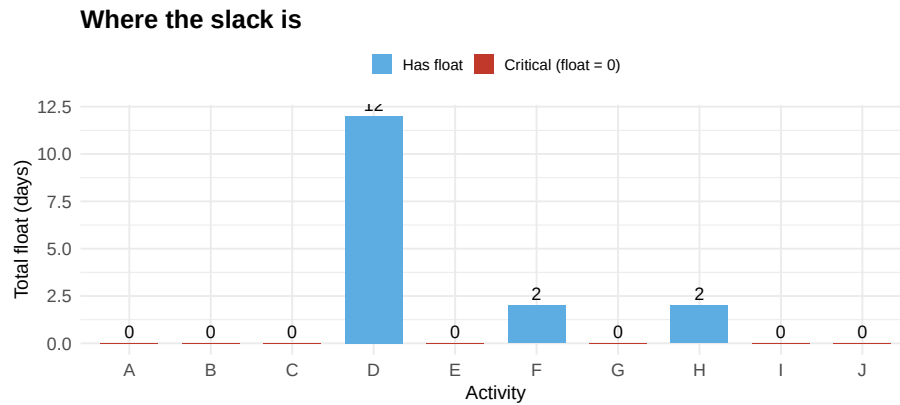


Figure 6.2: Total float per activity. The zero-float bars (A, B, C, E, G, I, J) are the critical path; D has 12 days of slack, F and H have 2.

$$t_e = \frac{o + 4m + p}{6}$$

6.6.1 Try it yourself: PERT durations

```
#| label: pert-te-06
tasks <- data.frame(
  id = c("A", "B", "C", "D", "E"),
  o = c(12, 7, 15, 9, 24),
  m = c(15, 10, 19, 13, 26),
  p = c(17, 13, 23, 16, 30)
)
tasks$te <- round((tasks$o + 4 * tasks$m + tasks$p) / 6, 2)
tasks$sd <- round((tasks$p - tasks$o) / 6, 2) # activity standard deviation

print(tasks, row.names = FALSE)
cat(sprintf("\nSum of expected durations along this chain: %.2f days\n", sum(tasks$te))
```

Why the weight of 4 on the most-likely value?

PERT assumes each activity duration follows a Beta distribution skewed toward the most-likely value, with the optimistic and pessimistic values as the tails. Weighting m by 4 and o , p by 1 each (dividing by 6) is the standard approximation of that distribution's mean. The activity standard deviation $(p - o)/6$ becomes important in Chapter 8, where you combine them to find the probability of finishing on time.

6.7 Shortening the critical path

Only work on the **critical path** shortens the project. Options, roughly in order of preference:

1. **Eliminate** critical tasks (is it truly needed?).
2. **Re-plan series paths to run in parallel.**
3. **Overlap** sequential tasks (start the successor before the predecessor fully finishes — *fast-tracking*).
4. **Shorten durations on critical activities** — prefer earlier-starting tasks, then the longest, then the easiest, then the ones **cheapest to speed up**.

6.7.1 Try it yourself: recompute the critical path after a change

```
#| label: path-duration-06
# All start-to-finish paths through the sub-project network, with activity durations.
dur <- c(A = 5, B = 8, C = 3, D = 4, E = 5, F = 7, G = 6, H = 8, I = 6, J = 5)

paths <- list(
  "A-B-C-E-G-I-J" = c("A", "B", "C", "E", "G", "I", "J"),
  "A-B-C-F-H-J"   = c("A", "B", "C", "F", "H", "J"),
  "A-D-G-I-J"      = c("A", "D", "G", "I", "J")
)

path_len <- sapply(paths, function(p) sum(dur[p]))
cat("Baseline path lengths:\n"); print(path_len)
cat(sprintf("Critical path: %s (%d days)\n\n", names(which.max(path_len)), max(path_len)))

# Now crash B from 8 to 5 days and re-check:
dur["B"] <- 5
path_len2 <- sapply(paths, function(p) sum(dur[p]))
cat("After crashing B by 3 days:\n"); print(path_len2)
cat(sprintf("New critical path: %s (%d days)\n", names(which.max(path_len2)), max(path_len2)))
cat("Note how the critical path can *shift* to a different chain once you crash enough.\n")
```

6.8 The downside of network diagrams

- Often **not drawn to scale** — time is just a number, which can create a false sense of how much time is available.
- Can be **too large and complex** to be useful.
- **Faulty construction logic** can oversimplify or misrepresent the work.
- Sometimes used for work they do not suit.
- **Positive bias** creeps into PERT estimates (people are optimistic).

6.9 Key Takeaways

- Build the **AON network** from a precedence table (durations + immediate predecessors); it flows left to right with no loops.
- **Forward pass:** $EF = ES + DUR$; at a **merge**, ES = the largest incoming EF . Gives the **project duration**.
- **Backward pass:** $LS = LF - DUR$; at a **burst**, LF = the smallest outgoing LS .
- **Float** = $LS - ES = LF - EF$. **Zero-float** activities form the **critical path** — the longest dependent chain, and the only one worth shortening.
- **PERT:** $t_e = (o + 4m + p)/6$, activity $SD = (p - o)/6$.
- Shorten the critical path by **eliminating, parallelizing, overlapping, or crashing** critical tasks — and re-check, because the critical path can shift.

6.10 Practice Problems

Problem 1: Solve this network

Activity	Predecessors	Duration
A	none	3
B	A	4
C	A	2
D	B	5
E	C	1
F	E	2
G	D, E	4
H	F, G	3
I	H	6

Find the project duration and the critical path.

Solution.

```
p1 <- data.frame(
  id = LETTERS[1:9],
  dur = c(3, 4, 2, 5, 1, 2, 4, 3, 6),
  pred = c("", "A", "A", "B", "C", "E", "D,E", "F,G", "H"),
  stringsAsFactors = FALSE
)
r1 <- cpm(p1)          # reuse the solver defined earlier in this chapter
print(r1, row.names = FALSE)
cat(sprintf("\nProject duration: %d days\n", max(r1$EF)))
cat("Critical path:", paste(r1$Activity[r1$Float == 0], collapse = " - "), "\n")
```

Activity	Dur	ES	EF	LS	LF	Float	Critical
A	3	0	3	0	3	0	<- critical
B	4	3	7	3	7	0	<- critical
C	2	3	5	9	11	6	
D	5	7	12	7	12	0	<- critical
E	1	5	6	11	12	6	
F	2	6	8	14	16	8	
G	4	12	16	12	16	0	<- critical
H	3	16	19	16	19	0	<- critical
I	6	19	25	19	25	0	<- critical

Project duration: 25 days

Critical path: A - B - D - G - H - I

Duration 25 days; critical path A-B-D-G-H-I. (The chain A-B-D-G is what pushes E, F, and C off the critical path — D's 5 days arriving at the G merge beat E's early finish by 6 days.)

Problem 2: PERT and the optimistic tail

An activity is estimated at $o = 6$, $m = 9$, $p = 24$ days.

- (a) What is t_e ? (b) Why is t_e noticeably larger than m ? (c) What is the activity standard deviation?

Solution. (a) $t_e = (6 + 36 + 24)/6 = 11$ days. (b) The pessimistic tail is far from the most-likely value (24 vs 9) while the optimistic tail is close (6 vs 9), so the distribution is **right-skewed** and its mean sits above m . (c) $SD = (24 - 6)/6 = 3$ days — a wide, risky activity.

Problem 3 (discussion): which task do you crash?

On the sub-project network, the critical path is A-B-C-E-G-I-J (38 days) and the sponsor needs 34. Crash data: B costs \$500/day to shorten (max 3 days), G costs \$1 200/day (max 2 days), E costs \$300/day (max 2 days). Which combination gets you to 34 days at least cost, and what do you check afterward?

Discussion points. You need 4 days. Cheapest first: E for 2 days (\$600), then B for 2 days (\$1 000) — total \$1 600 for 4 days. Avoid G at \$1 200/day. **Then re-run the network:** after removing 4 days from this chain, another path (e.g. A-B-C-F-H-J) may now be critical or co-critical, and crashing further would have to shorten *both*.

6.11 Further Resources

PERT / Three-Point Estimating Explained (Cornelius Fichtner):
https://www.youtube.com/watch?v=AQ_Wr1Me__8

Three-Point Estimates and PERT (PMC Lounge): <https://www.youtube.com/watch?v=u-9o-qa3HDg>

Chapter 7

Risk Management

7.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Distinguish** risk from uncertainty, and threats from opportunities.
2. **Classify** a risk and write a well-formed risk statement.
3. **Run** a qualitative analysis with a probability–impact matrix.
4. **Compute** expected monetary value (EMV) and solve a simple decision tree.
5. **Select** a response strategy: accept, minimize, share, transfer, or reserve.
6. **Size** a contingency reserve and distinguish task contingency from managerial contingency.
7. **Explain** how FMEA supports project risk management.

Before we begin: name one thing that could sink the RC-vehicle project

Now: how likely is it, and how bad would it be? You just did qualitative risk analysis. The rest of this chapter makes it systematic.

7.2 Introduction: the known unknowns

The RC-vehicle brief is a risk register waiting to be written. One candidate motor supplier is **unproven** and its staff are **negotiating a first labour contract**. The carbon-fibre supplier is **mid-restructuring** after a foreign acquisition. Choosing a **new assembly line** brings a **learning curve**. None of these is a certainty — but each is *identifiable*, and that is exactly what makes it a **risk** rather than an uncertainty.

Risk = the probability or threat of loss, injury, or any negative occurrence caused by internal or external vulnerabilities, that **can**

be reduced by preemptive action.

An **uncertainty** is when *neither* the probability *nor* the mode of occurrence is known. If you do not know about it, you cannot act on it.

The core equation:

$$\text{Risk} = \text{Probability of event} \times \text{Impact of event}$$

Probability is usually a percentage; impact is dollars or schedule delay. **Risk and opportunity are mirror images** — an opportunity is an unforeseen event with a *positive* outcome, handled with the same tools for the opposite end.

i Note

Industry snapshot. Aircraft makers such as Bombardier **minimize** part risk by forcing every significant supplier into direct contact with their quality department and auditing their processes. The European Space Agency and Airbus **share** the risk of a new airframe across nations because no one country can carry it. A fixed-price defense contract **transfers** budget-overflow risk to the vendor.

Expected Monetary Value (EMV) Explained Simply (Online PM Courses): https://www.youtube.com/watch?v=dD-Vygp_MiM

7.3 The four-stage process

1. **Risk identification** — determine the specific risk factors that can reasonably be expected to affect the project.
2. **Analysis of probability and consequences** — how likely, and how bad.
3. **Risk mitigation strategies** — steps to reduce the impact of the risks judged threatening enough to act on.
4. **Control and documentation** — track the risks and build a **knowledge base** for future projects from the lessons learned.

7.4 Risk identification and classification

Sort each identified risk into a class. A common (not exhaustive) list:

- **Financial** — material price moves, currency, budget cuts
- **Technology** — unproven design or process
- **Resources** — key people, equipment availability

- **Schedule** — dependency slips, long lead times
- **Weather, Political, Environmental**

Techniques: structured **brainstorming**, **interviewing** to get optimistic / most-likely / pessimistic views of an event, **probability distributions** to represent uncertain outcomes, and **expert judgment** to validate.

7.5 Qualitative analysis: the probability-impact matrix

Plot each risk by **likelihood** (x-axis) and **impact** (y-axis). The top-right is **red** (mitigation planned in advance), the middle band **yellow** (team decides whether to plan), the bottom-left **green** (just stay aware).

```
grid <- expand.grid(prob = 1:5, impact = 1:5)
grid$score <- grid$prob * grid$impact
grid$zone <- cut(grid$score, c(0, 6, 12, 25),
                 labels = c("Green", "Yellow", "Red"))

risks <- data.frame(
  risk = c("Unproven supplier can't deliver", "Carbon-fibre plant disruption",
           "New line learning curve overruns", "Labour action at supplier",
           "Steel price spike", "Minor drawing errors"),
  prob = c(3, 3, 4, 2, 3, 4),
  impact = c(5, 4, 3, 5, 2, 1)
)

ggplot(grid, aes(prob, impact)) +
  geom_tile(aes(fill = zone), color = "white", linewidth = 1) +
  scale_fill_manual(values = c(Green = "#A9DFBF", Yellow = "#F9E79F", Red = "#F5B7B1"),
                     name = "Zone") +
  geom_point(data = risks, aes(prob, impact), inherit.aes = FALSE,
            size = 3, color = "#1B4F72") +
  geom_text(data = risks, aes(prob, impact, label = risk), inherit.aes = FALSE,
           size = 2.7 * diagram_text_scale, vjust = -1, hjust = 0.5) +
  scale_x_continuous(breaks = 1:5, labels = c("Very low", "Low", "Medium", "High", "Very high"))
  scale_y_continuous(breaks = 1:5, labels = c("Negligible", "Minor", "Moderate", "Major", "Severe"))
  labs(title = "Probability x Impact", x = "Probability", y = "Impact") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 9.5 * diagram_text_scale)
  )
```

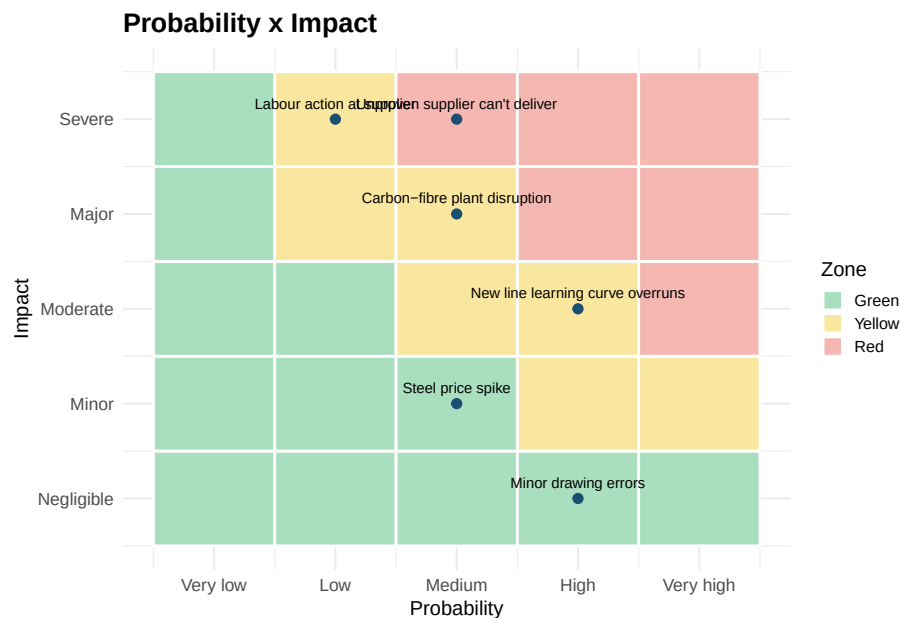


Figure 7.1: Probability-impact matrix for the RC-vehicle project. Red-zone risks (unproven supplier failure, carbon-fibre disruption) get a written mitigation plan before anything goes wrong.

7.6 Quantitative analysis

Tools: **sensitivity analysis** (which risks move the outcome most), **expected monetary value** (the probability-weighted average outcome), **decision trees** (lay out choices and chance events), and **Monte Carlo** simulation (run the network thousands of times with sampled durations/costs).

7.6.1 Worked Example 1: EMV of a single risk

The “new assembly line learning curve” risk: a **40 %** chance it adds **\$120 000** of rework and overtime; otherwise \$0.

```
p_occurs <- 0.40
impact    <- 120000

emv <- p_occurs * impact + (1 - p_occurs) * 0
cat(sprintf("EMV of this risk: 0.40 x %s = %s\n",
           format(impact, big.mark = " "), format(emv, big.mark = " ")))
cat("This is the amount you would budget in a risk reserve to 'cover' this one risk on
```

EMV of this risk: 0.40 x \$120 000 = \$48 000

This is the amount you would budget in a risk reserve to 'cover' this one risk on average.

7.6.2 Try it yourself: decision tree — buy the copper now or later?

You need **750 lb of copper** for the RC-vehicle wiring harness tooling. Buy it **now** and pay **\$100** storage, or buy it **later** at the summer price — which is **\$5.58/lb with 90 % probability**, or **\$4.50/lb with 10 % probability**.

```
#| label: decision-tree-07
lb <- 750
storage <- 100

# Branch: buy now (price locked, pay storage)
buy_now <- 5.58 * lb + storage

# Branch: buy later (two chance outcomes)
p_high <- 0.90; price_high <- 5.58
p_low  <- 0.10; price_low  <- 4.50
buy_later <- p_high * (price_high * lb) + p_low * (price_low * lb)

cat(sprintf("EMV, buy now   : %.2f\n", buy_now))
cat(sprintf("EMV, buy later: 0.90 x %.2f + 0.10 x %.2f = %.2f\n",
            price_high * lb, price_low * lb, buy_later))
cat(sprintf("\nChoose: %s (saves %.2f on average)\n",
            ifelse(buy_later < buy_now, "BUY LATER", "BUY NOW"),
            abs(buy_now - buy_later)))
```

What would flip the decision?

Raise the probability of the low summer price, widen the price gap, or cut the storage cost toward zero and “buy later” wins by more. Push storage cost up (a bonded warehouse, insurance, spoilage) or make the high price nearly certain and “buy now” wins. A decision tree makes the tipping point explicit instead of leaving it to gut feel.

7.7 Response strategies

Strategy	What it means	Example on our project
Accept	The risk is minor or unlikely enough to live with	Small drawing errors — caught in review, low cost

Strategy	What it means	Example on our project
Minimize	Reduce probability or impact	Audit the unproven supplier; put their staff in contact with your quality team
Share	Split the risk (and reward) with a partner	Co-develop the new control system with the supplier under a shared-savings deal
Transfer	Shift the risk to another party	Fixed-price contract (vendor eats overruns); liquidated damages for non-performance
Contingency reserve	Set aside funds/time for the risk if it hits	10–15 % reserve against the carbon-fibre and learning-curve risks
Other	Anything else fit for the situation	Dual-source the motor; pre-qualify a backup

For **opportunities**, the mirror strategies are **exploit**, **share**, **enhance**, and **accept**.

7.8 Contingency reserves

Two kinds:

- **Task (project) contingency** — offsets overruns on individual tasks/work packages. **Managed by the project manager.** As work completes, reserve for finished tasks is released and cannot be reused.
- **Managerial contingency** — a project-level buffer for scope changes forced by the customer, “acts of God,” and other unforeseeable events. **Managed by senior management, not the PM.**

In construction it is common to set aside **10–15 %** of the price. Other industries resist it — the customer may read a visible reserve as a sign of poor planning.

7.8.1 Try it yourself: does the mitigation pay for itself?

```
#| label: mitigation-value-07
# Risk: unproven supplier fails to deliver on time.
p_fail_no_action <- 0.35
impact_if_fail   <- 260000      # expedite alternative source + line downtime

# Mitigation: pre-qualify a backup supplier now.
```

```

mitigation_cost <- 22000
p_fail_with_action <- 0.10      # backup ready -> failure much less damaging
impact_with_action <- 90000     # switch to backup, smaller hit

emv_before <- p_fail_no_action * impact_if_fail
emv_after <- mitigation_cost + p_fail_with_action * impact_with_action

cat(sprintf("EMV, do nothing      : %s\n", format(round(emv_before), big.mark = " ")))
cat(sprintf("EMV, pre-qualify backup : %s (incl. %s spent now)\n",
            format(round(emv_after), big.mark = " "),
            format(mitigation_cost, big.mark = " ")))
cat(sprintf("\nExpected saving from acting: %s\n",
            format(round(emv_before - emv_after), big.mark = " ")))

```

7.9 FMEA as a risk tool

Failure Mode and Effects Analysis is often the first step of a reliability study: review as many components, assemblies, and subsystems as possible to identify **failure modes** (what can fail and how), their **causes**, and their **effects** (the consequences). It is a structured, bottom-up way to surface technical risks before they reach the customer — and it feeds the same probability–impact thinking used above (in FMEA terms, a **risk priority number** combining severity, occurrence, and detection).

7.10 Key Takeaways

- A **risk** is identifiable and can be acted on preemptively; an **uncertainty** cannot. **Risk = Probability × Impact**. Opportunities are the positive mirror.
- Four stages: **identify** → **analyze** → **mitigate** → **control & document** (build the knowledge base).
- **Qualitative**: plot risks on a **probability–impact matrix**; red-zone risks get a written plan *before* the event.
- **Quantitative**: **EMV** = probability × impact; **decision trees** compare choices with chance events; Monte Carlo for whole-network risk.
- Responses: **accept, minimize, share, transfer, reserve**. Pick per risk; ring-fence reserve against the biggest ones.
- **Task contingency** is the PM's; **managerial contingency** is management's, for the truly unforeseeable.
- **FMEA** systematically surfaces technical failure modes, causes, and effects.

7.11 Practice Problems

Problem 1: The road-trip risk register (from the course exercise)

Six friends borrow an older car and drive to an unfamiliar beach. The weather forecast is mixed with a chance of rain and high winds; one friend is a very picky eater; the owner plans to replace the car soon due to problems; there is a fireworks show and all-day music at a crowded venue. List the risks, classify each, and give a mitigation strategy.

Model answer (partial).

Risk	Class	Strategy
Get lost / arrive late	Schedule	<i>Minimize</i> — offline map, leave early
Car breaks down	Technology / Resources	<i>Minimize</i> (check oil, tires) + <i>transfer</i> (roadside assistance)
Rain / high winds spoil the day	Weather	<i>Accept</i> + <i>minimize</i> (pack a tarp, check the radar)
Picky eater has nothing to eat	Resources	<i>Minimize</i> — bring a known-safe option
No parking at a crowded venue	Resources	<i>Minimize</i> — arrive before the crowd
Injury in the crowd / fireworks	Environmental / Safety	<i>Minimize</i> — agree a meeting point, stay together

Problem 2: EMV of competing suppliers

Supplier X: \$140 000, 15 % chance of a 3-week slip costing \$45 000. Supplier Y: \$128 000, 35 % chance of a 3-week slip costing \$45 000. On expected cost alone, which is cheaper?

Solution.

```
X <- 140000 + 0.15 * 45000
Y <- 128000 + 0.35 * 45000
cat(sprintf("Supplier X expected cost: %s\n", format(X, big.mark = " ")))
cat(sprintf("Supplier Y expected cost: %s\n", format(Y, big.mark = " ")))
cat(sprintf("Cheaper on EMV: Supplier %s\n", ifelse(X < Y, "X", "Y")))
```

Supplier X expected cost: \$146 750

Supplier Y expected cost: \$143 750

Cheaper on EMV: Supplier Y

X: \$146 750. Y: \$143 750. Y is cheaper on expected cost — but note X's risk is *less likely*, which may matter more than the \$3 000 if a slip would blow the holiday launch window. EMV informs the decision; it does not make it.

Problem 3 (discussion): red-zone risk with no cheap mitigation

The “carbon-fibre plant disruption” risk sits in the red zone, but the supplier is the only qualified source and re-qualifying another would take four months you do not have. What are your options?

Discussion points. You cannot *minimize* the probability much, so work the **impact**: build a larger **contingency reserve** (time and money) against it; **transfer** part of it via contract terms (penalty clauses, guaranteed allocation); **share** it by co-investing in the supplier's modernization for priority supply; carry **safety stock** of the critical carbon-fibre parts; and design a **fallback** (an aluminium body variant) that could be substituted if the supply fails, accepting a heavier product rather than a missed launch.

7.12 Further Resources

Decision Tree Analysis for Calculating EMV (PMC Lounge): <https://www.youtube.com/watch?v=vmkdHT9VmZg>

Decision Tree and Expected Monetary Value (Cornelius Fichtner): <https://www.youtube.com/watch?v=uC2XBGpIcAs>

Chapter 8

Project Scheduling

8.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Read and build** a Gantt chart and explain what it adds over a network diagram.
2. **Distinguish** the four precedence relationships (FS, FF, SS, SF) and use leads and lags.
3. **Combine** PERT activity estimates into a project mean and standard deviation.
4. **Calculate** the probability of finishing by a target date.
5. **Compute** an activity's crash cost slope and build a **least-cost crashing** sequence.
6. **Compare** crashing and fast-tracking, and state the risks of each.

Before we begin: a schedule is a promise — how confident are you in it?

“The project finishes in 83 days” sounds precise. Is that a date you would bet your bonus on, or a coin flip? This chapter turns a single-number schedule into a probability you can actually stand behind.

8.2 Introduction: from a network to a calendar

The AON network from Chapter 6 tells you the project takes N days. The **schedule** puts those days on a calendar, assigns resources, and gives you something to track progress against. The classic tool is the **Gantt chart**, developed by Henry Gantt between 1910 and 1915 and used at scale for the first time on the **Hoover Dam** (1931–1936).

i Note

Industry snapshot. An automotive program's master schedule is a Gantt with thousands of rows rolled up to a one-page milestone view for the steering committee. A food-plant install schedule shows the trades stacked so the project manager can see where two crews need the same space on the same day. A defense schedule is baselined and every slip against it is reported.

3-Point Estimation for Project Schedule and Cost: https://www.youtube.com/watch?v=__YiAhMkBGbQ

8.3 Gantt charts

A Gantt chart is a **time-phased network**: it links each activity to a calendar. Linked to a schedule, it also **tracks progress against the plan**. Benefits: easy to read; shows the network *and* the schedule together; supports updating and control; useful for assigning resources; and easy to create.

```
g <- data.frame(
  task = c("A Design outer body", "B Design inner frame", "C Material selection",
           "D Procure motor/controller", "E Wiring harness",
           "F Prototype assembly", "G Drop & EMC test", "H Packaging & approvals"),
  start = c(0, 5, 5, 8, 15, 20, 30, 30),
  dur   = c(5, 8, 3, 12, 5, 10, 8, 12),
  crit  = c(TRUE, TRUE, FALSE, TRUE, TRUE, TRUE, TRUE, FALSE)
)
g$end <- g$start + g$dur
g$task <- factor(g$task, levels = rev(g$task))

ggplot(g) +
  geom_segment(aes(x = start, xend = end, y = task, yend = task, color = crit),
              linewidth = 8, lineend = "butt") +
  geom_text(aes(x = start + dur / 2, y = task, label = paste0(dur, "d")),
            color = "white", fontface = "bold", size = 3.1 * diagram_text_scale) +
  scale_color_manual(values = c("TRUE" = "#C0392B", "FALSE" = "#5DADE2"),
                     labels = c("TRUE" = "Critical path", "FALSE" = "Has float"),
                     name = NULL) +
  labs(title = "RC-vehicle sub-project schedule", x = "Project day", y = NULL) +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
```

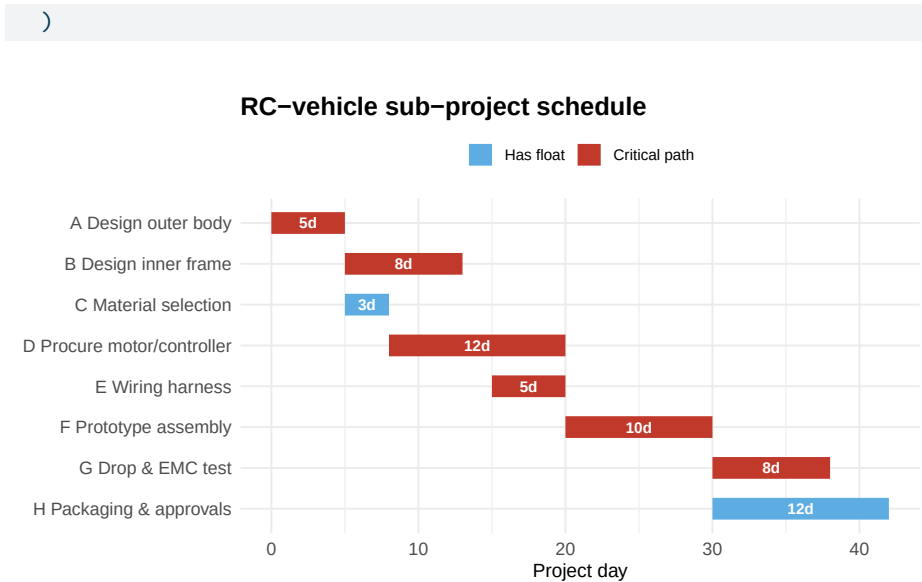


Figure 8.1: A Gantt chart for the RC-vehicle sub-project. Bars are activities; the dark segments show the critical path. Arrows (implied by stacking) show that assembly cannot start until design finishes.

8.4 Precedence relationships

Most dependencies are **finish-to-start**, but there are four:

Relationship	Meaning	Example
Finish-to-Start (FS)	B can't start until A finishes	Build the wall before you paint it
Finish-to-Finish (FF)	B can't finish until A finishes	Can't finish editing until writing is finished
Start-to-Start (SS)	B can't start until A starts (often simultaneous)	Start trimming edges when you start mowing
Start-to-Finish (SF)	A starting triggers B finishing (rare)	The new shift starting ends the old shift

A **lead** lets a successor start early (overlap); a **lag** forces a wait (e.g. “wait 2 days for the paint to cure” is FS + 2-day lag).

8.5 Probabilistic scheduling with PERT

A single duration hides uncertainty. Using three-point estimates (Chapter 6), for each activity:

$$t_e = \frac{o + 4m + p}{6}, \quad \sigma = \frac{p - o}{6}, \quad \sigma^2 = \left(\frac{p - o}{6}\right)^2$$

For the **project**, sum along the **critical path**:

$$\mu_{\text{project}} = \sum t_e, \quad \sigma_{\text{project}}^2 = \sum \sigma^2, \quad \sigma_{\text{project}} = \sqrt{\sigma_{\text{project}}^2}$$

8.5.1 Worked Example 1: project duration and its spread

The critical path is A–B–C–D–E with these three-point estimates:

```
cp <- data.frame(
  id = c("A", "B", "C", "D", "E"),
  o = c(12, 7, 15, 9, 24),
  m = c(15, 10, 19, 13, 26),
  p = c(17, 13, 23, 16, 30)
)
cp$te <- (cp$o + 4 * cp$m + cp$p) / 6
cp$var <- ((cp$p - cp$o) / 6)^2

mu <- sum(cp$te)
sigma <- sqrt(sum(cp$var))

cp_show <- cp
cp_show[c("te", "var")] <- lapply(cp_show[c("te", "var")], round, 2)
print(cp_show, row.names = FALSE)
cat(sprintf("\nProject mean duration : %.1f days\n", mu))
cat(sprintf("Project variance : %.2f\n", sum(cp$var)))
cat(sprintf("Project std deviation : %.2f days\n", sigma))
cat(sprintf("Rough 95%% range : %.0f to %.0f days\n", mu - 2 * sigma, mu + 2 * sigma))
```

```
id o m p te var
A 12 15 17 14.83 0.69
B 7 10 13 10.00 1.00
C 15 19 23 19.00 1.78
D 9 13 16 12.83 1.36
E 24 26 30 26.33 1.00
```

```
Project mean duration : 83.0 days
Project variance : 5.83
Project std deviation : 2.42 days
```

Rough 95% range : 78 to 88 days

8.5.2 Try it yourself: probability of hitting a target date

Standardize the target: $z = \frac{\text{target} - \mu}{\sigma}$, then read the normal cumulative probability.

```
#| label: prob-on-time-08
mu    <- 83.0      # from the worked example
sigma <- 2.42

for (target in c(80, 83, 86, 88)) {
  z <- (target - mu) / sigma
  cat(sprintf("Finish by day %2d:  z = %+5.2f  ->  P = %4.1f%%\n",
              target, z, 100 * pnorm(z)))
}

cat("\nWhat target gives 90% confidence?\n")
cat(sprintf("  day %.1f  (mu + %.2f x sigma)\n", mu + qnorm(0.90) * sigma, qnorm(0.90)))
```

Why quote “day 86 at 90 % confidence” instead of “day 83”?

Day 83 is the *mean* — by definition you have only about a 50 % chance of meeting it. Committing to the mean is committing to a coin flip. Adding a buffer to reach 85–90 % confidence is how you turn a schedule into a promise you can keep, and it is the honest version of the “padding” problem from Chapter 9 — one visible buffer at the project level, not hidden padding in every task.

8.6 Crashing the schedule

When you crash: the initial schedule was too optimistic; the market wants the product sooner; the project has slipped badly; or the contract rewards finishing early / punishes finishing late.

How you accelerate an activity: improve the productivity of existing resources (focus the team, review best practice); change the working method (swap drawing review for CAD model review); or add resources (overtime, extra shifts, expedited freight).

Warning

Caution on overtime. Research on engineering work found optimal marginal output at about **4 hours of overtime per week**; by **10 hours** of overtime the *marginal* extra output had fallen to **zero**. More hours is not more work.

Every activity has a **crash cost slope** — the cost to buy back one time unit:

$$\text{slope} = \frac{\text{crash cost} - \text{normal cost}}{\text{normal time} - \text{crash time}}$$

```
normal_t <- 8; normal_c <- 14000
crash_t <- 5; crash_c <- 23000
line <- data.frame(
  t = c(crash_t, normal_t),
  c = c(crash_c, normal_c)
)
ggplot(line, aes(t, c)) +
  geom_line(color = "#2E86C1", linewidth = 1.2) +
  geom_point(size = 4, color = "#1B4F72") +
  annotate("text", x = normal_t, y = normal_c, label = "Normal point\n(8 wk, $14 000)",
    hjust = 1.1, size = 3.2 * diagram_text_scale, lineheight = 0.9) +
  annotate("text", x = crash_t, y = crash_c, label = "Crash point\n(5 wk, $23 000)",
    hjust = -0.1, size = 3.2 * diagram_text_scale, lineheight = 0.9) +
  annotate("text", x = 6.5, y = 15500,
    label = "slope = $9 000 / 3 wk\n= $3 000 per week",
    size = 3.2 * diagram_text_scale, fontface = "italic", lineheight = 0.9) +
  scale_y_continuous(labels = scales::dollar_format(big.mark = " ")) +
  scale_x_reverse(breaks = 5:8) +
  labs(title = "Time-cost trade-off for crashing",
    x = "Activity duration (weeks) - shorter to the right",
    y = "Activity cost") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )
```

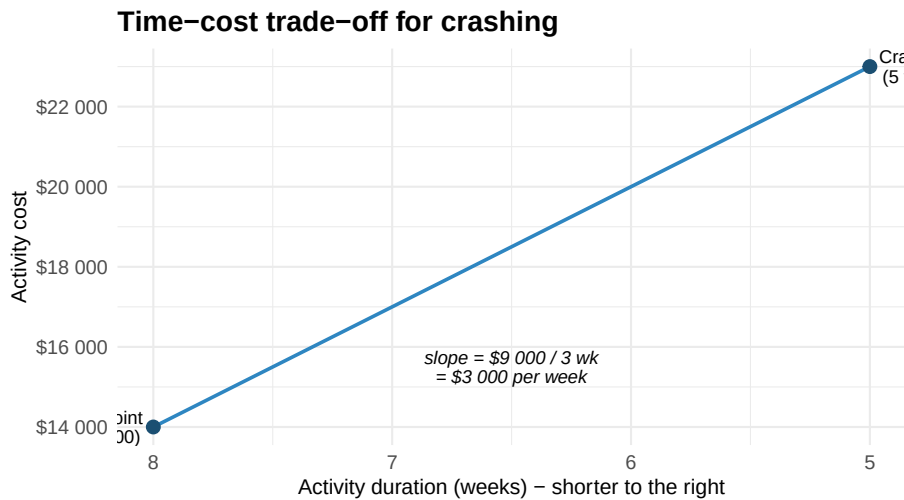


Figure 8.2: Time-cost trade-off for one activity. Between the normal and crash points, buying time gets steadily more expensive per day; past the crash point you cannot buy any more.

8.6.1 Try it yourself: least-cost crashing

To shorten the project, crash **critical-path** activities in **cheapest-slope- first** order, one time unit at a time, until you hit the target or run out of crashable critical tasks.

```
#| label: crash-sequence-08
# Critical-path activities: normal/crash time and cost.
act <- data.frame(
  id      = c("A", "F", "H"),
  normal_t = c(9, 6, 8),
  crash_t  = c(7, 5, 4),
  normal_c = c(5500, 4000, 3500),
  crash_c  = c(7550, 9000, 7550)
)
act$slope    <- (act$crash_c - act$normal_c) / (act$normal_t - act$crash_t)
act$max_crash <- act$normal_t - act$crash_t

cat("Crash cost per day:\n")
print(act[c("id", "slope", "max_crash")], row.names = FALSE)

target_reduction <- 4          # days to remove from the project
order_by_cost <- act[order(act$slope), ]

remaining <- target_reduction
```

```

total_cost <- 0
cat(sprintf("\nGoal: remove %d days, cheapest first\n", target_reduction))
for (i in seq_len(nrow(order_by_cost))) {
  take <- min(remaining, order_by_cost$max_crash[i])
  if (take <= 0) break
  cost <- take * order_by_cost$slope[i]
  total_cost <- total_cost + cost
  remaining <- remaining - take
  cat(sprintf("  crash %s by %d day(s) @ $%.0f/day = $%s\n",
              order_by_cost$id[i], take, order_by_cost$slope[i],
              format(cost, big.mark = " ")))
}
cat(sprintf("\nTotal crash cost for %d days: $%s\n",
            target_reduction - remaining, format(total_cost, big.mark = " ")))
if (remaining > 0) cat(sprintf("Still %d day(s) short -- no cheap critical task left to"))

```

8.7 Fast-tracking versus crashing

	Crashing	Fast-tracking
Method	Add resources (overtime, staff, expediting)	Overlap tasks that were sequential
Cost	Always adds cost	Little or no added cost
Risk	Diminishing returns; overtime burnout	Rework if the predecessor changes after the successor started

Both only help on the **critical path**, and after either one you must **re-check the network** — the critical path can move.

8.8 Key Takeaways

- A **Gantt chart** puts the network on a calendar and lets you track progress; it shows activities, dependencies, and resources at once.
- Four precedence types: **FS** (default), **FF**, **SS**, **SF** — with **leads** (overlap) and **lags** (forced wait).
- PERT project: $\mu = \sum t_e$ and $\sigma = \sqrt{\sum \sigma^2}$ along the **critical path**. The mean is a ~50 % date.
- Probability of on-time: $z = (\text{target} - \mu) / \sigma$, then the normal CDF. Commit to an ~85–90 % date, not the mean.
- **Crash cost slope** = (crash cost – normal cost) / (normal time – crash time). Crash **critical tasks, cheapest slope first**.

- **Crashing** buys time with money; **fast-tracking** buys it with rework risk. Re-check the critical path after either.

8.9 Practice Problems

Problem 1: Least-cost crash to a deadline

Project critical path A–F–H, currently 23 weeks; the contract needs 20. Crash data: A \$513/wk (max 4 wk), F \$1 500/wk (max 1 wk), H \$950/wk (max 2 wk). Find the least-cost way to remove 3 weeks.

Solution.

```
crash <- data.frame(
  id = c("A", "F", "H"),
  slope = c(513, 1500, 950),
  max = c(4, 1, 2)
)
need <- 3
crash <- crash[order(crash$slope), ]
left <- need; cost <- 0
for (i in seq_len(nrow(crash))) {
  take <- min(left, crash$max[i]); if (take <= 0) break
  cost <- cost + take * crash$slope[i]; left <- left - take
  cat(sprintf("Crash %s by %d wk @ $%d = $%s\n", crash$id[i], take, crash$slope[i],
    format(take * crash$slope[i], big.mark = " ")))
}
cat(sprintf("Total: $%s for %d weeks\n", format(cost, big.mark = " "), need - left))
```

Crash A by 3 wk @ \$513 = \$1 539

Total: \$1 539 for 3 weeks

Crash A by 3 weeks for \$1 539 — far cheaper than touching F or H.

Problem 2: Confidence in a date

A project's critical path has $\mu = 60$ days and $\sigma = 4$ days. The customer wants delivery in 64 days.

- (a) What is the probability of meeting day 64? (b) What date should you commit to for 95 % confidence?

Solution.

```
mu <- 60; sigma <- 4
cat(sprintf("P(finish by day 64) = %.1f%%\n", 100 * pnorm((64 - mu) / sigma)))
cat(sprintf("95% confident date = day %.1f\n", mu + qnorm(0.95) * sigma))
```

P(finish by day 64) = 84.1%

95% confident date = day 66.6

(a) ~84 %. (b) Day 66.6 — commit to day 67.

Problem 3 (discussion): crash or fast-track the RC-vehicle launch?

The retailer moves the launch three weeks earlier. Design is on the critical path and is followed by prototype assembly (also critical). You could add a second designer and pay overtime (crash), or start building the frame jig from the released frame drawings while the body design is still being finalized (fast-track). Which, and why?

Discussion points. Fast-tracking the jig against the *frame* drawings is low-cost and reasonably safe *if* the frame design is stable — but if the body design forces a late frame change, the jig is rework. Crashing design with a second modeller adds cost and has coordination overhead (two people on one model), but no rework risk. A common answer: fast-track the clearly stable parts, crash the rest, and hold a contingency for the rework risk you accepted.

8.10 Further Resources

PERT / Three-Point Estimation Technique: <https://www.youtube.com/watch?v=6B7Qtc3rrQ4>

Forward and Backward Pass -> Project Duration and Critical Path: <https://www.youtube.com/watch?v=4oDLMS11Exs>

Chapter 9

Cost Estimation and Budgeting

9.1 Learning Objectives

By the end of this chapter you will be able to:

1. **List** the common sources of project cost and **classify** any cost against the four cost dichotomies (direct/indirect, fixed/variable, normal/expedited, recurring/non-recurring).
2. **Calculate** direct labour cost from a rate and hours, and **apply** an indirect-cost percentage.
3. **Select** an estimating method (ballpark, parametric, feasibility, definitive) appropriate to the project stage, and **state** its expected accuracy.
4. **Explain** how planning horizon, duration, people, structure, padding, and organizational culture affect estimate quality.
5. **Compare** top-down and bottom-up budgeting and **build** a bottom-up budget from a work breakdown structure using **activity-based costing**.
6. **Construct** a time-phased budget and **plot** the cumulative cost baseline (the S-curve).
7. **Size** a contingency reserve and **distinguish** it from a management reserve.

Before we begin: you have been asked for “a number” before you have any details

A stakeholder corners you: *“Ballpark it — what will this project cost?”* You have a one-page concept and nothing else. What do you say, and how do you say it so the number does not follow you around for the next six months as a commitment? Hold that thought.

9.2 Introduction: the number that decides whether the project happens

Back to the running project. Your company must present a budget to **Senior Management before the Scope and Charter are approved**. The project brief hands you rough ranges to work from:

Table 9.1: Selected budget ranges from the RC-vehicle project brief

Item	Range
Project Manager (salary)	\$60 000 – \$80 000
Team members (salary, each)	\$30 000 – \$45 000
Sundry / overhead (per hour)	\$8.00 – \$15.00
Motor / controller	\$100 000 – \$175 000
Carbon fibre	\$120 000 – \$200 000
Aluminium	\$50 000 – \$115 000
Tooling	\$100 000 – \$180 000
Testing	\$25 000 – \$100 000
Government approval (CSA, CE, MTO)	\$2 000 – \$10 000
Packaging / shipping	\$5 000 – \$25 000

Cost estimation is the first step in building a budget, and the budget determines whether the project is feasible at all. Two principles run through everything below:

1. The more clearly you define each cost element up front, the fewer budgeting errors you make.
2. The more accurate your initial estimate, the more your budget reflects reality — and the less likely a painful overrun later.

i Note

Industry snapshot. A defense contractor bidding a **firm-fixed-price** contract must submit a *definitive* estimate ($\pm 5\%$): if the real cost comes in higher, the contractor eats the difference. A food company scoping a new retort line at the concept stage lives with a *ballpark* ($\pm 30\%$) and revises as quotes arrive. Same tools, very different tolerance for error — driven by where in the project you are standing.

The ABCs of Project Cost Management (Adriana Girdler): <https://www.youtube.com/watch?v=iEndSB5IgQM>

9.3 Where project cost comes from

Cost management starts in the **planning phase**, when you identify the project's human and material resources and lay out a **time-phased budget** that reflects when each will be used. The common sources of cost:

- **Labour** — the project team and anyone charging time to the project.
- **Materials** — everything consumed to build the deliverable.
- **Subcontractors** — external vendors, *and* internal support departments (test lab, drafting) that cross-charge.
- **Equipment and facilities** — purchased, leased, or allocated.
- **Travel.**
- **Shipping** — inbound and outbound, normal and expedited.

9.4 The eight cost types

Every cost gets sorted into **one or more** of eight categories, arranged as four either/or pairs. A single cost usually carries **two or three** labels at once.

```
cost_types <- data.frame(
  Dichotomy = c("Direct vs. Indirect", "", "Fixed vs. Variable", "",
                "Normal vs. Expedited", "", "Recurring vs. Non-recurring", ""),
  Type = c("Direct", "Indirect", "Fixed", "Variable",
           "Normal", "Expedited", "Recurring", "Non-recurring"),
  Meaning = c(
    "Clearly generated by the project (project labour, project materials, POs).",
    "Overhead: rent, property tax, benefits, utilities, depreciation, admin. Often applied as 20%",
    "Does not change with usage (a 6-month equipment lease at a flat fee).",
    "Proportional to usage (machine-hour charges, fuel, mileage).",
    "Incurred during normal operation and working hours; includes overtime that was already scheduled.",
    "Unplanned cost to go faster: rush freight, extra hires, premium to a supplier for priority.",
    "Happens again through the project life: utilities, logistics, ongoing labour, monthly maintenance.",
    "One-time, usually at the start or end: training, prototype tooling, initial market analysis."
  )
)
knitr::kable(cost_types, align = c("l", "l", "l"),
              caption = "The four cost dichotomies (eight types).") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                             full_width = FALSE) |>
  kableExtra::column_spec(2, bold = TRUE)
```

Table 9.2: The four cost dichotomies (eight types).

Dichotomy	Type	Meaning
Direct vs. Indirect	Direct	Clearly generated by the project (project labour, project materials, POs).

	Indirect	Overhead: rent, property tax, benefits, utilities, depreciation
Fixed vs. Variable	Fixed	Does not change with usage (a 6-month equipment lease)
	Variable	Proportional to usage (machine-hour charges, fuel, materials)
Normal vs. Expedited	Normal	Incurred during normal operation and working hours
	Expedited	Unplanned cost to go faster: rush freight, extra hire
Recurring vs. Non-recurring	Recurring	Happens again through the project life: utilities, logistics
	Non-recurring	One-time, usually at the start or end: training, prototyping

9.4.1 Direct labour cost

The core calculation is simple:

$$\text{Total Direct Labour Cost} = \text{Direct Labour Rate} \times \text{Hours Spent}$$

Indirect cost is harder to pin to a task, so many organizations apply it as a **percentage of direct cost**:

$$\text{Indirect Cost} = \text{Indirect Rate} \times \text{Direct Cost}, \quad \text{Indirect Rate} \approx 20\% - 50\%$$

9.4.2 Worked Example 1: fully-burdened cost of a work package

The “Design body & frame” work package needs **1 mechanical designer for 220 h** at \$46/h and **1 junior engineer for 120 h** at \$32/h. Direct materials for prototypes are \$4 500. The company applies indirect cost at **35 %** of direct cost.

```
designer_hours <- 220; designer_rate <- 46
junior_hours  <- 120; junior_rate  <- 32
direct_materials <- 4500
indirect_rate <- 0.35

direct_labour <- designer_hours * designer_rate + junior_hours * junior_rate
direct_cost   <- direct_labour + direct_materials
indirect_cost <- indirect_rate * direct_cost
total_cost    <- direct_cost + indirect_cost

cat("Direct labour   :", sprintf("%$s", format(direct_labour, big.mark = " ")), "\n")
cat("Direct materials:", sprintf("%$s", format(direct_materials, big.mark = " ")), "\n")
cat("Direct cost     :", sprintf("%$s", format(direct_cost, big.mark = " ")), "\n")
cat("Indirect (35%)  :", sprintf("%$s", format(indirect_cost, big.mark = " ")), "\n")
cat("-----\n")
cat("Fully-burdened work-package cost:",
    sprintf("%$s", format(round(total_cost), big.mark = " ")), "\n")
```

Direct labour : \$13 960
 Direct materials: \$4 500
 Direct cost : \$18 460
 Indirect (35%) : \$6 461

Fully-burdened work-package cost: \$24 921

Practice: classify the cost (from the MGMT-3105 budgeting case study)

For each item, give **up to three** labels from: A Direct, B Indirect, C Fixed, D Variable, E Recurring, F Non-recurring, G Normal, H Expedited.

#	Cost item	Answer
1	Technician's hourly wages for project assembly	A Direct, D Variable, G Normal
2	Office rent for company headquarters	B Indirect, C Fixed, E Recurring
3	One-time purchase of specialized testing software	A Direct, F Non-recurring
4	Electricity used by production equipment	D Variable, E Recurring, B Indirect
6	Hiring extra workers to meet a tight deadline	H Expedited, A Direct
9	Rush shipment of components to meet schedule	H Expedited, A Direct
12	Cost of the project manager's salary	C Fixed, E Recurring, A Direct
17	Routine monthly maintenance of equipment	F ... (<i>trick: this is E Recurring, B Indirect</i>)
23	Equipment lease for six months (flat fee)	B Indirect, C Fixed, E Recurring
40	One-time cost for prototype tooling	A Direct, F Non-recurring, C Fixed

The pattern: **anything charged straight to the project is Direct; overhead is Indirect; anything you pay to go faster is Expedited; anything paid once is Non-recurring.** Most items are genuinely two or three of these at once.

9.4.3 The dichotomies on the running project

- **Fixed vs. variable.** The 6-month lease on the CMM is **fixed** — the fee is the same whether you measure 50 parts or 5 000. The machine-hour charge on the rented moulding press is **variable**. Leasing an expensive machine is a real PM judgement call: will the usage justify the flat fee?
- **Normal vs. expedited.** Air-freighting circuit boards from the supplier to recover a slipped date is **expedited** and **direct**. The overtime you *planned* into the schedule from day one is still **normal**.
- **Recurring vs. non-recurring.** Prototype tooling and the initial market analysis are **non-recurring**. Utilities, logistics, and the team's salaries are **recurring** — and it matters which is which the moment you build a time-phased budget.

9.5 Estimating methods and their accuracy

The method you can use depends on how much you know, which depends on how far into the project you are.

```
methods <- data.frame(
  Method = c("Ballpark / order-of-magnitude", "Comparative / parametric",
    "Feasibility", "Definitive"),
  `When` = c(
    "Very early; scarce time/resources; deciding whether to even bid.",
    "A genuinely similar past project exists, with reliable archived actuals.",
    "Scope is clear and first supplier quotes are back (common in construction).",
    "Most design is complete; major POs can be written; detail is known."
  ),
  Accuracy = c("+/- 30 %", "+/- 15 %", "+/- 10 %", "+/- 5 %"),
  check.names = FALSE
)
knitr::kable(methods, align = c("l", "l", "c"),
  caption = "Estimating methods by project stage and typical accuracy.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) |>
  kableExtra::column_spec(3, bold = TRUE)
```

Table 9.3: Estimating methods by project stage and typical accuracy.

Method	When
Ballpark / order-of-magnitude	Very early; scarce time/resources; deciding whether to even bid.
Comparative / parametric	A genuinely similar past project exists, with reliable archived actuals.
Feasibility	Scope is clear and first supplier quotes are back (common in construction).
Definitive	Most design is complete; major POs can be written; detail is known.

This is the **cone of uncertainty**: the possible range around an estimate is widest at the concept stage and narrows as the project is defined.

```

cone <- data.frame(
  stage = factor(c("Ballpark", "Parametric", "Feasibility", "Definitive"),
    levels = c("Ballpark", "Parametric", "Feasibility", "Definitive")),
  x = 1:4,
  err = c(0.30, 0.15, 0.10, 0.05)
)
point_estimate <- 1.0 # $M

ggplot(cone, aes(x = x)) +
  geom_ribbon(aes(ymin = point_estimate * (1 - err),
    ymax = point_estimate * (1 + err)),
    fill = "#5DADE2", alpha = 0.35) +
  geom_line(aes(y = point_estimate * (1 + err)), linewidth = 1, color = "#2E86C1") +
  geom_line(aes(y = point_estimate * (1 - err)), linewidth = 1, color = "#2E86C1") +
  annotate("segment", x = 0.7, xend = 4.3, y = point_estimate, yend = point_estimate,
    linetype = "dashed", color = "grey30") +
  annotate("text", x = 4.3, y = point_estimate, label = "point estimate $1.0M",
    hjust = 1, vjust = -0.7, size = 3.3 * diagram_text_scale) +
  geom_point(aes(y = point_estimate * (1 + err)), size = 2.5, color = "#1B4F72") +
  geom_point(aes(y = point_estimate * (1 - err)), size = 2.5, color = "#1B4F72") +
  geom_text(aes(y = point_estimate * (1 + err),
    label = paste0("+", scales::percent(err, accuracy = 1))),
    vjust = -0.8, size = 3.2 * diagram_text_scale) +
  geom_text(aes(y = point_estimate * (1 - err),
    label = paste0("-", scales::percent(err, accuracy = 1))),
    vjust = 1.6, size = 3.2 * diagram_text_scale) +
  scale_x_continuous(breaks = 1:4, labels = levels(cone$stage)) +
  scale_y_continuous(labels = scales::dollar_format(suffix = "M", big.mark = " ")) +
  labs(title = "The cone of uncertainty", x = NULL, y = "Estimated project cost") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )

```

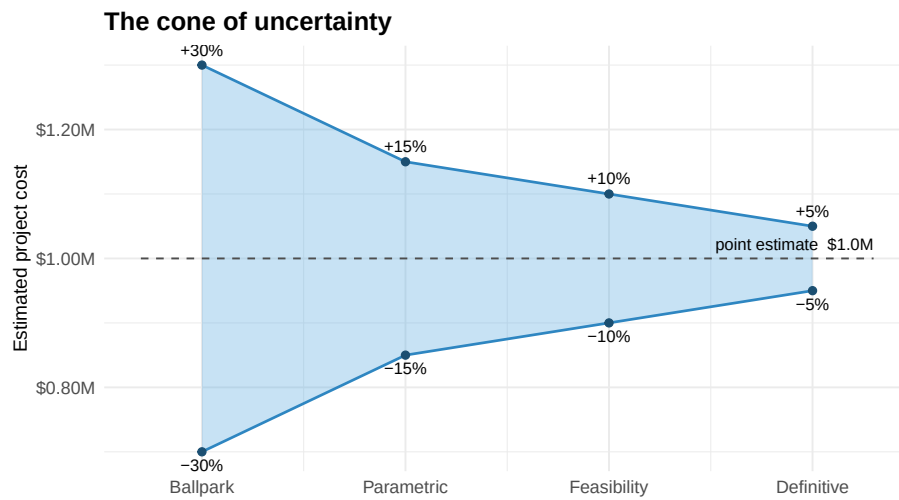


Figure 9.1: The cone of uncertainty. Early estimates can be off by +/- 30 %; the range narrows to +/- 5 % only once most design decisions are locked. The same \$1.0M point estimate means very different things at each stage.

9.5.1 Try it yourself: parametric estimate and the accuracy band

A **parametric** estimate uses a cost-estimating relationship (CER) from history — for example, “past RC-vehicle projects cost about \$14.50 per assembled part plus \$85 000 of fixed project cost.” The live cell builds the estimate and shows the band each method would put around it.

```
#| label: parametric-estimate-09
units      <- 120000
cost_per_unit_CER <- 14.50      # $/unit, from archived projects
fixed_project_cost<- 85000     # $, setup / PM / approvals

point_estimate <- units * cost_per_unit_CER + fixed_project_cost

bands <- data.frame(
  method  = c("Ballpark", "Parametric", "Feasibility", "Definitive"),
  accuracy = c(0.30,      0.15,      0.10,      0.05)
)

bands$low  <- round(point_estimate * (1 - bands$accuracy))
bands$high <- round(point_estimate * (1 + bands$accuracy))

cat(sprintf("Parametric point estimate: %s\n\n",
            format(round(point_estimate), big.mark = " ")))
for (i in seq_len(nrow(bands))) {
```

```

cat(sprintf("%-11s (+/- %2.0f%%):  %s  ...  %s\n",
           bands$method[i], bands$accuracy[i] * 100,
           format(bands$low[i],  big.mark = " "),
           format(bands$high[i], big.mark = " ")))
}

```

Why quote a range at all — why not just give “the number”?

Because a single number reads as a **commitment**. If you say “\$1.8M” at the concept stage and the definitive estimate later lands at \$2.0M, you have “overrun by \$200k” — even though \$2.0M was always inside the $\pm 30\%$ ballpark band. Quoting “\$1.3M–\$2.3M, ballpark” sets the expectation honestly and buys you room to refine. A clear distinction between an **estimate** and a **commitment** is one of the most important things a PM can establish early.

9.6 What makes estimates good or bad

Factor	Effect on estimate quality
Planning horizon	Estimates of near-term work are close to 100 % accurate; accuracy falls the further out the task sits.
Project duration	Long projects and new technology add uncertainty, especially if the scope statement is weak.
People	The closer the estimator’s experience matches the work, the better. Teams that have worked together before form faster. Staff turnover hurts.
Structure	Dedicated teams deliver faster but tie up people full-time. Matrix organizations are more cost-efficient but slower.
Padding	Estimators add a safety margin. If <i>everyone</i> pads a little, the project total is badly overstated and stops reflecting reality.
Organizational culture	Some cultures reward padding; others prize accuracy. The consequences an estimator faces for a miss shape the number they give.
Other	Equipment downtime, sickness, vacation, holidays, legal limits, and project priority all leak into the estimate.

9.6.1 The cost of padding

Padding feels prudent per task. Watch what it does across a whole project.

```
set.seed(2025)
n_tasks <- 40
true_mean <- 40 # hours, honest most-likely per task
# Honest estimates vary around the truth; padded estimates add 15% on top.
honest <- rnorm(n_tasks, true_mean, 6)
padded <- honest * 1.15

df <- data.frame(
  task = 1:n_tasks,
  honest_cum = cumsum(honest),
  padded_cum = cumsum(padded)
)

ggplot(df, aes(x = task)) +
  geom_line(aes(y = honest_cum, color = "Honest estimate"), linewidth = 1.1) +
  geom_line(aes(y = padded_cum, color = "Every task padded 15%"), linewidth = 1.1) +
  annotate("text", x = n_tasks, y = df$padded_cum[n_tasks],
    label = paste0("+", round(100 * (df$padded_cum[n_tasks] /
      df$honest_cum[n_tasks] - 1)), "% on the total"),
    hjust = 1, vjust = -0.6, size = 3.4 * diagram_text_scale) +
  scale_color_manual(values = c("Honest estimate" = "#27AE60",
    "Every task padded 15%" = "#C0392B")) +
  labs(title = "Per-task padding compounds into a meaningless total",
    x = "Cumulative task count", y = "Cumulative estimated hours",
    color = NULL) +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )
```

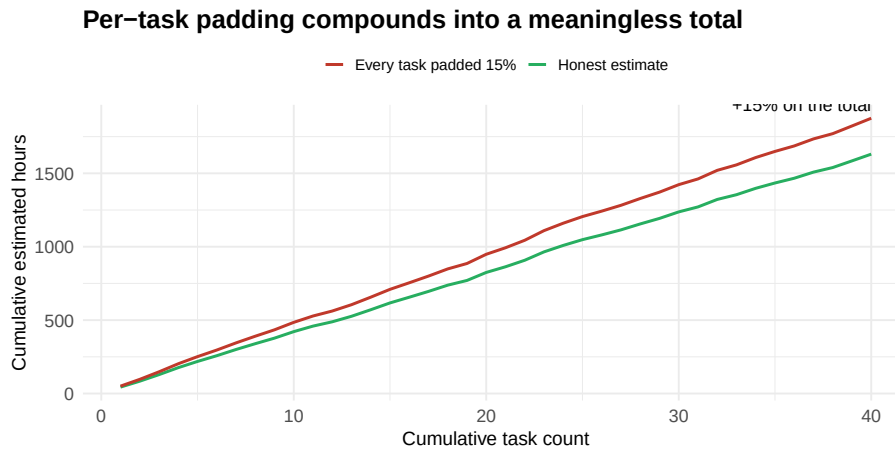


Figure 9.2: A 15 % ‘just to be safe’ pad added to every one of 40 tasks does not add 15 % to the project — it adds 15 % to the mean, but the padded total sits far above any realistic outcome because the pads do not all get used.

The fix is not “ban all safety margin” — it is to **remove padding from the work packages** and hold a single, visible **contingency reserve** at the project level (covered below).

9.6.2 Guidelines for estimating (cost and time)

- **Responsibility:** the person who will do the task makes the estimate (except for highly specialised work). It uses their experience and builds their buy-in.
- **Use several estimators:** independent estimates from knowledgeable people, reconciled, remove the extremes.
- **Assume normal conditions:** normal hours, normal resource levels — and write the assumptions down.
- **Define the time units** and the estimating base.
- **Estimate each task independently;** be careful with chains of dependent tasks, and keep one estimator per task.
- **Keep contingencies out of the work packages.** Contingency is a separate, managed fund.

9.7 Budgeting: top-down and bottom-up

The budget and the schedule are built **in tandem** — the budget decides whether the schedule’s milestones are reachable.

```
approaches <- data.frame(
  Aspect = c("How it starts", "Advantages", "Disadvantages"),
```

```

`Top-down` = c(
  "Senior management sets the overall budget and major work-package figures from experience",
  "Management aggregate estimates are often quite accurate; imposes budget discipline",
  "Creates friction: lower levels must fit management's fixed total. Turns into a zero-sum game",
),
`Bottom-up` = c(
  "Start from the WBS; apply direct and indirect cost to each activity; sum up to work package",
  "Forces detailed planning; improves PM--functional-manager communication; gives managers control",
  "Reduces management's control to oversight rather than initiation; can misalign priorities",
),
check.names = FALSE
)
knitr::kable(approaches, align = c("l", "l", "l"),
             caption = "Top-down vs. bottom-up budgeting.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                           full_width = FALSE) |>
  kableExtra::column_spec(1, bold = TRUE)

```

Aspect	Top-down
How it starts	Senior management sets the overall budget and major work-package figures from experience
Advantages	Management aggregate estimates are often quite accurate; imposes budget discipline
Disadvantages	Creates friction: lower levels must fit management's fixed total. Turns into a zero-sum game

9.8 Activity-based costing (ABC)

ABC treats a project as a series of **activities**, each of which **consumes resources**. Cost the activities and you have costed the project. Four steps:

1. Identify the activities that consume resources and assign costs to them.
2. Identify the **cost drivers** for each activity (personnel, materials, machine time...).
3. Compute a **cost rate per driver unit** (e.g. labour \$/hour).
4. **Assign cost** = rate $\times$ units the project uses. (*Example: an engineer at \$60/h working 80 h = \$4 800.*)

9.8.1 A preliminary ABC budget

Splitting each activity into its **direct** and **overhead** components:

```

abc <- data.frame(
  Activity = c("Survey", "Design", "Clear site", "Foundation", "Framing", "Plumb & w

```

```

  Direct    = c(3500, 7000, 3500, 6750, 8000, 3750),
  Overhead  = c(500, 1000, 500, 750, 2000, 1250)
)
abc$Total <- abc$Direct + abc$Overhead
abc <- rbind(abc, data.frame(Activity = "TOTAL",
                             Direct = sum(abc$Direct),
                             Overhead = sum(abc$Overhead),
                             Total = sum(abc$Total)))

knitr::kable(abc, align = c("l", "r", "r", "r"),
              format.args = list(big.mark = " "),
              caption = "Preliminary ABC budget: direct vs. overhead by activity.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                             full_width = FALSE) |>
  kableExtra::row_spec(nrow(abc), bold = TRUE)

```

Table 9.5: Preliminary ABC budget: direct vs. overhead by activity.

Activity	Direct	Overhead	Total
Survey	3 500	500	4 000
Design	7 000	1 000	8 000
Clear site	3 500	500	4 000
Foundation	6 750	750	7 500
Framing	8 000	2 000	10 000
Plumb & wire	3 750	1 250	5 000
TOTAL	32 500	6 000	38 500

9.8.2 Try it yourself: cost an activity, then a variance

The live cell costs a single activity from its drivers, then compares planned vs. actual to get the **variance**. Change the rates, hours, or actuals.

```

#| label: abc-activity-variance-09
# --- Step 3 & 4: cost the "Assemble & test prototype" activity from drivers ---
labour_rate <- 38; labour_hours <- 160
machine_rate <- 22; machine_hours <- 45
materials <- 2600
overhead_rate <- 0.30 # 30% of direct

direct <- labour_rate * labour_hours + machine_rate * machine_hours + materials
overhead <- overhead_rate * direct
planned <- direct + overhead

```

```

cat(sprintf("Direct   : %s\n", format(round(direct), big.mark = " ")))
cat(sprintf("Overhead : %s\n", format(round(overhead), big.mark = " ")))
cat(sprintf("Planned activity cost: %s\n\n", format(round(planned), big.mark = " ")))

# --- Variance: what actually happened ---
actual <- 11800
variance <- planned - actual          # positive = under budget (favourable)
pct <- 100 * variance / planned

verdict <- if (variance >= 0) "UNDER budget (favourable)" else "OVER budget (unfavourable)"
cat(sprintf("Actual   : %s\n", format(actual, big.mark = " ")))
cat(sprintf("Variance : %s (%.1f%%) -> %s\n",
            format(round(variance), big.mark = " "), pct, verdict))

```

Warning

Common mistake. Contingency is **not** built into an ABC budget. ABC costs the activities as they are planned to run; the contingency reserve is added *afterward*, on top of the total.

9.9 The time-phased budget and the S-curve

A **time-phased budget** spreads each activity's cost across the calendar periods when it will actually be spent. Summing down the columns gives **monthly planned spend**; the running total is the **cumulative cost baseline** — the S-curve you will track the project against in Chapter 12.

```

phased <- data.frame(
  month = factor(c("Jan", "Feb", "Mar", "Apr", "May"),
                 levels = c("Jan", "Feb", "Mar", "Apr", "May")),
  Survey      = c(4000, 3000, 0, 0, 0),
  Design      = c(0, 5000, 8000, 0, 0),
  ClearSite   = c(0, 1000, 4000, 2000, 0),
  Foundation  = c(0, 0, 3000, 4000, 0),
  Framing     = c(0, 0, 0, 3000, 4000),
  PlumbWire   = c(0, 0, 0, 0, 2000)
)
phased$monthly <- rowSums(phased[, -1])
phased$cumulative <- cumsum(phased$monthly)

scale_factor <- max(phased$cumulative) / max(phased$monthly)

ggplot(phased, aes(x = month)) +
  geom_col(aes(y = monthly), fill = "#5DADE2", width = 0.6) +

```

```

geom_line(aes(y = cumulative / scale_factor, group = 1),
          color = "#C0392B", linewidth = 1.2) +
geom_point(aes(y = cumulative / scale_factor), color = "#C0392B", size = 3) +
geom_text(aes(y = cumulative / scale_factor,
              label = scales::dollar(cumulative, big.mark = " ")),
          vjust = -1, size = 3.2 * diagram_text_scale, color = "#C0392B") +
scale_y_continuous(
  name = "Planned spend this month",
  labels = scales::dollar_format(big.mark = " "),
  sec.axis = sec_axis(~ . * scale_factor, name = "Cumulative baseline (S-curve)",
                      labels = scales::dollar_format(big.mark = " "))
) +
labs(title = "Time-phased budget and cumulative cost baseline", x = NULL) +
theme_minimal() +
theme(
  plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
  axis.title = element_text(size = 11 * diagram_text_scale),
  axis.text = element_text(size = 10 * diagram_text_scale)
)

```

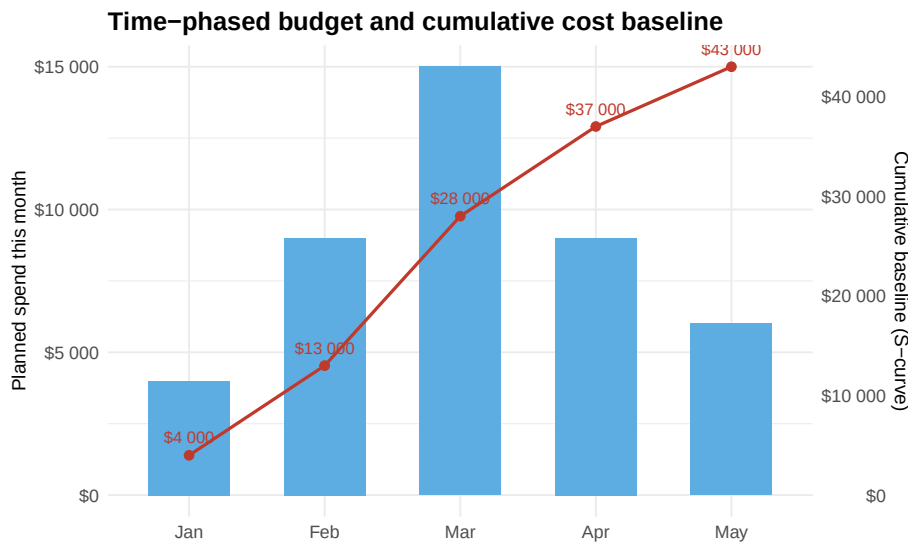


Figure 9.3: Time-phased budget for a six-activity project. Bars are planned spend per month; the line is the cumulative baseline (the S-curve). Spend is slow at first, steepest mid-project, and tapers at the end.

Matching the cost baseline to the schedule baseline is what lets you set **milestones for both schedule and budget** and later tell whether a project that has spent 60 % of its money is 60 %, 40 %, or 80 % done.

9.10 Contingency reserves

A **budget contingency** is extra money set aside to cover the uncertainty that every estimate carries — because even the best budget is only an estimate and something unforeseen always happens. Key points:

- Added **after all costs are identified**, as a cushion on top of the calculated total (commonly **10–15 %**; the MGMT-3105 Costs Chart uses **15 %**).
- **Not** included in an ABC budget — it sits on top.
- The **PM controls a project contingency fund**; separately, **top management holds a management reserve** for genuinely unknown-unknown events.

Why stakeholders push back: the percentage can look arbitrary, or feel like being asked to pay for the vendor's poor cost control. Decide early — and write into the scope statement — whether contingency applies across **all** tasks or is held for the tasks **critical** to project success.

```
reserves <- data.frame(
  Feature = c("Covers", "Owned by", "Part of the cost baseline?", "Typical size"),
  `Contingency reserve` = c(
    "Identified risks / \"known-unknowns\" (small scope adjustments, iteration, Murphy",
    "Project manager.",
    "Yes --- inside the baseline.",
    "10--15 % of estimated cost (project-dependent).",
  ),
  `Management reserve` = c(
    "Unidentified risks / \"unknown-unknowns.\",",
    "Senior management / sponsor.",
    "No --- outside the baseline; released by management.",
    "Set by organizational policy."
  ),
  check.names = FALSE
)
knitr::kable(reserves, align = c("l", "l", "l"),
  caption = "Contingency reserve vs. management reserve.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) |>
  kableExtra::column_spec(1, bold = TRUE)
```

Table 9.6: Contingency reserve vs. management reserve

Feature	Contingency reserve
Covers	Identified risks / "known-unknowns" (small scope adjustments, iteration, Murphy)
Owned by	Project manager.

Part of the cost baseline?	Yes — inside the baseline.
Typical size	10–15 % of estimated cost (project-dependent).

9.10.1 Try it yourself: build the RC-vehicle project budget

The live cell builds a rough **bottom-up** budget from the project brief's ranges (using a pick inside each range), rolls up labour and materials, and adds a contingency. Change the picks or the contingency rate and watch the total — then check it against what Senior Management is likely to approve.

```
#| label: rc-project-budget-09
# --- Labour (annualised salaries prorated to a 7-month project = 7/12) ---
months <- 7
pm_salary      <- 72000
n_team         <- 4
team_salary_each <- 38000
sundry_per_hour <- 11
sundry_hours   <- 1800          # total support hours booked to the project

labour <- (pm_salary + n_team * team_salary_each) * months / 12 +
  sundry_per_hour * sundry_hours

# --- Materials (one pick inside each brief range) ---
materials <- c(
  chassis_casting = 55000,
  fasteners       = 9000,
  motor_controller = 140000,
  carbon_fibre    = 0,          # not selected this time
  aluminium       = 85000,
  welding         = 25000,
  circuit_boards  = 32000,
  gov_approval    = 6000,
  testing         = 60000,
  tooling         = 150000,
  safety_equipment = 50000,
  packaging_shipping = 15000
)
material_total <- sum(materials)

subtotal      <- labour + material_total
contingency_r <- 0.15
contingency   <- contingency_r * subtotal
grand_total   <- subtotal + contingency

cat(sprintf("Labour subtotal      : $%s\n", format(round(labour),
big.mark = " ")))
```

```

cat(sprintf("Material subtotal      : %s\n", format(round(material_total),big.mark = "
cat(sprintf("Subtotal              : %s\n", format(round(subtotal),          big.mark = "
cat(sprintf("Contingency (%.0f%%)    : %s\n", contingency_r * 100,
          format(round(contingency), big.mark = " ")))
cat(sprintf("-----\nGRAND TOTAL      : %s\n",
          format(round(grand_total), big.mark = " ")))

```

How would top-down vs. bottom-up change this number?

The cell above is **bottom-up**: it sums the pieces and lets the total fall out. A **top-down** pass would start with a figure Senior Management is willing to fund — say “\$650k, no more” — and force each line to fit. In practice you do both: build bottom-up for defensibility, compare to the top-down ceiling, and negotiate the gap before the Charter is signed. If bottom-up comes in well over the ceiling, that is a **feasibility** finding, not a rounding problem.

9.11 Key Takeaways

- Cost management starts in **planning**. Sources: labour, materials, sub-contractors, equipment/facilities, travel, shipping.
- Every cost carries **two or three** of the eight labels: direct/indirect, fixed/variable, normal/expedited, recurring/non-recurring. **Direct labour cost = rate × hours**; indirect is often **20–50 % of direct**.
- Estimating method follows project stage: **ballpark ±30 %**, **parametric ±15 %**, **feasibility ±10 %**, **definitive ±5 %** — the **cone of uncertainty**. Always quote a **range**, and keep “estimate” separate from “commitment.”
- Estimate quality is driven by planning horizon, duration, people, structure, culture, and **padding** — which compounds into a meaningless total. Remove pads from work packages; hold one visible contingency.
- **Top-down** imposes discipline but breeds friction; **bottom-up** (via **activity-based costing** on the WBS) forces real planning. ABC = identify activities → cost drivers → rate per unit → rate × units.
- A **time-phased budget** produces the **cumulative cost baseline (S-curve)** you track against later.
- **Contingency reserve** (10–15 %, PM-owned, inside the baseline, covers known-unknowns) is distinct from the **management reserve** (sponsor-owned, outside the baseline, covers unknown-unknowns). Contingency is added **after** ABC, never inside it.

9.12 Practice Problems

Problem 1: Fully-burdened labour and an indirect-rate swing

A test campaign needs 2 technicians for 3 weeks (37.5 h/week each) at \$34/h,

plus \$6 200 of consumables. Compute the fully-burdened cost at an indirect rate of 25 %, and again at 45 %. How much does the indirect-rate assumption alone move the number?

Solution.

```
techs <- 2; weeks <- 3; hrs_week <- 37.5; rate <- 34
consumables <- 6200

direct_labour <- techs * weeks * hrs_week * rate
direct_cost <- direct_labour + consumables

for (ir in c(0.25, 0.45)) {
  total <- direct_cost * (1 + ir)
  cat(sprintf("Indirect %2.0f%%: direct $%s + indirect $%s = $%s\n",
              ir * 100,
              format(direct_cost, big.mark = " "),
              format(round(direct_cost * ir), big.mark = " "),
              format(round(total), big.mark = " ")))
}
cat(sprintf("\nSwing from the indirect-rate assumption alone: $%s\n",
            format(round(direct_cost * (0.45 - 0.25)), big.mark = " ")))
```

```
Indirect 25%: direct $13 850 + indirect $3 462 = $17 312
Indirect 45%: direct $13 850 + indirect $6 232 = $20 082
```

Swing from the indirect-rate assumption alone: \$2 770

The direct cost is fixed at \$21 500, but the burdened total moves by \$4 300 — 20 % of direct — purely on the overhead assumption. Always state the indirect rate you used.

Problem 2: Which estimating method, and what range?

For each situation, name the method and give the $\pm$ band around a \$500 000 point estimate.

- A retailer asks, in a hallway, roughly what a new product line will cost. You have a concept sketch.
- You have run three near-identical RC-vehicle launches in the last four years with clean cost records.
- All major POs are drafted and 90 % of the design is released for a firm-fixed-price defense bid.

Solution.

- (a) **Ballpark**, ± 30 % $\rightarrow$ \$350 000 – \$650 000.
- (b) **Comparative/parametric**, ± 15 % $\rightarrow$ \$425 000 – \$575 000.
- (c) **Definitive**, ± 5 % $\rightarrow$ \$475 000 – \$525 000. (Anything looser is a bad idea on a fixed-price bid.)

Problem 3: Variance analysis on the ABC budget

Using the preliminary ABC budget above (total planned \$63 250), the project finishes with these actuals: Survey \$4 250, Design \$8 000, Clear site \$3 500, Foundation \$8 500, Framing \$11 250, Plumb & wire \$5 150. Find the variance per activity and overall, and flag the activities that need a post-mortem.

Solution.

```
planned <- c(Survey = 4000, Design = 8000, `Clear site` = 4000,
             Foundation = 7500, Framing = 10000, `Plumb & wire` = 5000)
actual  <- c(4250, 8000, 3500, 8500, 11250, 5150)

variance <- planned - actual          # + = under budget
tab <- data.frame(Activity = names(planned),
                  Planned = as.integer(planned),
                  Actual = as.integer(actual),
                  Variance = as.integer(variance),
                  Pct = round(100 * variance / planned, 1))
print(tab, row.names = FALSE)

cat(sprintf("\nTOTAL planned %s actual %s variance %s (%.1f%%)\n",
            format(sum(planned), big.mark = " "),
            format(sum(actual), big.mark = " "),
            format(sum(variance), big.mark = " "),
            100 * sum(variance) / sum(planned)))
```

Activity	Planned	Actual	Variance	Pct
Survey	4000	4250	-250	-6.2
Design	8000	8000	0	0.0
Clear site	4000	3500	500	12.5
Foundation	7500	8500	-1000	-13.3
Framing	10000	11250	-1250	-12.5
Plumb & wire	5000	5150	-150	-3.0

TOTAL planned \$38 500 actual \$40 650 variance \$-2 150 (-5.6%)

Design and (roughly) Survey held. **Foundation (-13 %)** and **Framing (-13 %)** blew through and drove the whole project \$2 150 (3.4 %) over — those two get the post-mortem.

Problem 4 (discussion): sizing contingency for the RC-vehicle project

The project brief flags several live risks: one candidate motor supplier is unproven, that supplier's staff are negotiating a first labour contract, the carbon-fibre supplier is mid-restructuring after an acquisition, and a new assembly line would carry a learning curve. The subtotal budget is \$620 000.

Would you size the contingency at 10 %, 15 %, or 20 %? Would you spread it

across all tasks or ring-fence it for specific ones? Defend your answer.

Discussion points. With that many concentrated, plausible risks, **10 % is light**; 15 % is defensible and matches the course's Costs Chart; 20 % is arguable if the unproven-supplier and new-line risks are both realised. A strong answer **ring-fences** most of the reserve against the supplier and assembly-line line items (the tasks critical to success) rather than smearing it evenly, and states that split in the scope statement so stakeholders are not surprised when it is drawn down.

9.13 Further Resources

- **Cost estimating techniques (analogous, parametric, bottom-up):** curated explainer videos below.
- Preview of Chapter 12 — Project Control Systems: the S-curve you built here becomes the baseline for **earned value** analysis.

The Basics of Project Cost Management (ProjectManager): <https://www.youtube.com/watch?v=EyPFi0YO32M>

Cost Estimates - Key Concepts in Project Management (David McLachlan): <https://www.youtube.com/watch?v=hfn9VCAj4Dw>

How to Estimate Project Costs Accurately: <https://www.youtube.com/watch?v=2IxoWVjsphc>

What Is Earned Value? (Ricardo Vargas) - looking ahead to project control: <https://www.youtube.com/watch?v=x0EgZnXdunw>

Chapter 10

Negotiation and Conflict Management

10.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Identify** what a project manager negotiates and with whom.
2. **Prepare** for a negotiation: interests vs. positions, BATNA, reservation point, ZOPA.
3. **Apply** the principled-negotiation mindset (separate people from problem; focus on interests; win-win or no win).
4. **Determine** whether a zone of possible agreement exists and how wide it is.
5. **Classify** project conflict (goal-oriented, administrative, interpersonal) and its organizational vs. interpersonal causes.
6. **Select** a conflict-resolution method (confronting, compromising, withdrawal, smoothing, forcing) for a situation.
7. **Score** a multi-criteria supplier choice.

Before we begin: recall a negotiation you “won” but regretted

Did you win the point and lose the relationship? Most project negotiations are with people you will work with again next week. That constraint changes everything.

10.2 Introduction: the project manager negotiates every day

As a project manager your day is full of negotiation, inside and outside the organization. You negotiate for or with:

- **Resources** — material, time, equipment, money
- **Contractors** — costs, delivery schedules, quality of material, hours of work
- **Department managers** — manpower, schedules

On the RC-vehicle project the sharp negotiations are obvious: the **motor and control-system supply contract** (trusted MCS, now moving production to Mexico, vs. unproven Paragon four hours away with staff negotiating a first labour contract), and the recurring fight with **functional managers** over which engineers you get and for how long.

i Note

Industry snapshot. An automotive buyer negotiating a multi-year harness contract is really negotiating the *whole package* — price, tooling ownership, capacity guarantees, engineering support, payment terms. A defense subcontract negotiation is bounded by regulation (allowable cost, audit rights). In a food company, the PM negotiating a sanitation window with operations is negotiating *downtime*, which is money.

Project Management Negotiations: ZOPA, BATNA, and Win-Win (Parallel Project Training): <https://www.youtube.com/watch?v=MTnQX11Xey0>

10.3 The negotiation mindset

- **Separate the people from the problem.** Everyone has ego, fears, insecurities — do not tap into the other side's, or you lose control of both sides.
- **Stick to business.** Keep personalities out of it; a charm offensive fools no one.
- **Put yourself in their shoes.** Nobody likes a direct attack; it is counter-productive.
- **Win-win or no win.** Talk money **last** if possible; a negotiation is about the whole package, not just the number.

10.4 Getting ready

There are three stages — **before, during, after** — and **before should be the biggest**.

Gather information about the other side: their vital interests, goals, costs, strategies, positions, needs; their product strengths; concession patterns; past negotiating style; their **bottom line**; corporate deadlines and time constraints; their **budget**; geographic limits. Also know your own position, the players, and **who holds the power** — including people behind the scenes.

Develop strategy: aim high (with more than one win-win scenario in mind); be realistic; rank your objectives by priority; know your **bottom line**; decide whether deadlock or termination are tolerable outcomes; separate the **necessary** items from the **expendable**; have a few **red herrings** ready; plan how to help the other side sell the deal internally; find where your interests and theirs intersect.

Write the plan and an agenda: your interests and theirs; your desired outcomes and your estimate of theirs; your authority limits and theirs; any fact sheet either side needs. The agenda names the meeting's goal, is published in advance, and lists invitees and location.

10.4.1 The three questions — and who controls the room

Every negotiator should ask: **How much power do I have? What time pressure is there? Do I trust the other side?**

And: *who controls a conversation?* **Generally the person asking the questions**, because they set the direction. Aim to ask most of the questions — politely — and recognise when it is being done to you.

10.5 BATNA, reservation point, and ZOPA

- **BATNA** — your Best Alternative To a Negotiated Agreement. What you will do if this deal falls through. A strong BATNA is your real source of power.
- **Reservation point** — the worst deal you would still accept (a buyer's *maximum* price, a seller's *minimum* price), set by your BATNA.
- **ZOPA** — the Zone Of Possible Agreement: the overlap between the two reservation points. If there is no overlap, **no deal is possible** at those terms.

10.5.1 Worked Example 1: is there a ZOPA?

You are buying the motor/controller package. Your BATNA (Paragon's quote plus the cost of the risk it carries) puts your **reservation price at \$168 000**

— you would not pay MCS more than that. MCS's floor, given their costs and other customers, is **\$150 000**.

```
buyer_reservation <- 168000 # most you will pay
seller_reservation <- 150000 # least they will accept

if (seller_reservation <= buyer_reservation) {
  zopa_low <- seller_reservation
  zopa_high <- buyer_reservation
  cat(sprintf("ZOPA exists: $%s to $%s (width $%s)\n",
    format(zopa_low, big.mark = " "),
    format(zopa_high, big.mark = " "),
    format(zopa_high - zopa_low, big.mark = " ")))
  cat("Any agreed price in that band beats both sides' alternatives.\n")
} else {
  cat("No ZOPA -- the seller's floor is above the buyer's ceiling. Change the package")
}
```

```
ZOPA exists: $150 000 to $168 000 (width $18 000)
Any agreed price in that band beats both sides' alternatives.
```

10.5.2 Try it yourself: reservation points from a BATNA

```
## label: zopa-calc-10
# Buyer's side: BATNA is Paragon at $150k, but carrying $25k of extra risk cost.
paragon_price <- 150000
paragon_risk <- 25000
buyer_reservation <- paragon_price + paragon_risk # indifference point vs MCS

# Seller's side (MCS): cost to make + minimum acceptable margin.
mcs_cost <- 132000
mcs_margin <- 15000
seller_reservation <- mcs_cost + mcs_margin

cat(sprintf("Buyer reservation (max) : $%s\n", format(buyer_reservation, big.mark = " ")))
cat(sprintf("Seller reservation (min) : $%s\n", format(seller_reservation, big.mark = " ")))

if (seller_reservation <= buyer_reservation) {
  cat(sprintf("ZOPA width: $%s -- a deal is possible.\n",
    format(buyer_reservation - seller_reservation, big.mark = " ")))
  cat(sprintf("A midpoint settlement would be about $%s.\n",
    format(round((buyer_reservation + seller_reservation) / 2), big.mark = " ")))
} else {
  cat("No ZOPA. Improve your BATNA, expand the package, or walk away.\n")
}
```

Why does improving your BATNA move the whole negotiation?

Your reservation point is set by your BATNA. If you pre-qualify a credible second supplier (a better BATNA), your reservation price drops — you can credibly walk at a lower number — which shifts the ZOPA and the likely settlement in your favour, *without* saying a word at the table. The best negotiating move often happens before the meeting, in the form of a real alternative.

10.6 Language and tells

A few sample moves: “*My colleagues and I cannot settle for that*” (establish power early); “*What are you looking for in these negotiations?*” (make them open first); “*Do you have the authority to commit to that?*” (probe their authority); “*Let’s set that aside and come back to it*” (avoid a hard no).

And listen past the words. “*To be brutally honest...*” and “*To tell you the truth...*” often precede a lie. “*Off the record...*” — nothing is off the record; the negotiation is always on. Watch **congruence**: when words and body language disagree, believe the body language.

10.7 Conflict management

Conflict “*begins when you perceive that someone has frustrated, or is about to frustrate, a major concern of yours.*” It is **perceptual** (it doesn’t matter whether a wrong was actually done, only that it is perceived) and **dynamic** (the original cause may no longer matter, yet the conflict persists).

Three types on projects:

- **Goal-oriented** — disagreement over end results, scope outcomes, specs, priorities. Fuelled by vague goals that let people fill in their own interpretation (and hidden agendas).
- **Administrative** — who has authority or control over functions, tasks, and decisions. The **matrix** structure breeds this — everyone has two bosses.
- **Interpersonal** — personality differences: work ethic, behavioural style, ego, personalities that don’t mesh.

10.7.1 Three views of conflict

School	View	Response
Traditional	All conflict is bad	Suppress and eliminate it, fast
Behavioral	Conflict is natural and inevitable	Accept it and manage it well

School	View	Response
Interactionist	A <i>controlled</i> amount of conflict is <i>beneficial</i>	Encourage some — it prevents stagnation and drives innovation; too little = inertia, too much = chaos

```

cs <- data.frame(
  source = c("Schedules", "Project priorities", "Resources", "Technical opinions",
            "Administrative procedures", "Cost", "Personality"),
  intensity = 7:1
)
cs$source <- factor(cs$source, levels = rev(cs$source))
ggplot(cs, aes(intensity, source)) +
  geom_col(fill = "#5DADE2", width = 0.68) +
  geom_text(aes(label = intensity), hjust = -0.4, size = 3.6 * diagram_text_scale) +
  scale_x_continuous(limits = c(0, 8), expand = expansion(mult = c(0, 0.02))) +
  labs(title = "What project teams actually fight about",
       x = "Relative intensity (7 = highest)", y = NULL) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )

```

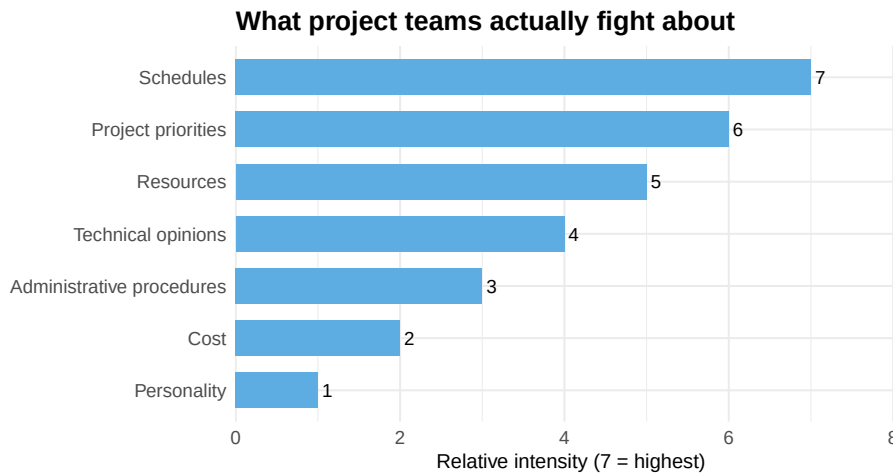


Figure 10.1: Sources of project conflict ranked by typical intensity (from PMBOK data). Schedules, priorities, and resources dominate; personality is last – it is usually the first thing blamed and rarely the real cause.

Organizational causes: reward systems (functional managers hoard their best people), scarce resources, uncertainty (weak matrix — unclear who is in charge), different mindsets among departments. **Interpersonal causes:** misconceptions about others’ behaviour, faulty communication (often *style*, not content), personal grudges.

“It only becomes personal when the root cause of the conflict is not resolved. Personality is very frequently the first diagnosis” — and usually the wrong one.

10.7.3 Conflict-resolution methods (PMBOK)

Method	What it is	Best when
Confronting (problem-solving)	Solve the real, underlying problem	There is time and goodwill; the durable fix
Compromising	Each side gives something; partial satisfaction for both	Positions are far apart and a deadline looms
Withdrawal / avoidance	Retreat or postpone	The issue is trivial, or emotions need to cool
Smoothing	Emphasize agreement, play down differences	Preserving the relationship matters more right now

Method	What it is	Best when
Forcing	Push one view at the other's expense	A true emergency, or a matter of safety/ethics

10.7.4 Try it yourself: score the supplier choice

Multi-criteria scoring turns “MCS feels safer” into a defensible number.

```
#| label: supplier-scoring-10
criteria <- c("Quality record", "Delivery reliability", "Price",
             "Technical capability", "Supply-chain risk")
weight    <- c(0.30, 0.25, 0.15, 0.15, 0.15) # must sum to 1

# Score each supplier 1 (poor) to 5 (excellent) on each criterion.
mcs       <- c(5, 4, 3, 3, 2) # trusted, but production moving to Mexico
paragon   <- c(3, 3, 4, 5, 3) # unproven, new tech, plant 4 h away, labour talks

mcs_total    <- sum(weight * mcs)
paragon_total <- sum(weight * paragon)

tab <- data.frame(Criterion = criteria, Weight = weight,
                  MCS = mcs, Paragon = paragon)
print(tab, row.names = FALSE)
cat(sprintf("\nWeighted score MCS: %.2f Paragon: %.2f\n", mcs_total, paragon_total))
cat(sprintf("Recommended: %s\n", ifelse(mcs_total >= paragon_total, "MCS", "Paragon")))
cat("Change the weights to your project's priorities and watch the answer move.\n")
```

10.8 Key Takeaways

- PMs negotiate constantly — for resources, with contractors, with functional managers. **Before** is the biggest stage; prepare more than you think you need.
- **Separate people from the problem**, focus on **interests** not positions, aim for **win-win or no win**, and talk money last.
- **BATNA** sets your **reservation point**; the overlap of both reservation points is the **ZOPA**. No ZOPA, no deal — improve your BATNA or change the package.
- The person **asking the questions** usually controls the conversation.
- Project conflict is **perceptual and dynamic**. Types: **goal-oriented**, **administrative**, **interpersonal**. The **interactionist** view holds that a controlled amount of conflict is healthy.
- Teams fight most about **schedules, priorities, and resources** — **personality** is last and is usually a misdiagnosis of an unresolved root cause.

- Match the resolution method — **confronting, compromising, withdrawal, smoothing, forcing** — to the situation; confronting is the durable fix.

10.9 Practice Problems

Problem 1: Name the cause of conflict

Which cause does each statement point to?

1. “I can’t talk to her after what her son said to my son.” → **Personal grudge** (interpersonal).
2. “Sales management here are lazy — they always leave by 4:00.” → **Different mindsets among departments** (organizational).
3. “She barely looked at my report; she’s already decided to ignore it.” → **Misconceptions about others’ behaviour** (interpersonal).
4. “Our computers are slow because their division always gets the best ones.” → **Scarce resources** (organizational).
5. “The worse your project does at year-end, the better mine looks.” → **Reward systems** (organizational).
6. “Why should I do that for the PM if my functional manager didn’t tell me?” → **Uncertainty / two bosses** (organizational, matrix).

Problem 2: ZOPA and a walk-away

You will pay at most \$95 000 for a tooling package (your BATNA: build it in-house for \$98 000). The vendor’s floor is \$102 000.

- (a) Is there a ZOPA? (b) What are your options?

Solution.

```
buyer_max <- 95000
seller_min <- 102000
cat(ifelse(seller_min <= buyer_max, "ZOPA exists", "No ZOPA"), "\n")
cat(sprintf("Gap to close: %s\n", format(seller_min - buyer_max, big.mark = " ")))
```

No ZOPA

Gap to close: \$7 000

- (a) No — the vendor’s floor is \$7 000 above your ceiling. (b) Exercise your BATNA (build in-house at \$98 000), expand the package so the vendor’s per-unit economics improve, split tooling ownership, or find a third vendor.

Problem 3 (discussion): the functional manager skips your stand-up

The maintenance supervisor on your project has missed the first two morning meetings. How could this signal *each* of: reward systems, scarce resources, uncertainty, different mindsets, misconceptions, faulty communication, personal grudge?

Discussion points. *Reward systems:* their boss doesn't value project work, so they deprioritize it. *Scarce resources:* they are triple-booked and the meeting lost. *Uncertainty:* they are not sure the meeting applies to them. *Different mindsets:* maintenance culture sees planning meetings as talk, not work. *Misconceptions:* they think you have already sidelined their input. *Faulty communication:* the invite never reached them, or the time was ambiguous. *Grudge:* a past project soured them on you. The point: **do not jump to the grudge** — confront it by asking, and you will usually find one of the structural causes.

10.10 Further Resources

Negotiation Strategy: ZOPA (Mediator Academy): <https://www.youtube.com/watch?v=JEoFHePrp6k>

BATNA and ZOPA in Negotiations: <https://www.youtube.com/watch?v=kJWF2oYzTq0>

Chapter 11

Resource Management

11.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Name** the constraint types on a project and focus on the resource constraints.
2. **Build** a resource histogram and spot over-allocation.
3. **Apply** resource **leveling** and the heuristics that decide which task moves first.
4. **Distinguish** time-constrained from resource-constrained scheduling.
5. **Apply** resource **smoothing** within available float.
6. **Explain** the multi-project resource problem and the priority rules used to resolve it.
7. **Quantify** the cost of multitasking on effective capacity.

Before we begin: your plan assumed the person was free — were they?

Almost every first-draft project schedule is built as if the organization has unlimited trained people standing by. It never does. This chapter is what happens when the plan meets the staffing reality.

11.2 Introduction: the optimism of the first schedule

The RC-vehicle schedule from Chapter 8 quietly assumed every resource was available exactly when needed. Reality: there is **one CMM** for validation, the **millwrights** are shared with production, the design engineers are **borrowed** from a functional department, and the new assembly line has a **learning curve**. *Building plans under the assumption of unlimited resources inevitably leads to*

schedule slippage and extra cost as reality overtakes the optimism.

Projects run inside three kinds of constraint — **technical**, **physical**, and **resource** — and this chapter is about the last one: **people, materials, space, money, equipment**.

i Note

Industry snapshot. An automotive company with one set of climate chambers watches validation projects stack up behind that single resource — the classic bottleneck. A food plant schedules every project around the *one* weekend sanitation window. A defense program's cleared personnel are a fixed pool, so every new task is a reallocation, not an addition.

Resource Allocation and Leveling: Tutorial and Example (AF Math & Engineering): <https://www.youtube.com/watch?v=JIUXah6fsLI>

11.3 Resource loading and the histogram

A **resource histogram** plots how much of a resource is required over time. Stack each activity's demand day by day and the over-allocations jump out — provided you remember the chart assumes each person is available **100 % of the time**, which is rarely true.

```
days <- 1:14
before <- c(6,6,6,6,6, 9,9,9,9,10, 8,10,9,7)
after  <- c(6,6,6,6,6, 9,9,9,9,9, 9,9,9,7)
df <- data.frame(
  day = rep(days, 2),
  people = c(before, after),
  state = rep(c("Before leveling", "After leveling"), each = length(days))
)
df$state <- factor(df$state, levels = c("Before leveling", "After leveling"))

ggplot(df, aes(day, people)) +
  geom_col(aes(fill = people > 9), width = 0.75) +
  geom_hline(yintercept = 9, linetype = "dashed", color = "#C0392B", linewidth = 0.9) +
  annotate("text", x = 2, y = 9.6, label = "limit = 9/day",
    color = "#C0392B", size = 3 * diagram_text_scale, hjust = 0) +
  scale_fill_manual(values = c("FALSE" = "#5DADE2", "TRUE" = "#C0392B"), guide = "none") +
  facet_wrap(~state, ncol = 1) +
  scale_x_continuous(breaks = days) +
  labs(title = "Leveling removes the over-allocation peak",
    x = "Project day", y = "People required") +
  theme_minimal() +
  theme(
```

```

plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
strip.text = element_text(size = 11 * diagram_text_scale, face = "bold"),
axis.title = element_text(size = 11 * diagram_text_scale),
axis.text = element_text(size = 9.5 * diagram_text_scale)
)

```

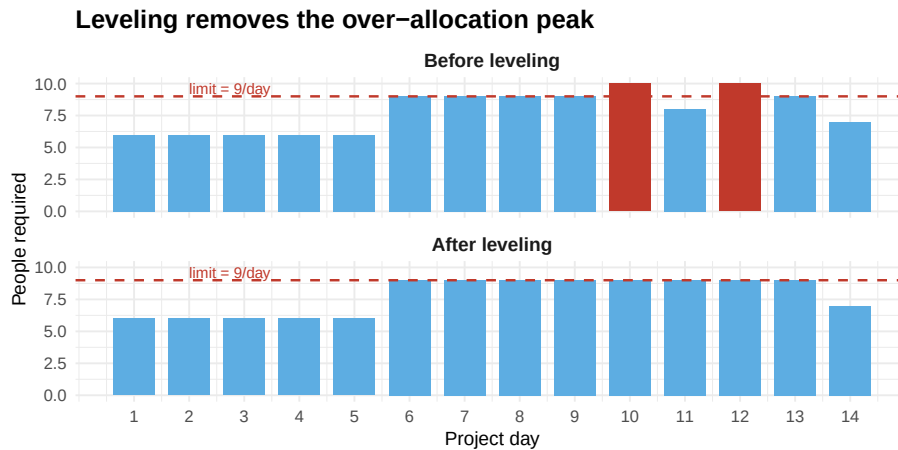


Figure 11.1: Resource histogram before and after leveling, against a limit of 9 people/day. Leveling shifts a task with float so the peak on day 12 no longer breaches the limit – without extending the project.

11.4 Resource leveling

PMBOK: *a technique in which start and finish dates are adjusted based on resource constraints, with the goal of balancing demand for resources against the available supply.*

Two objectives: (1) make sure resources are available at the right time, and (2) let each activity run with the **smoothest possible transitions** across resource-usage levels.

When you must choose which activity to move, work through them in this order:

1. Activities with the **smallest float**
2. Activities with the **smallest duration**
3. Activities with the **lowest ID** (earliest in the WBS)
4. Activities with the **most successors**
5. Activities requiring the **most resources**

11.4.1 Worked Example 1: which task do you move?

Day 12 needs 10 people against a limit of 9. Four activities are running that day:

```
running_day12 <- data.frame(
  task = c("D", "E", "H", "J"),
  float = c(12, 0, 0, 5),
  duration = c(6, 6, 7, 3),
  successors = c(1, 2, 1, 0),
  people = c(3, 3, 3, 1)
)
print(running_day12, row.names = FALSE)

# Only tasks with float can move without extending the project.
movable <- running_day12[running_day12$float > 0, ]
# Among movable tasks, prefer smallest float, then smallest duration.
pick <- movable[order(movable$float, movable$duration), ][1, ]
cat(sprintf("\nMovable tasks: %s\n", paste(movable$task, collapse = ", ")))
cat(sprintf("Move first: task %s (float %d, duration %d) -- shift it later within its slack\n",
            pick$task, pick$float, pick$duration))
cat("E and H are on the critical path (float 0) -- moving them would extend the project\n")
```

task	float	duration	successors	people
D	12	6	1	3
E	0	6	2	3
H	0	7	1	3
J	5	3	0	1

Movable tasks: D, J

Move first: task J (float 5, duration 3) -- shift it later within its slack.

E and H are on the critical path (float 0) -- moving them would extend the project.

11.5 Time-constrained vs. resource-constrained

- **Time-constrained:** the end date is fixed; you add or reassign resources to protect it (this is closer to *smoothing*).
- **Resource-constrained:** the resource limit is fixed; if leveling within float is not enough, the **project duration extends** (undesirable, but sometimes unavoidable).

11.6 Resource smoothing

Smoothing is done **after** leveling and **never extends the project duration**. It uses available **float** to even out resource usage — filling the valleys from the peaks.

11.6.1 Try it yourself: smooth a resource within its float

```

#| label: smoothing-11
# Weekly demand for one resource, and the resource's weekly limit.
weekly_demand <- c(52, 30, 61, 24, 45, 20, 58)
limit <- 45

peak <- max(weekly_demand)
total_work <- sum(weekly_demand)
weeks <- length(weekly_demand)
ideal_flat <- total_work / weeks

cat(sprintf("Peak week demand    : %d (limit %d -> %d over)\n", peak, limit, max(0, peak - limit)))
cat(sprintf("Total work          : %d resource-weeks over %d weeks\n", total_work, weeks))
cat(sprintf("Perfectly smooth    : %.1f per week\n", ideal_flat))

if (ideal_flat <= limit) {
  cat("\nThe work FITS within the limit if spread evenly -> smoothing can solve this\n")
  cat("by shifting tasks into their float, no schedule change needed.\n")
} else {
  cat(sprintf("\nEven perfectly smoothed (%.1f/wk) still exceeds the limit -> this needs\n", ideal_flat))
  cat("leveling (extend duration) or more resource, not just smoothing.\n")
}

```

11.7 Splitting a task

Splitting means interrupting continuous work on an activity, moving that resource to another activity, then returning. It helps *only* when there are no large **start-up or shutdown costs**, and **only some tasks can be split** at all (you cannot pause a concrete pour).

11.8 The multi-project resource problem

Organizations rarely buy new resources for each project — they **share** them across projects. The core challenge is **interdependency**: a resource decision in one project ripples into others. Common problems:

- Inefficient use of resources
- Resource **bottlenecks**
- **Ripple effects**
- Heightened pressure on people to **multitask**

Any multi-project scheduling system tries to minimize three things:

1. **Schedule slippage** — often carrying real financial penalty clauses per day late.

2. **Resource utilization** losses — the unused-capacity gap; the goal is to fill it with other projects' work.
3. **In-process inventory (WIP)** — work waiting on an unavailable resource, stacked behind a bottleneck (e.g. one shared climate chamber).

```

wk <- 1:14
used <- c(3,4,5,6,6,7,7,6,5,5,4,3,3,2)
cap <- 8
df <- data.frame(wk, used, unused = cap - used)
df_long <- tidyr::pivot_longer(df, c(used, unused), names_to = "kind", values_to = "res")
df_long$kind <- factor(df_long$kind, levels = c("unused", "used"),
                      labels = c("Unused capacity", "Used by this project"))

ggplot(df_long, aes(wk, res, fill = kind)) +
  geom_col(width = 0.8) +
  geom_hline(yintercept = cap, linetype = "dashed", color = "grey30") +
  scale_fill_manual(values = c("Used by this project" = "#5DADE2",
                              "Unused capacity" = "#F5CBA7"), name = NULL) +
  scale_x_continuous(breaks = wk) +
  labs(title = "Where is the idle capacity?",
       x = "Project week", y = "Resource units") +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )

```

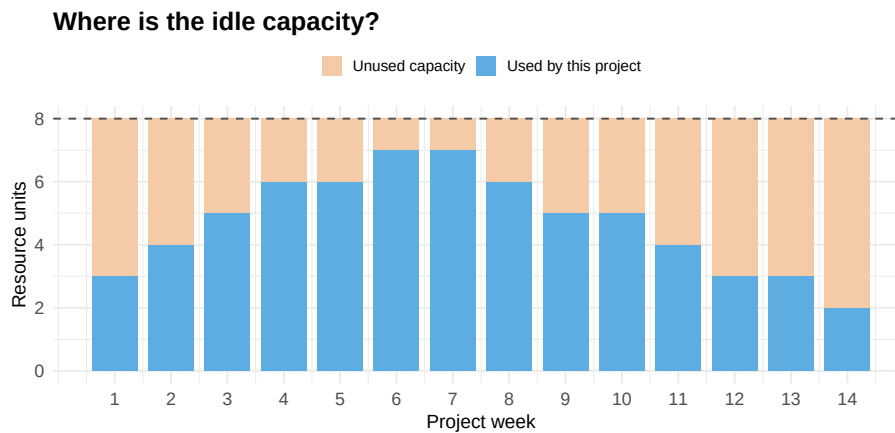


Figure 11.2: Resource utilization across a project. The shaded area is capacity the project does not use – in a multi-project shop it should be filled with concurrent projects’ work rather than left idle.

11.8.1 Priority rules for allocating shared resources

Rule	Logic
First in line (first come, first served)	Simplest — follow the master calendar; ignores priority and commercial value
Greatest resource demand	Schedule the most resource-hungry project first so bottlenecks surface early
Greatest resource utilization (most resource <i>types</i>)	Schedule the project that uses the widest mix of resources first, keeping all of them busy
Minimum late finish (earliest finish first)	Prioritize the project with the earliest late-finish date
Mathematical programming	Optimize (minimize total time / utilization loss / lateness); powerful but complex and slow for large problems

11.8.2 Try it yourself: “40 hours is not one week’s work”

```
#| label: multitask-11
nominal_hours <- 40
# Each additional concurrent project imposes a context-switching loss.
```

```

switch_loss_per_extra_project <- 0.15    # 15% of capacity lost per extra project

for (projects in 1:4) {
  extras <- projects - 1
  effective <- nominal_hours * (1 - switch_loss_per_extra_project * extras)
  effective <- max(effective, 0)
  cat(sprintf("On %d project(s): %.0f 'nominal' hours -> %.1f effective hours (%.0f%%\n",
    projects, nominal_hours, effective, 100 * effective / nominal_hours))
}
cat("\nAssigning a person to a project at '40 hours' while they carry two others\n")
cat("is really assigning maybe 28 -- which is how resource-leveled plans still slip.\n")

```

11.9 Sustainability and resources

Resource management is not only about people-hours. The **circular-economy** thinking from Chapter 1 applies directly: minimize raw material use, prefer recyclable and local inputs, design the project so equipment and tooling can be reused on the next program, and track material and energy as managed resources with their own KPIs — not just labour and money.

11.10 Key Takeaways

- Projects run inside **technical, physical, and resource** constraints; resources are people, materials, space, money, equipment.
- A **resource histogram** exposes **over-allocation** (against a 100 %-availability assumption).
- **Leveling** shifts tasks within their **float** to respect a resource limit; move tasks in order of **smallest float** → **smallest duration** → **lowest ID** → **most successors** → **most resources**. Critical-path tasks have no float, so moving them extends the project.
- **Time-constrained** protects the date with resources; **resource-constrained** may extend the date.
- **Smoothing** happens after leveling, uses float, and **never** extends duration.
- **Splitting** helps only when start-up/shutdown costs are low and the task can be split.
- Multi-project sharing creates **bottlenecks, ripple effects, and multi-tasking pressure**; minimize **slippage, utilization loss, and WIP**, and pick a **priority rule** deliberately.
- **Multitasking is not free** — “40 hours” of nominal capacity is far less once someone carries several projects.

11.11 Practice Problems

Problem 1: Level to the limit

Day 10 requires 10 people; the limit is 9. Running that day: task C (float 4, duration 5, 4 people), task D (float 0, duration 6, 3 people), task F (float 0, duration 6, 3 people). Which task moves, and what is the risk if none can?

Solution.

```
d10 <- data.frame(task = c("C", "D", "F"),
                  float = c(4, 0, 0), dur = c(5, 6, 6), ppl = c(4, 3, 3))
mov <- d10[d10$float > 0, ]
cat("Only task with float:", mov$task, "\n")
cat("Move C into its 4 days of slack -> day-10 demand drops to 6, under the limit.\n")
cat("If C could not move (no split, hard dependency), the project duration extends.\n")
```

Only task with float: C

Move C into its 4 days of slack -> day-10 demand drops to 6, under the limit.

If C could not move (no split, hard dependency), the project duration extends.

Problem 2: Does this need leveling or smoothing?

A resource has a 40 h/week limit. Weekly demand over 6 weeks is 55, 20, 48, 15, 50, 22 h. Total 210 h.

Solution.

```
demand <- c(55, 20, 48, 15, 50, 22); limit <- 40
flat <- sum(demand) / length(demand)
cat(sprintf("Even spread: %.1f h/week vs limit %d\n", flat, limit))
cat(ifelse(flat <= limit,
           "Fits if spread evenly -> smoothing (shift within float) can solve it.\n",
           "Still over even when flat -> leveling (extend) or add resource.\n"))
```

Even spread: 35.0 h/week vs limit 40

Fits if spread evenly -> smoothing (shift within float) can solve it.

35 h/week average is under the 40 h limit, so **smoothing** within float can solve it without extending the project.

Problem 3 (discussion): the shared CMM bottleneck

Three projects, including yours, all need the single CMM for final validation in the same two-week window. Using the priority rules, how would you decide the order, and what could reduce the bottleneck?

Discussion points. Apply a rule and be explicit about it: **minimum late finish** (whichever project has the tightest deadline goes first) is defensible if one launch date is immovable; **greatest resource demand** surfaces the bottleneck earliest. Reduce it by: staggering the validation tasks into each project's float,

renting a second CMM for the peak, outsourcing some measurement to a metrology lab, or moving non-CMM checks off the critical window so the CMM time is used only for what truly needs it.

11.12 Further Resources

What Is Resource Leveling in Project Management?: <https://www.youtube.com/watch?v=iSGWSGfG65U>

Chapter 12

Project Control Systems

12.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Describe** the monitoring-and-control loop and the performance measurement baseline.
2. **Read** the three earned-value curves: planned value (PV), earned value (EV), and actual cost (AC).
3. **Compute** schedule and cost variance and the SPI / CPI performance indices.
4. **Forecast** with EAC, ETC, VAC, and TCPI, and interpret them.
5. **Run** integrated change control so the baseline stays meaningful.
6. **Report** status honestly and recognize the “watermelon” trap.

Before we begin: the project has spent 60 % of its money — is it 60 % done?

Almost certainly not. “Percent of budget spent” and “percent of work finished” are different numbers, and the gap between them is the whole subject of this chapter.

12.2 Introduction: from a plan to a control system

You have planned the RC-vehicle project — scope, WBS, schedule, budget. It is now **week 8**. The sponsor asks the only question that matters at this stage: **are we on track, and if not, how far off, and what will it cost to finish?**

Answering that needs a **control system**: a closed loop that **measures** progress, **compares** it to the baseline, **analyzes** the variance, and drives a corrective **action** — then measures again.

```

loop <- data.frame(
  x = c(0, 2.2, 4.4, 6.6),
  y = 0,
  label = c("MEASURE\nprogress & cost", "COMPARE\nto baseline",
            "ANALYZE\nvariance", "ACT\ncorrective action")
)
ggplot(loop, aes(x, y)) +
  geom_label(aes(label = label), size = 3 * diagram_text_scale, lineheight = 0.9,
            fill = "#D6EAF8", label.size = 0, fontface = "bold") +
  annotate("segment", x = c(0.95, 3.15, 5.35), xend = c(1.55, 3.75, 5.95),
          y = 0, yend = 0, arrow = arrow(length = unit(0.18, "cm")), color = "grey40") +
  annotate("curve", x = 6.6, y = -0.35, xend = 0, yend = -0.35,
          curvature = 0.25, arrow = arrow(length = unit(0.18, "cm")), color = "grey40") +
  annotate("text", x = 3.3, y = -0.62, label = "repeat every reporting period",
          size = 2.8 * diagram_text_scale, fontface = "italic", color = "grey45") +
  coord_cartesian(xlim = c(-0.9, 7.5), ylim = c(-0.9, 0.6)) +
  theme_void()

```

Warning: The `label.size` argument of `geom\_label()` is deprecated as of ggplot2 3.5.0. Please use the `linewidth` argument instead.

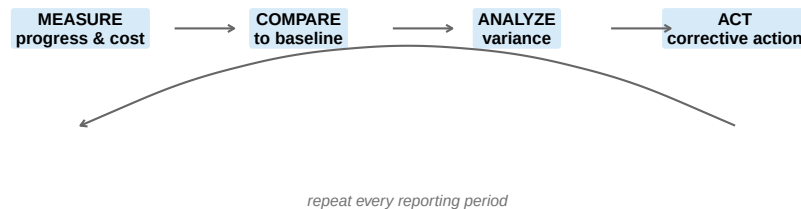


Figure 12.1: The monitoring-and-control loop. Without the ‘compare to baseline’ step, status reporting is just activity; with it, you get variance you can act on.

The reference for “compare” is the **performance measurement baseline** — the approved **scope + schedule + cost** locked in at the end of planning (Chapter 9’s S-curve is its cost dimension).

i Note

Industry snapshot. Earned value management is **contractually mandated** on large U.S. defense programs (EIA-748). Automotive programs run a lighter version at the milestone level. A food-plant capital project tracks committed vs. spent vs. installed. The math below is the same at every scale.

What Is Earned Value? (Ricardo Vargas): <https://www.youtube.com/watch?v=x0EgZnXdunw>

12.3 The three curves

Earned value management tracks three cumulative quantities against time:

Term	Also called	Meaning
PV — Planned Value	BCWS	The budgeted cost of the work scheduled to be done by now
EV — Earned Value	BCWP	The budgeted cost of the work actually done by now
AC — Actual Cost	ACWP	What has actually been spent to do that work

BAC (Budget At Completion) is the total baseline budget — where the PV curve ends.

```
cutoff <- 8
wk <- seq(0, cutoff, by = 0.25)
# S-shaped ramps scaled to pass through the week-8 values in Worked Example 1.
shape <- function(t, k) (t / cutoff)^k
df <- data.frame(
  wk = wk,
  PV = 217000 * shape(wk, 1.8),
  EV = 186000 * shape(wk, 2.0),
  AC = 210000 * shape(wk, 1.7)
)

ggplot() +
  geom_line(data = df, aes(wk, PV, color = "PV (planned value)"), linewidth = 1.4) +
  geom_line(data = df, aes(wk, EV, color = "EV (earned value)"), linewidth = 1.4) +
  geom_line(data = df, aes(wk, AC, color = "AC (actual cost)"), linewidth = 1.4) +
  geom_vline(xintercept = cutoff, linetype = "dotted", color = "grey40") +
```

```

geom_point(data = data.frame(m = c("PV", "EV", "AC"), y = c(217000, 186000, 210000))
  aes(x = cutoff, y = y, color = m), size = 3, show.legend = FALSE) +
annotate("text", x = cutoff + 0.12, y = 228000, hjust = 0,
  label = "PV $217 000", size = 3 * diagram_text_scale, color = "#2E86C1") +
annotate("text", x = cutoff + 0.12, y = 205000, hjust = 0,
  label = "AC $210 000", size = 3 * diagram_text_scale, color = "#C0392B") +
annotate("text", x = cutoff + 0.12, y = 178000, hjust = 0,
  label = "EV $186 000", size = 3 * diagram_text_scale, color = "#27AE60") +
annotate("text", x = cutoff - 0.25, y = 45000, hjust = 1,
  label = "status date\n(week 8)", size = 3 * diagram_text_scale, lineheight = 1.2) +
scale_color_manual(values = c("PV (planned value)" = "#2E86C1",
  "EV (earned value)" = "#27AE60",
  "AC (actual cost)" = "#C0392B",
  "PV" = "#2E86C1", "EV" = "#27AE60", "AC" = "#C0392B"),
  breaks = c("PV (planned value)", "EV (earned value)", "AC (actual cost)"),
  name = NULL) +
scale_y_continuous(labels = scales::dollar_format(big.mark = " ")) +
coord_cartesian(xlim = c(0, 9.7)) +
labs(title = "Earned value: planned vs. earned vs. actual",
  x = "Project week", y = "Cumulative cost") +
theme_minimal() +
theme(
  legend.position = "top",
  plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
  axis.title = element_text(size = 11 * diagram_text_scale),
  axis.text = element_text(size = 10 * diagram_text_scale)
)

```

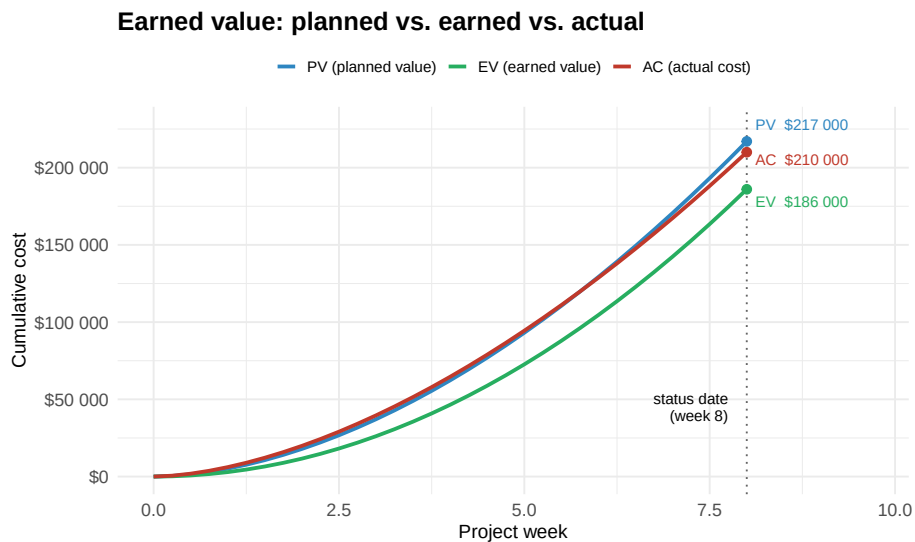


Figure 12.2: The three curves through week 8 of the RC-vehicle project (BAC = \$620 000, off the top of this view). By the status date EV sits below PV – behind schedule – and AC sits above EV – over cost. The vertical gaps at week 8 are the schedule and cost variances.

12.4 The six core metrics

$$\text{Schedule Variance } SV = EV - PV$$

$$\text{Cost Variance } CV = EV - AC$$

$$\text{Schedule Performance Index } SPI = EV/PV$$

$$\text{Cost Performance Index } CPI = EV/AC$$

Reading	Schedule	Cost
Ahead / under	$SV > 0, SPI > 1$	$CV > 0, CPI > 1$
On plan	$SV = 0, SPI = 1$	$CV = 0, CPI = 1$
Behind / over	$SV < 0, SPI < 1$	$CV < 0, CPI < 1$

12.4.1 Worked Example 1: the RC-vehicle project at week 8

Baseline budget **BAC = \$620 000**. By week 8 the plan said **35 %** of the work should be done; **30 %** is actually done; **\$210 000** has been spent.

```

BAC <- 620000
planned_pct <- 0.35
actual_pct <- 0.30
AC <- 210000

PV <- planned_pct * BAC
EV <- actual_pct * BAC

SV <- EV - PV
CV <- EV - AC
SPI <- EV / PV
CPI <- EV / AC

cat(sprintf("PV = $%s      EV = $%s      AC = $%s\n",
            format(PV, big.mark = " "), format(EV, big.mark = " "),
            format(AC, big.mark = " ")))
cat(sprintf("SV = $%s (%s schedule)\n", format(SV, big.mark = " "),
            ifelse(SV < 0, "behind", "ahead")))
cat(sprintf("CV = $%s (%s budget)\n", format(CV, big.mark = " "),
            ifelse(CV < 0, "over", "under")))
cat(sprintf("SPI = %.3f (getting %.0f cents of schedule per planned dollar)\n",
            SPI, SPI * 100))
cat(sprintf("CPI = %.3f (getting %.0f cents of work per dollar spent)\n",
            CPI, CPI * 100))

```

```

PV = $217 000      EV = $186 000      AC = $210 000
SV = $-31 000 (behind schedule)
CV = $-24 000 (over budget)
SPI = 0.857 (getting 86 cents of schedule per planned dollar)
CPI = 0.886 (getting 89 cents of work per dollar spent)

```

The project is **behind schedule and over cost** — and this is visible in week 8, while there is still time to act, instead of as a surprise at the end.

12.5 Forecasting

Forecast	Formula	Meaning
EAC (Estimate At Completion)	BAC/CPI	Total cost if current cost efficiency holds
EAC (schedule too)	$AC + \frac{BAC - EV}{CPI \times SPI}$	If both cost <i>and</i> schedule performance persist
EAC (one-off overrun)	$AC + (BAC - EV)$	If the overrun so far was a one-time event

Forecast	Formula	Meaning
ETC (Estimate To Complete)	$EAC - AC$	Money still needed from here
VAC (Variance At Completion)	$BAC - EAC$	Projected total over/under budget
TCPI (To-Complete Performance Index)	$\frac{BAC - EV}{BAC - AC}$	The cost efficiency the <i>remaining</i> work must hit to still land on BAC

12.5.1 Try it yourself: EAC scenarios

```
#| label: eac-scenarios-12
BAC <- 620000
EV  <- 186000
AC  <- 210000
PV  <- 217000
CPI <- EV / AC
SPI <- EV / PV

eac_cpi      <- BAC / CPI
eac_oneoff   <- AC + (BAC - EV)
eac_cost_sched <- AC + (BAC - EV) / (CPI * SPI)

cat(sprintf("CPI = %.3f   SPI = %.3f\n\n", CPI, SPI))
cat(sprintf("EAC, cost efficiency continues : %s\n", format(round(eac_cpi), big.mark = " ")))
cat(sprintf("EAC, overrun was a one-off      : %s\n", format(round(eac_oneoff), big.mark = " ")))
cat(sprintf("EAC, cost AND schedule persist : %s\n", format(round(eac_cost_sched), big.mark = " ")))

etc  <- eac_cpi - AC
vac  <- BAC - eac_cpi
tcpi <- (BAC - EV) / (BAC - AC)
cat(sprintf("\nETC (money still needed) : %s\n", format(round(etc), big.mark = " ")))
cat(sprintf("VAC (projected overrun)   : %s\n", format(round(vac), big.mark = " ")))
cat(sprintf("TCPI (efficiency needed) : %.3f ", tcpi))
cat(ifelse(tcpi > 1.10, "-- that is a big ask; the BAC is likely unrealistic now.\n",
  ifelse(tcpi > 1.0, "-- recoverable, but the team must run leaner than planned.\n",
    "-- achievable.\n")))
```

Which EAC formula should you actually report?

It depends on *why* you are over. If a one-time event caused it (a failed test you have since fixed), $AC + (BAC - EV)$ is fair. If the overrun reflects how the team is fundamentally working (rates, productivity), BAC/CPI is the honest number. If you are *also* behind schedule and recovery will cost overtime, the

$CPI \times SPI$ version is the most conservative. Report the one your evidence supports — and say which assumption you made.

12.6 Reading the S-curve honestly

- **Spend is not progress.** A project can be on budget ($AC = PV$) and badly behind ($EV < PV$) — it just has not done the work yet.
- **The “watermelon” report** — green on the outside, red on the inside. High-level status shows green because spend matches plan, while the earned value underneath is red.
- **Leading vs. lagging indicators.** SPI and CPI are somewhat lagging. Leading signals — rising defect find-rate, slipping sub-milestones, key staff leaving — warn you earlier. Watch both.

12.7 Integrated change control

Control only works against a **meaningful baseline**. Every change request goes through the same gate as in Chapter 3: assess the impact on scope, schedule, and cost; approve or reject; and if approved, **re-baseline** so future variance is measured against the new plan — not the old one you can no longer hit. **Configuration management** keeps the product definition (drawings, specs, software versions) in step with the approved baseline.

12.8 Status reporting

The course’s **Project Manager Report** and **Team Member Report** templates are the reporting instruments: they roll up progress, cost, schedule, risks, and issues on a fixed cadence. Good reporting is **honest** (report the red while it is still fixable), **consistent** (same metrics every period so trends are visible), and **actionable** (every variance has an owner and a next step). A dashboard that only ever shows green is not a control system — it is a comfort blanket.

12.9 Key Takeaways

- **Control** is a closed loop: **measure** → **compare to baseline** → **analyze variance** → **act**, every reporting period.
- Track **PV** (work scheduled), **EV** (work done), **AC** (money spent), all in budget dollars, against **BAC**.
- $SV = EV - PV$, $CV = EV - AC$, $SPI = EV/PV$, $CPI = EV/AC$. Below 1 / negative = behind / over.
- Forecast with **EAC** (BAC/CPI is the common honest form), **ETC** = $EAC - AC$, **VAC** = $BAC - EAC$, **TCPI** = $(BAC - EV)/(BAC - AC)$.

- **Spend is not progress.** Beware the **watermelon** report; watch **leading** indicators, not just SPI/CPI.
- Keep the **baseline meaningful** through integrated change control and configuration management.
- Report **honestly, consistently, and actionably** — every variance gets an owner.

12.10 Practice Problems

Problem 1: Full EVM snapshot

BAC = \$400 000. At the status date: PV = \$180 000, EV = \$150 000, AC = \$165 000. Compute SV, CV, SPI, CPI, EAC (BAC/CPI), ETC, VAC, TCPI, and state the project's condition in one sentence.

Solution.

```
BAC <- 400000; PV <- 180000; EV <- 150000; AC <- 165000
SV <- EV - PV; CV <- EV - AC; SPI <- EV / PV; CPI <- EV / AC
EAC <- BAC / CPI; ETC <- EAC - AC; VAC <- BAC - EAC
TCPI <- (BAC - EV) / (BAC - AC)
cat(sprintf("SV=$%s CV=$%s SPI=%.3f CPI=%.3f\n",
            format(SV, big.mark=" "), format(CV, big.mark=" "), SPI, CPI))
cat(sprintf("EAC=$%s ETC=$%s VAC=$%s TCPI=%.3f\n",
            format(round(EAC), big.mark=" "), format(round(ETC), big.mark=" "),
            format(round(VAC), big.mark=" "), TCPI))
```

```
SV=$-30 000 CV=$-15 000 SPI=0.833 CPI=0.909
EAC=$440 000 ETC=$275 000 VAC=$-40 000 TCPI=1.064
```

Behind schedule (SPI 0.83) and over cost (CPI 0.91); projected to finish about \$40 000 over budget, and the remaining work must run at TCPI 1.06 to still hit BAC.

Problem 2: On budget but in trouble

A project reports PV = \$500 000, EV = \$400 000, AC = \$400 000. The sponsor sees “spend exactly matches earned value — CPI = 1.0 — we’re fine.” Are they?

Answer. Cost efficiency is fine (CPI = 1.0), but **SPI = EV/PV = 0.80** — the project has completed only 80 % of the work it should have by now. It is **20 % behind schedule** and will either finish late or need to spend more (overtime, added staff) to catch up, which will then push CPI below 1. This is the classic watermelon: the cost view is green, the schedule view is red.

Problem 3 (discussion): re-baseline or not?

At week 8 the RC-vehicle project is forecast (EAC) at \$700 000 against a \$620 000 BAC, mostly because the retailer added a second body colourway (an ap-

proved scope change) and the new line's learning curve ran longer than planned. Should you re-baseline?

Discussion points. Split the two causes. The **approved scope change** is a legitimate reason to re-baseline — the extra colourway was authorised, so the budget and schedule should be formally updated and future variance measured against the new plan. The **learning-curve overrun** is *performance* against the plan, not a change to the plan — that stays as variance you must recover or explain. Re-baselining to bury performance problems (“we’ll just move the goalposts”) destroys the value of the control system; re-baselining for authorised changes keeps it honest.

12.11 Further Resources

Earned Value Management Explained: PV, EV, AC, CPI, SPI: <https://www.youtube.com/watch?v=EEIu8-87nAY>

Earned Value Analysis Calculations Tutorial (CPI, SPI, CV, SV, EAC, ETC, TCPI): <https://www.youtube.com/watch?v=7DLyZjsrG14>

Chapter 13

Project Closeout and Termination

13.1 Learning Objectives

By the end of this chapter you will be able to:

1. **State** what project termination provides and distinguish its types — extinction, addition, integration, starvation.
2. **Work through** the seven elements of closeout.
3. **Explain** what commonly prevents effective closeout.
4. **Recognize** the early warning signs of project failure.
5. **Apply** the cancel/continue decision using sunk cost, cost-to-complete, and remaining benefit.
6. **Reconcile** a completed project: planned vs. actual schedule and cost.
7. **Draw** the common lessons from famous project failures.

Before we begin: which is harder — starting a project or finishing one?

Starting has energy and a blank page. Finishing has a tired team, a client eager to move on, and a next project already calling. Closeout is *“the most difficult phase to complete effectively and has the most impact on future projects.”*

13.2 Introduction: planning for extinction

A defining feature of a project is a **finite life**. When you plan a project, you are also *“planning for its extinction.”* The RC-vehicle prototype project will end — and if it succeeds, its team likely rolls straight into supporting the 100 000-unit production run. That is a real, named way for a project to end, not an afterthought.

Project termination provides for:

- **Acceptance** of the project by the sponsor
- **Completion** of the project records
- **Final revision and issue** of documentation to its as-built condition
- **Retention** of the essential documentation — above all, the **lessons learned**

Unlock the Secrets to Easy Project Closure (Association for Project Management): <https://www.youtube.com/watch?v=0HVtpstAT7M>

```
tt <- data.frame(
  type = c("Extinction", "Addition", "Integration", "Starvation"),
  x = 1:4,
  desc = c("Stopped -- success or failure.\nBudget audited, team\nreassigned, assets d
    "Institutionalized as a\npermanent part of the\norganization (project team\n
    "Resources and team\nreabsorbed into the existing\nstructure / functional\n
    "Budget quietly cut until\nthe project cannot survive.\nOften political -- a
)
ggplot(tt, aes(x, 0)) +
  geom_label(aes(label = type), nudge_y = 0.55, size = 3.6 * diagram_text_scale,
    fontface = "bold", fill = c("#A9DFBF", "#A9DFBF", "#A9DFBF", "#F5B7B1"),
    label.size = 0) +
  geom_text(aes(label = desc), size = 2.6 * diagram_text_scale, lineheight = 0.95,
    vjust = 1) +
  coord_cartesian(xlim = c(0.4, 4.6), ylim = c(-1.1, 1)) +
  theme_void()
```

Warning: The `label.size` argument of `geom\_label()` is deprecated as of ggplot2 3.5.0. Please use the `linewidth` argument instead.

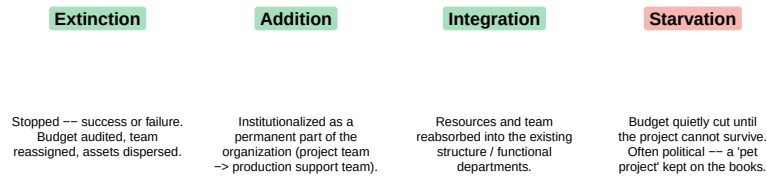


Figure 13.1: Four ways a project ends. Extinction, addition, and integration are normal; starvation is a slow, deliberate neglect.

- **Extinction** — the project is stopped, successfully or not. The final budget is audited, team members get new assignments, material assets are dispersed or transferred per company/contract terms.
- **Addition** — a successful project is made a **formal part of the organization**. An automaker that has launched a new vehicle keeps the project team on as a **production support team**.
- **Integration** — the team and resources are **reabsorbed** into the existing structure. Common but complicated — and organizations often **lose key people** here, who leave for fresh project challenges rather than return to the old department.
- **Starvation** — not really a termination act but “*a willful form of neglect*”: the budget is decreased slowly until the project cannot remain viable. Kept “on the books” for political reasons, a powerful sponsor’s pet project, or so it can be reactivated when budgets recover.

13.4 The seven elements of closeout

```
close <- data.frame(
  `#` = 1:7,
  Element = c("Verify scope", "Closeout contracts", "Closeout administration",
    "Conduct a lessons-learned review", "Build a project history file",
    "Create the final project report", "Celebrate success"),
  What = c(
    "Compare to the original scope statement; resolve discrepancies; confirm every deliverable is",
    "Resolve all open issues with the customer; return or destroy proprietary documents as directed",
    "Return all loaned personnel; complete performance evaluations.",
    "Survey everyone who was on the project; hold the meeting; be critical -- dissect what went wrong",
    "Document planned vs. actual schedule and cost (with reasons), approved changes, meeting minutes"
```

```

    "Document overall success, project organization, recommended changes for similar p
    "Involve everyone who played a role; hold it away from the workplace; personally r
  ),
  check.names = FALSE
)
knitr::kable(close, align = c("c", "l", "l"),
              caption = "The seven elements of project closeout.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                            full_width = FALSE) |>
  kableExtra::column_spec(2, bold = TRUE)

```

#	Element	What
1	Verify scope	Compare to the original scope statement; resolve discrep
2	Closeout contracts	Resolve all open issues with the customer; return or dest
3	Closeout administration	Return all loaned personnel; complete performance evalu
4	Conduct a lessons-learned review	Survey everyone who was on the project; hold the meetin
5	Build a project history file	Document planned vs. actual schedule and cost (with re
6	Create the final project report	Document overall success, project organization, recomm
7	Celebrate success	Involve everyone who played a role; hold it away from th

13.4.1 What prevents effective closeout

- Getting the project **signed off** discourages the other closeout activities.
- The **assumed urgency** of every project pressures the team to cut corners on the back end.
- Closeout is given **low priority** and easily ignored.
- **Lessons learned** is dismissed as a bookkeeping chore.
- People assume that because every project is unique, **carryover is minimal**.
- People just want to **move on** to the next challenge.

13.5 Project failure

Failure = the project does not deliver what was required, judged against the **acceptance criteria** of the sponsor and customer, and can be triggered by changes in **scope, cost, or time**.

Early warning signs:

- Lack of viable **commercial objectives**
- Lack of sufficient **decision-making authority**

- **Market volatility**
- **Low priority** assigned to the project

When to consider cancelling:

- Costs **exceed the business benefits**
- The project no longer meets **strategic-fit** criteria
- **Deadlines** keep being missed
- **Technology evolves** beyond the project's scope

13.5.1 Worked Example 1: continue or cancel?

A project has already spent **\$480 000** (sunk). Finishing it needs another **\$260 000**. The product, if delivered, is now worth an estimated **\$300 000** in benefit — the market moved while you were building.

```
sunk          <- 480000      # already spent -- IRRELEVANT to the decision
cost_to_finish <- 260000
remaining_benefit <- 300000

net_if_continue <- remaining_benefit - cost_to_finish
cat(sprintf("Sunk cost (ignore it)          : %s\n", format(sunk, big.mark = " ")))
cat(sprintf("Cost to finish                  : %s\n", format(cost_to_finish, big.mark = " ")))
cat(sprintf("Benefit if finished              : %s\n", format(remaining_benefit, big.mark = " ")))
cat(sprintf("Net from *here*                  : %s\n", format(net_if_continue, big.mark = " ")))
cat(ifelse(net_if_continue > 0,
           "\n-> CONTINUE: the remaining benefit still beats the remaining cost.\n",
           "\n-> CANCEL: finishing loses money from here, regardless of the $480k already gone.\n"))
```

```
Sunk cost (ignore it)          : $480 000
Cost to finish                  : $260 000
Benefit if finished              : $3e+05
Net from *here*                  : $40 000
```

-> CONTINUE: the remaining benefit still beats the remaining cost.

The \$480 000 is a **sunk cost** — it is spent whether you continue or not, so it must not enter the decision. From here, spending \$260 000 to get \$300 000 is still worth \$40 000. (The sunk-cost fallacy — “we’ve come too far to stop” — is what keeps doomed projects alive.)

13.5.2 Try it yourself: the cancel decision under uncertainty

```
#| label: cancel-decision-13b
sunk          <- 480000
cost_to_finish <- 260000
```

```
# The remaining benefit is uncertain -- try a range of market outcomes.
for (benefit in c(150000, 250000, 300000, 400000)) {
  net <- benefit - cost_to_finish
  verdict <- ifelse(net > 0, "CONTINUE", "CANCEL")
  cat(sprintf("If remaining benefit = %s -> net from here %+d -> %s\n",
              format(benefit, big.mark = " "), net, verdict))
}
cat(sprintf("\nBreak-even benefit: %s. Below that, finishing destroys value\n",
            format(cost_to_finish, big.mark = " ")))
cat("-- no matter how much has already been spent.\n")
```

```
ov <- data.frame(
  project = c("Channel Tunnel", "Canadian Firearms Registry", "Project Taurus (LSE)"),
  multiple = c(1.85, 8.4, 132)
)
ov$project <- factor(ov$project, levels = rev(ov$project))
ggplot(ov, aes(multiple, project)) +
  geom_col(fill = "#C0392B", width = 0.6) +
  geom_text(aes(label = paste0(multiple, "x")), hjust = -0.2,
            size = 3.8 * diagram_text_scale) +
  scale_x_continuous(trans = "log10", limits = c(1, 300),
                    breaks = c(1, 2, 5, 10, 30, 100, 300),
                    expand = expansion(mult = c(0, 0.05))) +
  labs(title = "Final cost as a multiple of the original budget (log scale)",
       x = "Overrun multiple", y = NULL) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 13 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )
```

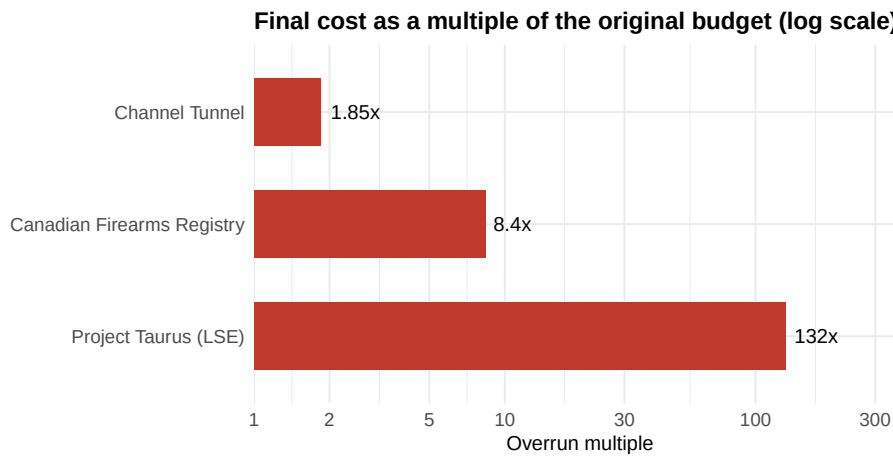


Figure 13.2: Cost overrun as a multiple of the original budget for three well-documented failures. Project Taurus (London Stock Exchange) ran to roughly 132 times its first budget before being abandoned after 11 years.

- **Bank of America “MasterNet”** (1984) — trust-department automation. Over budget by \$35M, rising toward \$80M; a 1988 failure meant loss of control over billions in trust accounts, all eventually recovered, but the **entire 255-person department was let go** as depositors fled. *Cause: scope and complexity understood far too late.*
- **Future Imagery Architecture** (US spy satellites, 1997) — bid won by a contractor that had never built an imaging satellite; internal estimate already predicted a \$4B overrun. **Testing was cut to save money**; a subcontractor’s undisclosed manufacturing change caused gyroscope defects undiscovered for three years; costs reached ~\$18B by 2005. *Cause: unqualified bidder, cut testing, uncontrolled change.*
- **Canadian Firearms Registry** (1995) — estimated at \$119M to implement; a 2002 audit projected ~\$1B. *Causes: scope not understood, no plan for inter-agency cooperation, inconsistent data criteria, cost cover-up, no measurement of whether the registry actually worked.*
- **Channel Tunnel** (budgeted 1979, opened 1994, one year late) — final cost roughly **double** the original budget.
- **Project Taurus** (London Stock Exchange paperless settlement, 1983) — abandoned after **11 years** at **~132 times** its original budget after switching to an incompatible US system that clashed with British finance law.

The recurring threads: scope not truly understood; optimism and cost concealment; cutting testing to save money; no plan for the interfaces between organizations; and pressing on past every off-ramp because of sunk cost.

13.6.1 Try it yourself: the final project reconciliation

```
#| label: reconciliation-13
# Planned vs actual for the RC-vehicle prototype project at closeout.
planned_weeks <- 30;      actual_weeks <- 34
planned_cost  <- 620000;  actual_cost  <- 703000
planned_scope_items <- 12; delivered_items <- 12  # all deliverables met

sched_var_pct <- 100 * (actual_weeks - planned_weeks) / planned_weeks
cost_var_pct  <- 100 * (actual_cost - planned_cost) / planned_cost

cat(sprintf("Schedule : planned %d wk, actual %d wk  (%+.1f%%)\n",
            planned_weeks, actual_weeks, sched_var_pct))
cat(sprintf("Cost      : planned $%s, actual $%s  (%+.1f%%)\n",
            format(planned_cost, big.mark = " "),
            format(actual_cost, big.mark = " "), cost_var_pct))
cat(sprintf("Scope    : %d of %d deliverables accepted\n",
            delivered_items, planned_scope_items))

verdict <- if (delivered_items == planned_scope_items && cost_var_pct < 20 && sched_var_pct < 20) {
  "SUCCESS with overruns -- document the causes in the history file and lessons learned"
} else {
  "TROUBLED -- the lessons-learned review is the priority deliverable."
}
cat("\n", verdict, "\n", sep = "")
```

13.7 Closing out the RC-vehicle project

- **Verify scope** against the signed scope statement: all body options delivered, drop and EMC tests passed, compliance approvals in hand, retailer satisfaction confirmed.
- **Closeout contracts**: settle final invoices with MCS/Paragon and the carbon-fibre supplier; return the customer's proprietary artwork.
- **Administration**: return the borrowed design engineers and millwrights to their functional managers with completed performance reviews.
- **Lessons learned**: the new-line learning curve and the unproven-supplier risk are the two entries that matter most for the next program.
- **Termination by addition**: the prototype team transitions into the **production support team** for the 100 000-unit run.
- **Celebrate** — away from the plant, with everyone who contributed.

13.8 Key Takeaways

- Termination provides **sponsor acceptance, complete records, as-built documentation, and retained lessons learned**.
- Types: **extinction** (stopped), **addition** (becomes permanent), **integra-**

tion (reabsorbed), **starvation** (slow deliberate neglect).

- Work the **seven elements**: verify scope, close contracts, close administration, lessons learned, history file, final report, celebrate.
- Closeout fails because sign-off feels like “done”, the next project is louder, and lessons learned is treated as bookkeeping. Resist all three.
- **Cancel** when the **cost to complete exceeds the remaining benefit** — the money already spent is **sunk** and irrelevant to that decision.
- Famous failures share the same causes: **scope not understood, cost concealed, testing cut, interfaces unplanned, sunk-cost momentum**.

13.9 Practice Problems

Problem 1: Sunk cost trap

A project has consumed \$1.2M of a \$1.5M budget. Finishing needs \$500 000 (the last \$300 000 of budget plus a \$200 000 overrun). The delivered product is now worth \$350 000. The sponsor says “we’ve spent \$1.2M — we can’t stop now.” Evaluate.

Solution.

```
cost_to_finish <- 500000
remaining_benefit <- 350000
cat(sprintf("Net from here: %s - %s = %s\n",
           format(remaining_benefit, big.mark = " "),
           format(cost_to_finish, big.mark = " "),
           format(remaining_benefit - cost_to_finish, big.mark = " ")))
```

Net from here: \$350 000 - \$5e+05 = \$-150 000

Finishing turns \$500 000 of new spend into \$350 000 of value — a \$150 000 loss **from today**. The \$1.2M is gone either way. Cancel (or de-scope until the remaining cost drops below \$350 000). “We can’t stop now” is the sunk-cost fallacy.

Problem 2: Which termination type?

1. The RC-vehicle prototype team becomes the production line’s support team. → **Addition**.
2. A cancelled R&D project’s engineers return to the design department. → **Integration**.
3. A failed pilot is shut down, its budget audited, its test rig sold. → **Extinction**.
4. A VP’s favourite project keeps a \$1/year budget so it stays “active” on paper. → **Starvation**.

Problem 3 (discussion): the team wants to skip lessons learned

The RC-vehicle project delivered late and over budget but the product passed. The exhausted team wants to skip the lessons-learned review and start on the production ramp. As PM, what do you do and why?

Discussion points. This is exactly the moment the review matters most — a troubled project is where the carryover value is highest, and the next program will hit the *same* new-line learning curve and *same* supplier risk. Keep it short and structured, focus on **what went wrong and why** (not blame), capture 2–3 concrete, assigned actions, and put it in the history file before the team disperses and the memory fades. Skipping it guarantees the next project repeats the overrun.

13.10 Further Resources

What Is Project Management? (Association for Project Management): <https://www.youtube.com/watch?v=Jk-JwtScIlw>

Bibliography